Supplementary Clinical-Evidence Corroboration

1 Complete Original-Cohort Estimator Matrix

Table 1 reports every prespecified original-cohort dataset–exposure comparison for CausalForestDML, LinearDML, and CausalPFN. Values are retained irrespective of magnitude, direction, or cross-estimator agreement. The table reports mean model-estimated CATE summaries over each analyzed sample; these are not ground-truth causal effects.

Table 1: Complete original-cohort estimator summaries (CF-DML denotes CausalForestDML). Values are mean model-estimated CATE summaries over the analyzed samples; display precision is six decimal places.

Dataset	Exposure	CF-DML	LinearDML	CausalPFN	Sign agreement
MIMIC-III	Cardiac strain	+0.220356	+0.187634	+0.259327	all three
MIMIC-III	Inflammation/sepsis	+0.161179	+0.161334	+0.148544	all three
MIMIC-III	Hepatic/coag. dysfunction	+0.097598	+0.083017	+0.096846	all three
MIMIC-III	Renal dysfunction	+0.091290	+0.088595	+0.086972	all three
MIMIC-III	Global severity	+0.062101	+0.080153	+0.073050	all three
MIMIC-III	Respiratory failure	+0.036133	+0.039098	+0.049766	all three
MIMIC-III	Neurologic dysfunction	+0.034253	+0.031550	+0.034993	all three
MIMIC-III	Shock	+0.020994	+0.021246	+0.031645	all three
MIMIC-III	Metabolic derangement	+0.019743	+0.017989	+0.031323	all three
PhysioNet 2012	Renal dysfunction	+0.120027	+0.090622	+0.110669	all three
PhysioNet 2012	Cardiac injury/strain	+0.111831	+0.122033	+0.119413	all three
PhysioNet 2012	Global severity	+0.107988	+0.106740	+0.105039	all three
PhysioNet 2012	Hepatic dysfunction	+0.090527	+0.060184	+0.106843	all three
PhysioNet 2012	Neurologic dysfunction	+0.081636	+0.075728	+0.077049	all three
PhysioNet 2012	Metabolic derangement	+0.077704	+0.079738	+0.090734	all three
PhysioNet 2012	Inflammation/sepsis burden	+0.071800	+0.070872	+0.067480	all three
PhysioNet 2012	Respiratory failure	+0.064750	+0.062660	+0.062984	all three
PhysioNet 2012	Coag./heme dysfunction	+0.010794	+0.004136	+0.025428	all three
PhysioNet 2012	Shock	-0.013849	-0.026944	+0.004122	disagreement

2 Clinical Evidence Used to Construct the Proxy Exposures

Table 2 documents the 19 admitted proxy exposures used by CliniCause: nine in MIMIC-III and ten in PhysioNet 2012. Only sources explicitly recoverable from the preserved ChatGPT responses are called *ChatGPT-supplied*. Later literature checks are identified separately and must not be interpreted as original prompt provenance. The rules below are analytical proxy definitions rather than chart-adjudicated diagnoses; a cited source may support the broad clinical construct or an individual clause without validating the complete local Boolean composite.

Table 2: Proxy-definition evidence for all admitted CliniCause exposures. Each row reports the implemented source columns, exact cutoff and aggregation rule, clinical interpretation, and the preserved ChatGPT source evidence with a bounded support paraphrase and DOI or stable URL.

Dataset/proxy	Source columns	Exact cutoff, combination, and aggregation rule	Clinical interpretation	ChatGPT-supplied support, DOI/URL, and evidence qualification
MIMIC-III M01 Inflammation/ sepsis	Temperature min/max; WBC min/max; lactate max; culture ordered/positive; suspected infection; antibiotic started/any.	Five indicators: temperature $\geq$ 38.3 or $<$ 36; WBC $>$ 12 or $<$ 4; lactate $>$ 2; culture or suspected-infection evidence; antibiotic or suspected-infection evidence. Positive when score $\geq$ 2 and a culture, antibiotic, temperature, or WBC anchor is present. Whole available ICU record summaries; missing clauses do not fire.	Inflammation or suspected-infection/sepsis burden; not confirmed sepsis.	Original ChatGPT source: Sepsis-3, doi:10.1001/jama.2016.0287. It supports infection-related dysregulated host response and organ dysfunction, but not this five-clause score or its exact threshold.
MIMIC-III M02 Global severity	MAP/SBP; vasopressor; SpO_2; P/F ratio; ventilation; GCS/RASS; creatinine/urine; lactate/pH/bicarbonate; temperature/WBC; bilirubin/platelets/INR.	Seven Boolean domains: circulatory, respiratory, neurologic, renal, metabolic, inflammatory, and hepatic/coagulation. Positive when at least three domains are abnormal. Whole-record extrema/first values and any-flags; urine uses first 24 h and minimum rolling 6 h summaries; missing clauses do not fire.	Multi-domain acute physiological severity.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`. Later verification: original SOFA paper, doi:10.1007/BF01709751. These support six organ domains and daily worst values, not the local seven-domain Boolean score or cutoff of three.
MIMIC-III M03 Shock	MAP/SBP; sustained MAP $<$ 70 flag; vasopressor; lactate; 24 h and rolling 6 h urine; pH; base excess.	Five indicators: MAP $<$ 65, SBP $\leq$ 90, or sustained MAP $<$ 70; vasopressor; lactate $>$ 2; urine $<$ 0.5 mL/kg/h over 6 h or $<$ 500 mL/24 h; pH $<$ 7.30 or base excess $\leq$ -5. Positive when score $\geq$ 2, or vasopressor plus hypoperfusion, acidosis, or hypotension. Whole-record aggregation; missing clauses do not fire.	Circulatory shock or clinically meaningful hemodynamic instability.	Original ChatGPT sources: Sepsis-3, doi:10.1001/jama.2016.0287, and the MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`. They support vasopressor-dependent hypotension, lactate $>$ 2, and cardiovascular-dysfunction concepts, not the project's broader any-two composite.
MIMIC-III M04 Respiratory failure	SpO_2; P/F ratio; FiO_2; invasive/non-invasive ventilation; respiratory rate; $PaCO_2$; pH.	Five indicators: SpO_2 $<$ 92; P/F $\leq$ 300; respiratory support or FiO_2 $\geq$ 0.5; respiratory rate $\geq$ 30 or $\leq$ 8; $PaCO_2$ $\geq$ 50 with pH $<$ 7.30. Positive when score $\geq$ 2, or support plus hypoxemia, low P/F, or ventilatory failure. Whole-record aggregation; missing clauses do not fire.	Hypoxemic or ventilatory failure and/or substantial respiratory support.	Original ChatGPT source: MSD Berlin-definition table, `https://www.msdmanuals.com/professional/multimedia/table/berlin-definition-of-ards`. Later verification: Berlin definition, doi:10.1001/jama.2012.5669. It supports P/F severity categories under PEEP/CPAP together with timing, imaging, and edema-origin requirements; the project proxy is broader and is not an ARDS diagnosis.
MIMIC-III M05 Renal dysfunction	Creatinine max/delta; BUN; 24 h and rolling 6 h urine; potassium; bicarbonate; dialysis/CRRT.	Five indicators: creatinine $\geq$ 2 or rise $\geq$ 0.3; BUN $\geq$ 40; urine $<$ 0.5 mL/kg/h over 6 h or $<$ 500 mL/24 h; potassium $\geq$ 5.5 or bicarbonate $<$ 18; dialysis. Positive when score $\geq$ 2 or dialysis is present. Whole-record aggregation; missing clauses do not fire.	Acute or clinically important renal dysfunction.	Original ChatGPT source: MSD KDIGO staging table, `https://www.msdmanuals.com/professional/multimedia/table/staging-criteria-for-acute-kidney-injury-kdigo-2012`. Later verification: KDIGO AKI guideline, `https://kdigo.org/guidelines/acute-kidney-injury/`. These support a 0.3 mg/dL creatinine rise, baseline-relative change, and timed urine criteria, not the BUN/electrolyte composite or its any-two rule.

Dataset/proxy	Source columns	Exact cutoff, combination, and aggregation rule	Clinical interpretation	ChatGPT-supplied support, DOI/URL, and evidence qualification
MIMIC-III M06 Hepatic/coag. dysfunction	Bilirubin; AST/ALT; platelets; INR; prolonged PT/PTT; albumin.	Five indicators: bilirubin $\geq$ 2; AST or ALT $\geq$ 120; platelets $<$ 100; INR $\geq$ 1.5 or prolonged PT/PTT; albumin $<$ 2.5. Positive when score $\geq$ 2. Whole-record aggregation; missing clauses do not fire.	Combined hepatic injury/reserve and coagulation dysfunction.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; later verification used the original SOFA paper, doi:10.1007/BF01709751. SOFA supports bilirubin and platelet domains separately, not this combined five-indicator score.
MIMIC-III M07 Neurologic dysfunction	GCS; abnormal GCS components; RASS; pupil/focal-neurologic abnormality; sedation/intubation context.	GCS $\leq$ 8 contributes 2; otherwise GCS $\leq$ 13 contributes 1. Abnormal GCS component, RASS $\leq$ -3 or $\geq$ 2, and pupil/focal abnormality each contribute 1. Positive when score $\geq$ 2; sedation/intubation with an isolated one-point score is forced negative. Whole-record aggregation; missing clauses do not fire.	Depressed consciousness or focal/behavioral neurologic dysfunction.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; later verification: doi:10.1007/BF01709751. SOFA supports GCS as the neurologic domain, not the RASS/focal-sign weighting or sedation exclusion.
MIMIC-III M08 Metabolic derangement	pH; bicarbonate; base excess; lactate; potassium; sodium; glucose.	Six domains: pH $<$ 7.30 or $>$ 7.55; bicarbonate $<$ 18 or $>$ 35 or base excess $\leq$ -5; lactate $>$ 2; potassium $<$ 3 or $\geq$ 5.5; sodium $<$ 130 or $>$ 150; glucose $<$ 70 or $>$ 250. Positive when at least two domains are abnormal. Whole-record aggregation; missing clauses do not fire.	Acid-base, lactate, electrolyte, or glucose derangement.	No proxy-specific original ChatGPT source or DOI/URL was recovered. No locally inspected paper validates the exact six-domain score, its cutoffs, or the any-two combination.
MIMIC-III M09 Cardiac strain	Troponin flags and maxima; CK-MB; arrhythmia; heart rate; MAP/SBP; first care unit; cardiac-surgery context.	Five indicators: abnormal troponin, with fallback T $\geq$ 0.1 or I $\geq$ 0.4; abnormal CK-MB; arrhythmia or HR $>$ 150/$<$ 40; HR $>$ 130 or $<$ 40 with hypotension; CCU/CSRU or cardiac-surgery context. Positive when score $\geq$ 2 and a troponin or rhythm anchor is present. Whole-record aggregation; missing clauses do not fire.	Myocardial injury, rhythm instability, or severe cardiac strain.	No proxy-specific original ChatGPT source was recovered. A later myocardial-injury source was identified, doi:10.1161/CIR.0000000000000617, but it was not preserved as original ChatGPT provenance and its full text was not available for this audit. The assay-specific thresholds and composite remain unverified.
PhysioNet P01 Global severity	MAP/NIMAP/SBP; ventilation; P/F or PaO_2/FiO_2 approximation; SaO_2; respiratory rate; GCS; creatinine/BUN/urine; bilirubin/platelets; lactate/pH/HCO_3; troponin/HR.	Seven domains: hemodynamic, respiratory, neurologic, renal, hepatic/coagulation, metabolic, and cardiac. Positive when score $\geq$ 3, or score $\geq$ 2 plus lactate $\geq$ 4, pH $<$ 7.20, ventilation, GCS $\leq$ 8, or MAP $<$ 60. Whole first-48 h record using min/max/mean/first/last summaries; missing clauses do not fire.	Multi-domain first-48 h acute physiological severity.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; later verification: doi:10.1007/BF01709751. SOFA supports six organ domains and ordinal thresholds, not the local seven-domain score or critical override.
PhysioNet P02 Shock	MAP/NIMAP/SBP; lactate; heart rate; urine summary; pH; bicarbonate.	Six indicators: MAP or NIMAP $<$ 70; SBP or non-invasive SBP $\leq$ 90; lactate $>$ 2; HR $\geq$ 110; urine $<$ 500 mL; pH $<$ 7.30 or bicarbonate $<$ 18. Positive when score $\geq$ 2, or low MAP with lactate $\geq$ 4. Whole first-48 h record; missing clauses do not fire.	Circulatory shock or hemodynamic instability without vasopressor data.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; later verification: doi:10.1007/BF01709751. It supports MAP $<$ 70 as cardiovascular dysfunction, not the complete any-two score or lactate override.
PhysioNet P03 Respiratory failure	Ventilation; P/F or PaO_2/FiO_2 approximation; SaO_2/PaO_2; respiratory rate; $PaCO_2$; pH.	Five indicators: ventilation; P/F $<$ 300; SaO_2 $<$ 92 or PaO_2 $<$ 60; respiratory rate $\geq$ 22 or $<$ 8; $PaCO_2$ $\geq$ 50, or $PaCO_2$ $\geq$ 45 with pH $<$ 7.30. Positive when score $\geq$ 2, or ventilation plus P/F $<$ 300. Whole first-48 h record; missing clauses do not fire.	Hypoxemic or ventilatory respiratory failure/support.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`, supporting P/F oxygenation scoring. Later verification: Berlin definition, doi:10.1001/jama.2012.5669. Neither source validates the full composite, and the proxy is not an ARDS diagnosis.
PhysioNet P04 Renal dysfunction	Creatinine first/max; BUN; urine summary; potassium; bicarbonate.	Five indicators: creatinine $\geq$ 2; creatinine rise $\geq$ 0.3; BUN $\geq$ 40; urine $<$ 500 mL; potassium $\geq$ 5.5 or bicarbonate $<$ 18. Positive when score $\geq$ 2 or creatinine $\geq$ 3.5. Whole first-48 h record; missing clauses do not fire.	Acute or clinically important renal dysfunction.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`. Later verification: KDIGO AKI guideline, `https://kdigo.org/guidelines/acute-kidney-injury/`. These support creatinine/urine domains and the 0.3 rise, not the local full score or sufficient condition.

Dataset/proxy	Source columns	Exact cutoff, combination, and aggregation rule	Clinical interpretation	ChatGPT-supplied support, DOI/URL, and evidence qualification
PhysioNet P05 Hepatic dysfunction	Bilirubin; AST/ALT; alkaline phosphatase; albumin; platelets.	Weighted score: bilirubin ≥ 2 contributes 2; AST or ALT ≥ 200, ALP ≥ 250, albumin < 2.5, and platelets < 100 each contribute 1. Positive when score ≥ 2. Whole first-48 h record; missing clauses do not fire.	Cholestatic/hepatocellular injury or impaired hepatic reserve.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; later verification: doi:10.1007/BF01709751. SOFA supports bilirubin as a liver-domain marker, not the enzyme, albumin, platelet weighting, or the local threshold combination.
PhysioNet P06 Coagulation/ hematologic	Platelets first/min; hematocrit min/max; WBC min/max.	Weighted score: platelets < 100 contributes 2; platelets < 150, hematocrit < 25 or > 55, WBC < 4 or > 20, and platelet drop ≥ 50 each contribute 1. Positive when score ≥ 2. Whole first-48 h record; missing clauses do not fire.	Thrombocytopenia, anemia/polycythemia, or hematologic stress.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`, supporting platelet thresholds. Later verification used an ISTH DIC source, doi:10.1055/s-0037-1616068, which uses a different multi-test algorithm. Neither validates the HCT/WBC additions or local weighting.
PhysioNet P07 Inflammation/ sepsis	Temperature; WBC; heart rate; respiratory rate; $PaCO_2$; lactate; platelets.	Six indicators: temperature > 38.3 or < 36; WBC > 12 or < 4; HR > 90; respiratory rate > 20 or $PaCO_2 < 32$; lactate > 2; platelets < 150. Positive when score ≥ 3, or abnormal temperature/WBC plus lactate and score ≥ 2. Whole first-48 h record; missing clauses do not fire.	Systemic inflammatory or sepsis-like host response; not confirmed sepsis.	Original ChatGPT source: Sepsis-3, doi:10.1001/jama.2016.0287. It supports infection-related organ dysfunction and supersedes a simple inflammation-only definition; the local record lacks an infection anchor, and the SIRS-like score is not Sepsis-3.
PhysioNet P08 Neurologic dysfunction	GCS; sodium; glucose; $PaCO_2$; SaO_2; pH.	GCS ≤ 12 contributes 2; GCS < 15, sodium < 130 or > 150 or glucose < 70 or > 300, and $PaCO_2 \geq 50$/$SaO_2 < 90$/pH < 7.25 each contribute 1. Positive when score ≥ 2. Whole first-48 h record; missing clauses do not fire.	Depressed consciousness or neurologic dysfunction with metabolic/gas contributors.	Original ChatGPT source: MSD SOFA table, `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; later verification: doi:10.1007/BF01709751. SOFA supports GCS as a neurologic domain, not the added electrolyte and gas clauses or their weighting.
PhysioNet P09 Cardiac injury/ strain	Troponin I/T; ICU type; heart rate; MAP/SBP; lactate.	Weighted score: troponin I > 0.1 or T > 0.01 contributes 2; ICU type 1/2, HR ≥ 130 or < 50, MAP < 70 or SBP ≤ 90, and lactate > 2 each contribute 1. Positive when score ≥ 2. Whole first-48 h record; missing clauses do not fire.	Myocardial injury, ischemia, or severe cardiac strain.	No proxy-specific original ChatGPT source was recovered. A later myocardial-injury source was identified, doi:10.1161/CIR.0000000000000617, but it was not preserved as original ChatGPT provenance and its full text was unavailable. Assay thresholds, ICU-type contribution, and the composite remain unverified.
PhysioNet P10 Metabolic derangement	pH; bicarbonate; lactate; sodium; potassium; magnesium; glucose; $PaCO_2$.	Six domains: pH < 7.30 or > 7.50; bicarbonate < 18 or > 32; lactate > 2; sodium < 130 or > 150, potassium < 3 or > 5.5, or magnesium < 0.6 or > 1.2; glucose < 70 or > 250; $PaCO_2 < 32$ or > 50. Positive when score ≥ 2, or pH < 7.20, lactate ≥ 4, or potassium ≥ 6. Whole first-48 h record; missing clauses do not fire.	Acid-base, lactate, electrolyte, or glucose derangement.	No proxy-specific original ChatGPT source or DOI/URL was recovered. No locally inspected paper validates this exact multi-domain score, the domain grouping, or the critical overrides.

3 Clinical-Evidence Corroboration

3.1 Scope and Interpretation

Table 3 reports the complete numerical comparison between the original-cohort CliniCause estimates and mortality contrasts identified in the clinical literature. None of the cited studies provides a fully commensurate ground-truth causal effect for the same proxy definition, ICU population, adjustment strategy, and in-hospital mortality outcome used by CliniCause. Most comparators are propensity-score-matched or unadjusted mortality differences. Two studies use stronger causal methods — targeted minimum loss-based estimation for ARDS and a marginal structural model for delirium—but their constructs and outcome horizons remain different from the corresponding CliniCause analyses. The comparisons therefore assess direction and approximate scale; they do not score CliniCause against known causal truth.

Table 3: Complete comparison of CliniCause mean model-estimated CATE ranges with numerical mortality contrasts reported in clinical studies. Values are percentage points (pp). * denotes a matched observational contrast, ** an unadjusted observed contrast, and † a mortality horizon other than the in-hospital outcome used by CliniCause. Published values may differ in population, construct severity, exposure definition, adjustment strategy, and outcome horizon.

Clinical construct	MIMIC-III (pp)	PhysioNet (pp)	Published mortality contrast (pp)
Renal dysfunction / AKI	8.7–9.1	9.1–12.0	8.1*† [6]
Hepatic dysfunction / bilirubin[a]	8.3–9.8	6.0–10.7	7.4* [14]
Cardiac strain / injury	18.8–25.9	11.2–12.2	17.0** [3]; 17.9* [9]
Inflammation / sepsis burden	14.9–16.1	6.7–7.2	5.0* [11]; 9.1* [5]; 8.9*† [2]
Global severity	6.2–8.0	10.5–10.8	No sufficiently comparable numerical contrast identified
Shock	2.1–3.2	−2.7–0.4	11.1**† [8]
Coagulation / hematologic dysfunction[b]	—	0.4–2.5	19.5* [12]; 15.3** [1]
Respiratory dysfunction	3.6–5.0	6.3–6.5	15.0† [13]; 10.2–21.0*† [10]
Neurologic dysfunction	3.2–3.5	7.6–8.2	0.9† [7]
Metabolic dysfunction / hyperlactatemia	1.8–3.1	7.8–9.1	31.0** [4]

[a]The MIMIC-III proxy combines hepatic and coagulation dysfunction. [b]MIMIC-III has no separate coagulation/hematologic proxy. CliniCause ranges span CausalForestDML, LinearDML, and CausalPFN on the original cohorts.

3.2 Closest Comparisons

The renal/AKI comparison provides the closest correspondence in construct and scale. A propensity-score-matched study reports 8.1 pp of attributable 30-day mortality, compared with CliniCause ranges of 8.7–9.1 pp in MIMIC-III and 9.1–12.0 pp in PhysioNet. The hepatic estimates of 6.0–10.7 pp also contain the matched bilirubin contrast of 7.4 pp, and the cardiac estimates of 11.2–25.9 pp bracket the published 17.0- and 17.9-pp contrasts. For inflammation/sepsis burden, the PhysioNet range of 6.7–7.2 pp lies near published matched contrasts of 5.0–9.1 pp, whereas the MIMIC-III range of 14.9–16.1 pp is larger.

The bilirubin comparator was produced by an independent research team using MIMIC-III and the same 2-mg/dL bilirubin threshold represented within the CliniCause hepatic/coagulation rule. This is an external analysis of the same source database, not independent-population replication. Moreover, the CliniCause proxy combines bilirubin with additional hepatic and coagulation signals, so the two exposures are not identical.

3.3 Discrepancies and Possible Explanations

Shock is the clearest disagreement. The literature comparator reports an 11.1-pp mortality difference between septic shock and sepsis without shock, whereas CliniCause estimates 2.1–3.2 pp in MIMIC-III and −2.7–0.4 pp in PhysioNet. The PhysioNet result is therefore both substantially smaller and, for two estimators, opposite in direction.

Several other constructs differ materially in scale. Published ARDS effects or matched contrasts of 10.2–21.0 pp exceed the CliniCause respiratory ranges of 3.6–6.5 pp. Severe-thrombocytopenia contrasts of 15.3–19.5 pp exceed the PhysioNet coagulation range of 0.4–2.5 pp, and the 31.0-pp unadjusted lactic-acidosis contrast is larger than the CliniCause metabolic ranges of 1.8–9.1 pp. In the opposite direction, the marginal-structural-model estimate of 0.9 pp for delirium-attributable mortality is below the CliniCause neurologic ranges of 3.2–8.2 pp. No sufficiently comparable numerical contrast was identified for global severity.

These differences do not isolate a single failure mechanism. The published studies often examine narrower or more severe clinical states: ARDS includes radiographic and hypoxemia requirements, severe thrombocytopenia applies a low platelet threshold, and lactic acidosis is more extreme than mild hyperlactatemia. Their study populations include sepsis-only, surgical, COVID-19, and older vasopressor-treated cohorts, and their endpoints include ICU, 28-day, 30-day, 90-day, and post-discharge mortality. Differences may also arise from CliniCause proxy thresholds, temporal windows, treatment-mediated measurements, missingness, DAG structure, mediator adjustment, overlap, or estimator behavior.

4 Prompts for Resource Validation

Approximate agreement provides bounded evidence that several knowledge-derived representations recover clinically plausible directions and scales. It does not establish that the proxies are literal interventions, that the graph is correct, or that the estimated effects are causal truths. Conversely, large disagreements are retained as diagnostic findings rather than removed as unfavorable results. Shock, coagulation, respiratory, neurologic, and metabolic proxies are priorities for analyses that vary in definitions, thresholds, temporal windows, and adjustment sets. Such analyses are required before discrepancies can be attributed to either representation design or estimator behavior.

“‘text Conduct a targeted, source-verified literature search for clinical evidence corresponding to the following CliniCause proxy:

PROXY / CLINICAL CONSTRUCT: [INSERT CONSTRUCT NAME]

CLINICAUSE OPERATIONAL DEFINITION: [INSERT PROXY VARIABLES, CUTOFFS, AND TEMPORAL RULES]

CLINICAUSE OUTCOME: In-hospital mortality.

CLINICAUSE ESTIMATED EFFECT RANGES: - MIMIC-III: [INSERT RANGE] percentage points - PhysioNet 2012: [INSERT RANGE] percentage points

GOAL Identify peer-reviewed studies in adult ICU or closely comparable critically ill populations that estimate the mortality effect associated with the nearest recognized clinical state. The published exposure need not be identical to the CliniCause proxy, but all differences in definition, population, and mortality horizon must be stated explicitly.

PRIORITIZE EVIDENCE IN THIS ORDER 1. Explicit causal or attributable-effect estimates using target-trial emulation, g-computation, inverse-probability weighting, matching, doubly robust estimation, instrumental variables, or another clearly stated causal design. 2. Adjusted or propensity-score-matched mortality contrasts. 3. Unadjusted mortality differences, only when no stronger numerical evidence is available.

Prefer studies reporting absolute mortality risks or risk differences in percentage points. Retain odds ratios and hazard ratios only as contextual evidence; do not convert them into percentage-point effects without sufficient reported baseline risks and a justified calculation.

FOR EACH CANDIDATE STUDY, REPORT - Full title, authors, year, and peer-reviewed venue - DOI, PMID, and stable primary-source URL - ICU population, country, dataset, and sample size - Exact exposure definition and cutoff - Comparator definition - Mortality outcome and time horizon - Study design and adjustment strategy - Exact estimand and effect scale - Exposed and comparator mortality risks, if reported - Absolute mortality difference in percentage points, calculated only when supported by the reported values - Confidence interval and p-value, when available - A short exact quotation supporting the numerical claim, with table, section, or page location - Whether the result is directly comparable, partially comparable, or not comparable with the CliniCause proxy, with a brief reason - Whether the source database is independent of MIMIC-III and PhysioNet 2012

VERIFICATION RULES Use the original peer-reviewed paper or official publisher/PubMed record. Do not treat search snippets, reviews, preprints, or LLM summaries as primary verification. Do not invent missing values, quotations, page numbers, causal claims, or confidence intervals. Clearly label anything that could not be verified from the primary source. Distinguish explicit causal estimates from matched, adjusted-associational, and unadjusted contrasts.

Finish with: 1. A compact evidence table. 2. The strongest defensible numerical comparator, if one exists. 3. A comparison with the CliniCause ranges that emphasizes differences in construct, population, adjustment, and outcome horizon. 4. An explicit statement if no directly comparable causal estimate was found. “‘

5 Dataset-Specific Causal DAGs

Figures 1 and 2 show the dataset-specific directed acyclic graphs used by the causal-analysis pipeline. Blue nodes denote background or case-mix variables, orange nodes denote the latent clinical proxy variables, and green nodes denote observed measurements, care-process variables, and the mortality outcome.

The graphs were produced during the ChatGPT-assisted representation-design process and subsequently encoded in the project software. They were not learned from the observational data. Their role in the pipeline is to represent the assumed causal ordering and to support graph-based identification of candidate adjustment variables. The graph structures should therefore be interpreted as elicited modeling assumptions rather than empirically established causal ground truth. In particular, the preserved elicitation record does not provide an independently verified literature citation for every individual directed edge.

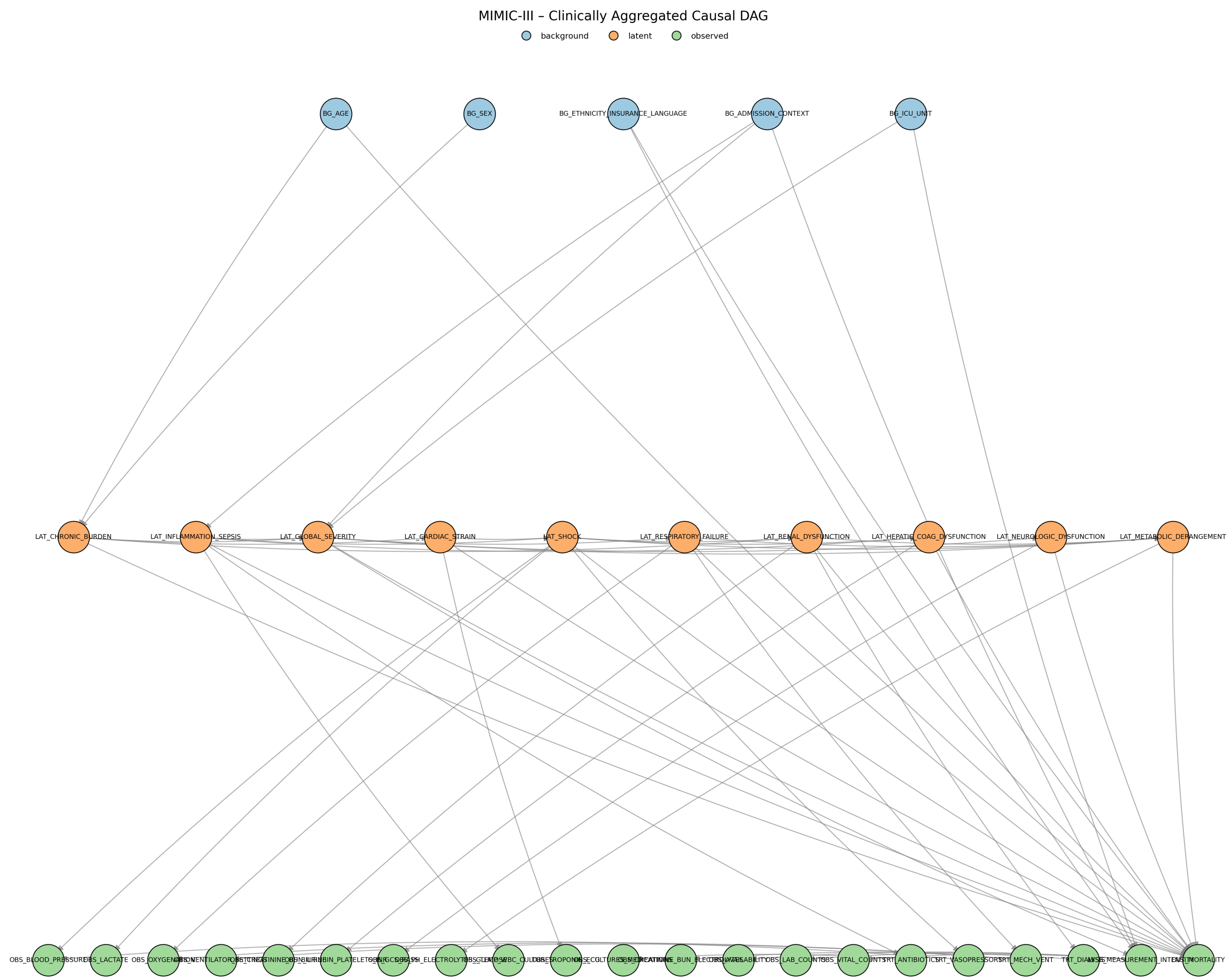

Figure 1: MIMIC-III dataset-specific causal DAG. Background and case-mix variables are shown in blue, latent clinical proxy variables in orange, and observed measurements, care-process variables, and mortality in green. The DAG was elicited as a high-level clinical representation and encoded for graph-based causal adjustment; it was not learned from the data or independently validated edge by edge.

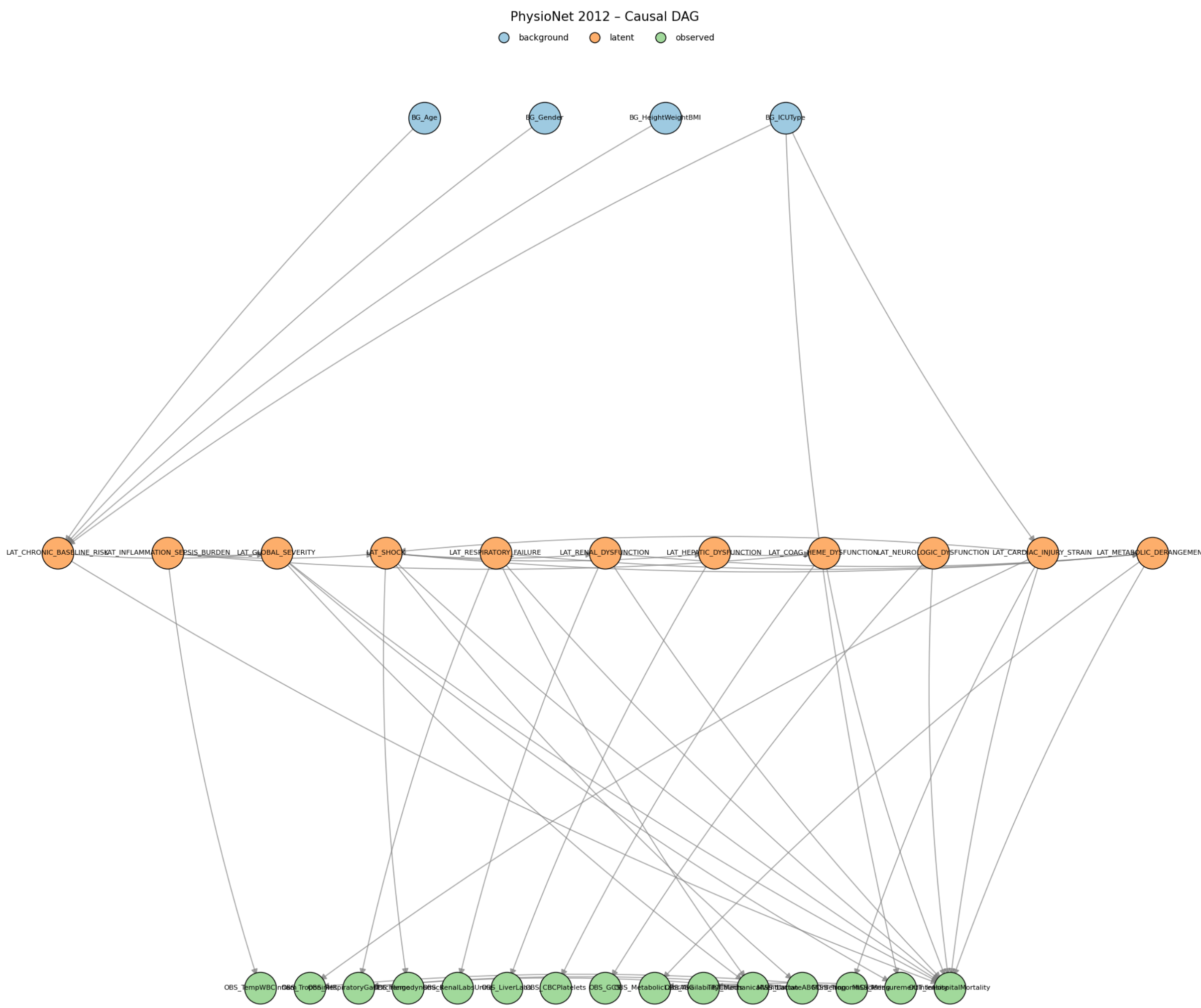

Figure 2: PhysioNet 2012 dataset-specific causal DAG. Background and case-mix variables are shown in blue, latent clinical proxy variables in orange, and observed measurements and mortality in green. The DAG was elicited as a high-level clinical representation and encoded for graph-based causal adjustment; it was not learned from the data or independently validated edge by edge.

6 Computational Environment and Code Provenance

6.1 Execution environment and hardware

All experiments were executed through Python command-line programs orchestrated by the CliniCause router, with Bash wrappers and Slurm used for server submission. The submitted Slurm configuration requested one GPU, eight CPU cores, 29 GB of system memory, and a maximum wall time of 24 hours. The job ran inside an Apptainer container with the project Conda environment mounted into the container. The Slurm request did not require a particular GPU model; depending on scheduler placement, the allocated accelerator was an NVIDIA RTX 6000 Ada Generation GPU or an NVIDIA A100 SXM4 GPU with 80 GB of device memory. The exact assigned accelerator and software-visible device state were recorded in the job log using `nvidia-smi`.

The submitted one-GPU configuration set `strats-max-concurrent=1`. Consequently, when both datasets were requested, the router could maintain separate dataset processes and output trees, but GPU-bound stages were protected by a process-safe lock and used the single assigned accelerator serially.

The router additionally supports a parallel two-GPU configuration. In this mode, it launches PhysioNet and MIMIC as separate child processes, assigns each process a distinct GPU through `CUDA_VISIBLE_DEVICES`, and permits two concurrent GPU stages. The corresponding expanded server allocation uses two GPUs, 16 CPU cores, and 55 GB of system memory. CPU-core and RAM allocations are specified by Slurm; the router controls process creation, GPU assignment, locking, logging, and separation of dataset-specific artifacts.

6.2 Predictive-model implementations

The STraTS, TCN, GRU, and GRU-D implementations originated from the public STraTS repository:

`https://github.com/sindhura97/STraTS`

The upstream code was adapted for the present study. The original supervised task used a single binary mortality target. In our implementation, the target columns are read from the dataset-specific latent-proxy CSV file, and the prediction head produces a vector with one logit per proxy variable. The output dimension is therefore the number of proxy targets rather than one mortality logit. All four architectures share this modified prediction interface and are trained using element-wise weighted binary cross-entropy with logits. The per-target positive-class weights are estimated from the training partition, and sigmoid probabilities are thresholded at 0.5 when binary proxy predictions are exported.

The encoder implementations and their upstream provenance were retained, while the multi-target proxy-prediction interface, split and artifact contracts, checkpoint metadata, prediction export, and integration into CliniCause were implemented for this study.

6.3 Causal-analysis implementations

The CliniCause orchestration layer and the project-specific causal pipeline were implemented in Python for this study. This code covers dataset adaptation, rule-based proxy generation, majority voting, dataset-specific DAG construction, graph-based adjustment-set selection, matching diagnostics, causal-effect estimation, result aggregation, permutation analyses, artifact validation, and run-level provenance.

The matching implementation is project code contained in `src/matching_causal_effect.py`. Matching is executed as a diagnostic stage within each causal run and its outputs are stored under that run's matching directory. It is not a separate estimator family, and no AIPW estimator was used.

The two double-machine-learning estimators were imported from version 0.16.0 of the EconML library, specifically:

- `econml.dml.CausalForestDML`;
- `econml.dml.LinearDML`.

The project configures these estimator classes and their nuisance models, selects treatment-specific adjustment variables from the dataset DAG, fits the estimators, and writes the treatment-level CATE outputs and diagnostics. The underlying EconML estimator implementations were not reimplemented by us.

The CausalPFN estimator was imported as `causalpfn.CATEEstimator` from CausalPFN version 0.1.4. Its upstream implementation is available at:

`https://github.com/vdblm/CausalPFN`

Our pipeline supplies the selected confounders and effect modifiers to the pretrained CausalPFN estimator, invokes its fitting and CATE-estimation interface, and integrates the returned estimates into the same run-specific output and provenance structure used for the EconML estimators.

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Archived ChatGPT Elicitation Materials

The following materials reproduce the ChatGPT elicitation artifacts used during the representation-design stage. They are included separately from the authored supplementary analyses and are preserved without substantive editing. The general protocol was instantiated separately for MIMIC-III and PhysioNet 2012. The resulting outputs document the proposed latent variables, operational proxy rules, measurement model, missingness analysis, causal graph, and implementation specifications. These archived outputs record the elicitation process; they do not establish clinical validation or causal ground truth.

General Latent-Causal-Model Elicitation Prompt

The following reusable prompt was applied separately to the two datasets after setting its dataset-name and official-description global variables.

```
You are an expert clinical causal inference researcher, medical ML scientist, and causal representation learning specialist.

Your task is to construct the most clinically plausible latent-variable causal DAG for an irregular medical time-series dataset.

--------------------------------------------
GLOBAL VARIABLES - CONFIGURE BEFORE USE
--------------------------------------------

DATASET-NAME = ENTER-DATASET-NAME-HERE
DATASET-DESCRIPTION-LINK = ENTER-OFFICIAL-DATASET-DESCRIPTION-URL-HERE

Use DATASET-NAME and DATASET-DESCRIPTION-LINK as global variables throughout this prompt.
Do not hard-code any dataset-specific name elsewhere in the prompt.
The only external source that may be referenced directly by URL is DATASET-DESCRIPTION-LINK.

--------------------------------------------
GLOBAL RESEARCH MISSION - READ BEFORE DOING ANYTHING
--------------------------------------------

Your final goal is NOT merely to construct rule-based decision trees.

Your final goal is to construct a research-backed latent-variable causal model for DATASET-NAME.

The decision trees are only an implementation mechanism for converting observed measurements into binary proxy labels for latent
clinical states. They are not the scientific endpoint.

The scientific endpoint is:

1. A clinically plausible latent causal ontology:
   - What are the true hidden clinical states behind the observed measurements?
   - Which latent states are meaningful causal variables?
   - Which observed variables are only noisy manifestations of these states?

2. A clinically and causally justified latent DAG:
   - How do background variables affect latent clinical states?
   - How do latent states affect each other?
   - How do latent states affect mortality?
   - How do latent states generate observed labs, vitals, and charted values?
   - How does missingness/measurement intensity arise from latent severity and clinical decision-making?

3. A causal-inference-ready graph:
   - Which variables are confounders?
   - Which variables are mediators?
   - Which variables are colliders?
   - Which variables should or should not be adjusted for when estimating effects on mortality?
   - Which latent states are outcome-proximal severity markers rather than valid adjustment variables?

4. A practical operationalization layer:
   - Only after the latent variables are selected and justified, create rule-based binary decision trees.
   - These trees produce implementable proxy labels such as LAT_SHOCK, LAT_RENAL_DYSFUNCTION, and LAT_GLOBAL_SEVERITY.
   - The trees are tools for implementation, not the main research contribution.

Priority order:
1. Research-backed latent clinical states.
2. Research-backed latent causal DAG.
3. Measurement model from latent states to observed variables.
4. Missingness and irregular-sampling model.
5. Confounding/mediation/collider analysis.
6. Rule-based decision trees for implementation.

If there is any conflict between making a simple decision tree and preserving causal correctness, preserve causal correctness.

Core objective:
Construct a clinically grounded latent causal representation of the dataset, including:
1. A set of causal latent variables representing underlying patient clinical states.
2. Formal definitions of these latent variables using available observed measurements and metadata.
3. Rule-based binary decision trees that convert observational/meta data into binary latent-variable labels.
4. A directed acyclic graph (DAG) over background variables, latent clinical states, observed measurements, measurement/missingness
processes, and mortality.
```

5. A clear explanation of how the latent clinical states causally affect mortality.
6. A clear explanation of how the observed variables are generated as manifestations, measurements, or proxies of the latent clinical states.

Important conceptual rule:
The observed variables in the dataset are not the main causal variables. They are clinical measurements, test results, charted values, or proxies. The causal variables of interest are latent clinical states such as shock, respiratory failure, renal dysfunction, hepatic dysfunction, inflammation/sepsis burden, coagulation dysfunction, metabolic derangement, neurologic dysfunction, cardiac injury, global severity, chronic baseline risk, or other dataset-appropriate latent states.

Do not construct a naive causal graph where every lab/vital directly causes every other lab/vital. Instead, construct a causal representation in which:
- Background/static variables influence latent patient states.
- Latent patient states influence observed measurements.
- Latent patient states influence mortality.
- Observed measurements are mainly manifestations of latent physiology, not primary causal mechanisms.
- Measurement frequency and missingness may themselves be informative and influenced by latent severity, clinical decision-making, or care intensity.

Your goal is to approximate the true underlying clinical causal structure as well as possible using deep research, medical reasoning, causal inference principles, and the official dataset documentation. Do not merely produce a correlation graph.

PHASE 1 - Dataset audit

First, open and study DATASET-DESCRIPTION-LINK carefully.

Extract:
1. The complete list of available variables.
2. Which variables are static/background variables.
3. Which variables are time-varying observed measurements.
4. Which variables are outcomes, especially mortality.
5. Which variables are interventions or care-process variables, if any.
6. Units, clinical meaning, valid ranges, and measurement windows when available.
7. Known dataset collection rules and missingness patterns.
8. Whether the dataset contains ICU type, admission type, care unit, demographics, diagnoses, chart events, labs, vitals, medications, procedures, or outcomes.

Then summarize the dataset in a compact audit table.

Do not proceed to latent-variable construction until the dataset audit is complete.

PHASE 2 - Deep Research protocol and research goals

Perform deep research before constructing latent variables, decision trees, or the DAG.

The purpose of Deep Research is not to collect thresholds for decision trees.

The purpose of Deep Research is to infer the most plausible latent clinical causal structure behind the dataset.

Your research must answer the following research questions:

RQ1 - Latent-state discovery:
What hidden physiological or clinical states plausibly generate the observed measurements in DATASET-NAME?

Examples:
- Shock or hemodynamic instability may generate abnormal blood pressure, heart rate, lactate, urine output, and vasopressor-related signals if available.
- Respiratory failure may generate abnormal oxygenation, respiratory rate, blood gas, ventilation, and saturation signals.
- Renal dysfunction may generate abnormal creatinine, BUN, urine output, potassium, bicarbonate, and acid-base patterns.
- Inflammation/sepsis burden may generate fever/hypothermia, WBC abnormalities, lactate elevation, shock, and organ dysfunction.
- Global severity may represent multi-system physiological instability rather than a single measurement.

RQ2 - Measurement model:
For each observed variable, determine whether it is:
- a background/static cause,
- an observed proxy of a latent state,
- a treatment/intervention/care-process variable,
- a measurement/missingness-process variable,
- an outcome,
- or irrelevant/noisy for the causal model.

Do not treat observed labs and vitals as the primary causal variables unless they are true interventions.

RQ3 - Latent causal DAG:
Determine the causal relationships among:
- background variables,
- latent clinical states,
- mortality,
- observed measurements,
- and missingness/measurement processes.

The graph should represent a clinical data-generating process:
background risk → latent physiology → observed measurements and mortality.

RQ4 - Mortality mechanism:
Explain how mortality is caused through latent clinical states, not through raw measurements directly.

For example:
- chronic baseline risk → vulnerability → global severity → organ dysfunction → mortality
- acute insult → inflammation/sepsis → shock → renal/metabolic failure → mortality
- respiratory failure → hypoxemia/acidosis → global severity → mortality

Adapt these mechanisms to the actual dataset variables.

RQ5 - Missingness and irregular sampling:
Because DATASET-NAME is an irregular clinical time-series dataset, research whether missingness is likely:
- MCAR,
- MAR,
- MNAR,
- or caused by care intensity / clinician suspicion / latent severity.

Treat measurement frequency and missingness as potentially informative, not as random noise.

RQ6 - Operational proxy construction:
Only after RQ1-RQ5 are answered, construct binary rule-based decision trees for the approved latent variables.

The decision trees should approximate the latent states using available observed variables, but they must not replace the causal DAG.

Research sources to use:
1. DATASET-DESCRIPTION-LINK.
2. Dataset-specific papers, benchmark papers, and preprocessing conventions.
3. ICU severity literature: SOFA, APACHE, SAPS, sepsis, shock, organ failure, respiratory failure, renal failure, coagulation, hepatic dysfunction, neurologic dysfunction, metabolic derangement.
4. Causal inference literature: SCMs, DAGs, latent variables, backdoor adjustment, mediation, colliders, potential outcomes.
5. Causal representation learning literature: observed variables as functions/proxies of latent causal variables.
6. Clinical EHR/time-series literature: irregular sampling, informative missingness, measurement intensity, care processes.
7. Existing clinical causal graphs or phenotype networks, if available.

For every proposed latent variable, decision rule, and DAG edge, provide:
- clinical mechanism,
- dataset support,
- causal direction,
- whether it risks leakage,
- whether it is a confounder, mediator, collider, or measurement proxy,
- confidence level,
- and whether it should be kept, optional, or rejected.

Do not proceed from research to final DAG until you can explain why the selected latent variables are the right causal level for this dataset.

PHASE 3 - Core causal ontology

Use the following node types:

A. Background/static variables
Examples:
- Age
- Sex/gender
- Ethnicity/race, if available
- Height, weight, BMI
- ICU type, admission type, first care unit
- Chronic disease burden or baseline vulnerability, if derivable

Background variables may cause latent clinical states and mortality risk, but they are generally not caused by ICU measurements.

B. Latent clinical state variables
These are the main causal variables.

Possible candidate latent states include, but are not limited to:
- Chronic baseline risk / chronic burden
- Acute insult / acute physiological stress
- Global severity of illness
- Circulatory shock / hemodynamic instability
- Respiratory failure / pulmonary dysfunction
- Renal dysfunction / kidney injury
- Hepatic dysfunction / liver injury
- Coagulation or hematologic dysfunction
- Inflammation / infection / sepsis burden
- Neurologic dysfunction
- Cardiac injury / cardiac strain
- Metabolic derangement / acid-base-electrolyte disorder
- Nutritional, fluid balance, or volume-status burden, if supported by variables
- Treatment/care intensity, if the dataset contains care-process variables

Do not force this list. Adapt the latent variables to the actual variables in DATASET-NAME.

Each latent variable must have:
1. A clinical definition.
2. A causal interpretation.
3. Observable indicators from the dataset.

4. Suggested proxy rule or thresholding logic.
5. Expected direction of abnormality.
6. Confidence level.
7. Notes on possible ambiguity.

C. Observed measurement variables
These include labs, vitals, charted measurements, and test results.

Treat these as manifestations or noisy measurements of latent states, for example:
- Shock may manifest through low blood pressure, high lactate, tachycardia, vasopressor use if available.
- Respiratory failure may manifest through PaO2, FiO2, oxygen saturation, respiratory rate, ventilation variables if available.
- Renal dysfunction may manifest through creatinine, BUN, urine output, potassium, bicarbonate.
- Hepatic dysfunction may manifest through bilirubin, AST, ALT, albumin.
- Coagulation dysfunction may manifest through platelets, INR/PT/PTT if available.
- Inflammation/infection may manifest through WBC, temperature, lactate, cultures/antibiotics if available.
- Cardiac injury may manifest through troponin, heart rate, blood pressure, ECG-related variables if available.
- Metabolic derangement may manifest through glucose, bicarbonate, pH, electrolytes, anion gap if derivable.

An observed variable may be an indicator of more than one latent state. Allow multi-parent measurement edges when clinically justified.

D. Mortality outcome
Mortality should be treated as an outcome node.
Latent states may cause mortality directly.
Background variables may affect mortality directly or indirectly through latent states.
Observed measurements should generally not directly cause mortality unless the variable is truly an intervention or treatment exposure.
Labs and vitals usually indicate the severity state rather than causing death themselves.

E. Measurement and missingness process
Because DATASET-NAME is an irregular medical dataset, include a measurement process if relevant:
- Measurement intensity
- Test ordering process
- Missingness indicator
- Observation availability
- Care intensity

These may be influenced by latent severity, clinician suspicion, ICU type, care unit, admission type, or background risk.
Do not assume missingness is MCAR.
Discuss whether missingness is likely MAR or MNAR.
If missingness is clinically informative, include it as a separate observation-process node, not as a biological causal state.

PHASE 4 - Latent variable selection and definition

Before constructing the DAG, define the latent variables that will appear in the DAG.

This phase must happen before DAG construction.

Your task in this phase:
1. Propose candidate latent clinical variables based on the dataset audit.
2. Map available observed and metadata variables to candidate latent variables.
3. Remove candidate latent variables that are not sufficiently supported by the dataset.
4. Merge overlapping latent variables if the dataset cannot distinguish them reliably.
5. Mark weakly measured but clinically important latent variables as optional.
6. Finalize the latent variable set before constructing causal edges.

For every latent variable, define it operationally from observed data.

For each latent variable produce:
1. Name
2. Clinical meaning
3. Why it is latent rather than directly observed
4. Observable indicators
5. Metadata indicators, if relevant
6. Suggested binary definition
7. Suggested continuous severity score, if useful
8. Thresholds or quantile-based fallback rules
9. Missingness handling
10. Directionality: higher value means worse/better?
11. Whether this latent variable should be used as:
    - prediction target,
    - confounder,
    - mediator,
    - outcome-proximal severity marker,
    - treatment-effect modifier.
12. Confidence: high / medium / low.
13. Whether to keep, merge, make optional, or reject.

Definitions must be clinically meaningful but also implementable on DATASET-NAME.

Do not define latent variables from a single measurement unless unavoidable.
Prefer multiple indicators and majority-vote or score-based definitions.
If the dataset lacks sufficient indicators for a latent state, either:
- remove the latent variable,
- mark it as weakly measured,
- merge it with another latent variable,
- or define it as optional.

Do not use mortality or post-outcome information to define any latent variable.
Do not allow outcome leakage.
Do not use future measurements if the latent label is intended to represent an earlier clinical state.

At the end of this phase, provide a final approved latent-variable list.
Only these approved latent variables may be used as latent nodes in the DAG.

IMPORTANT - Decision trees are not the final goal

The rule-based binary decision trees are only an operational layer.

They are used to create binary proxy labels for latent variables so that the thesis pipeline can generate latent_tags.csv and use those labels in downstream modeling.

Do not optimize the entire answer around the decision trees.

A correct answer with clinically meaningful latent variables and a strong DAG is better than a large collection of simplistic decision trees.

The decision trees must serve the causal model, not define the causal model.

Before writing any tree, explicitly state:
- which latent clinical state the tree approximates,
- why this state belongs in the DAG,
- which observed variables are manifestations of that state,
- and how the proxy label may differ from the true unobserved clinical state.

PHASE 5 - Rule-based binary decision trees for latent labels

After the latent-variable set is finalized, construct a rule-based binary decision tree for each approved latent variable.

Use a self-contained score-based tree structure.

The desired structure is:

```
1. Define several clinically meaningful domain indicators.
2. Each domain indicator is a Boolean condition based on observed measurements or metadata.
3. Combine domain indicators into a score:
   score = sum(domain indicators)
4. Apply a threshold:
   if score >= THRESHOLD:
       LATENT_VARIABLE = 1
   else:
       LATENT_VARIABLE = 0
```

Example score-based structure:

```
- Circulatory domain:
  MAP_first < 70 OR SysABP_first <= 100
- Neurologic domain:
  GCS_first < 15
- Respiratory domain:
  RespRate_first >= 22 OR SaO2_first < 92
- Metabolic domain:
  Lactate_first > 2 OR pH_first < 7.3 OR HCO3_first < 18
- Renal domain:
  Creatinine_first >= 2
- score = sum(domain indicators)
- score >= 2 gives Severity = 1
- otherwise Severity = 0
```

For each latent variable, create a similar binary decision tree.

Requirements:
1. The output of every tree must be binary: 0 or 1.
2. The rules must use only observed variables and metadata available in DATASET-NAME.
3. Use clinically interpretable thresholds when possible.
4. If standard clinical thresholds are not available or not dataset-appropriate, propose quantile-based fallback thresholds.
5. Each tree should use multiple indicators when possible.
6. A tree may use first value, worst value, mean value, slope, variability, measurement frequency, or missingness indicators, but you must justify each choice.
7. Avoid using mortality, survival time, discharge disposition, or outcome-derived variables.
8. Avoid using future values if the latent variable is meant to represent early-state physiology.
9. Use missingness explicitly when clinically informative, but do not let missingness dominate the latent definition unless justified.
10. Mark rules as high / medium / low confidence.

For each decision tree, provide:
1. Latent variable name.
2. Clinical interpretation of label 1.
3. Input variables.
4. Domain indicators.
5. Scoring function.
6. Binary threshold.
7. Missingness policy.
8. Pseudocode.

```
9. A compact Mermaid flowchart.
10. Clinical rationale.
11. Potential false positives and false negatives.
12. Confidence level.

The decision trees should be designed so they can be directly implemented in code to generate a latent_tags.csv output.

Use naming convention:
LAT_<LATENT_NAME> = 0/1

For example:
LAT_SHOCK = 1 means the patient has evidence of circulatory shock or clinically meaningful hemodynamic instability.
LAT_RENAL_DYSFUNCTION = 1 means the patient has evidence of acute or clinically meaningful renal dysfunction.
LAT_GLOBAL_SEVERITY = 1 means the patient has evidence of multi-domain severity according to the score-based tree.

After creating the decision trees, verify:
- Every approved latent variable has a binary tree.
- Every tree uses only allowed variables.
- No tree uses mortality or future outcome information.
- Every threshold is clinically or statistically justified.
- Every tree can be implemented on irregular time-series data.

Only after this verification, proceed to DAG construction.

--------------------------------------------
PHASE 6 - DAG construction rules
--------------------------------------------

Construct the DAG only after:
1. The dataset variables were audited.
2. The latent variables were selected and finalized.
3. The binary decision trees were constructed and validated.

The final graph must be a DAG.

Rules:
1. No directed cycles.
2. No edge from mortality back to clinical states.
3. No edge from future measurements to earlier latent states.
4. No automatic observed-variable-to-observed-variable causal chains unless there is a true clinical mechanism.
5. Prefer latent-state-mediated explanations over direct lab-to-lab causation.
6. Use temporal and clinical ordering:
   - Background risk precedes acute illness.
   - Acute insult precedes or contributes to global severity.
   - Global severity may contribute to organ dysfunction.
   - Organ dysfunction contributes to mortality.
   - Organ dysfunction produces observed abnormal measurements.
7. Allow clinically justified latent-to-latent edges, such as:
   - Chronic risk → acute vulnerability
   - Acute insult → global severity
   - Global severity → shock
   - Shock → renal dysfunction
   - Shock → hepatic dysfunction
   - Shock → metabolic derangement
   - Inflammation/sepsis → shock
   - Inflammation/sepsis → coagulation dysfunction
   - Respiratory failure → metabolic derangement
   - Renal dysfunction → metabolic derangement
   - Cardiac injury → shock
8. Avoid over-dense DAGs. Use sparse, clinically meaningful edges.
9. Each edge must have a mechanism and confidence score.
10. If an edge is plausible but uncertain, mark it as optional or low-confidence.
11. Do not confuse measurement with intervention.
12. Do not confuse association with causation.
13. Do not use the mortality label to define latent variables in a way that leaks the outcome.
14. Observed variables should usually be children of latent states, not parents of latent states.
15. Binary latent labels generated by the decision trees are operational proxies for latent states, not the biological states
themselves. Make this distinction explicit.

DAG construction order:
1. Add background/static nodes.
2. Add approved latent clinical-state nodes.
3. Add mortality outcome node.
4. Add observed measurement nodes as manifestations of latent states.
5. Add measurement/missingness process nodes if needed.
6. Add background-to-latent edges.
7. Add latent-to-latent edges.
8. Add latent-to-mortality edges.
9. Add latent-to-observed measurement edges.
10. Add latent/background-to-measurement-process edges.
11. Check acyclicity.
12. Check clinical plausibility.
13. Check whether the graph is sparse enough.

--------------------------------------------
PHASE 7 - Iterative validation procedure
--------------------------------------------
```

Do not produce the final answer in one pass.

Use the following internal process:

Round 1:
Create an initial list of candidate latent variables from the official dataset variables.

Round 2:
Map every observed variable to one or more latent clinical states. Identify unassigned variables and decide whether they are irrelevant, background, outcome, treatment, or measurement-process variables.

Round 3:
Finalize the latent-variable set. Remove, merge, or mark optional latent variables that are poorly supported.

Round 4:
Construct rule-based binary decision trees for every approved latent variable using observed and metadata variables.

Round 5:
Validate the decision trees:
- Do the rules use only available variables?
- Are thresholds clinically meaningful?
- Is missingness handled correctly?
- Is there outcome leakage?
- Are the trees implementable on irregular time-series data?
- Are false positives/false negatives acceptable?

Round 6:
Construct a first-pass latent DAG using only approved latent variables. Check for cycles, clinically implausible edges, and over-dense structure.

Round 7:
Research again. Compare the graph to known clinical severity frameworks and EHR causal/phenotype-network literature. Revise latent variables, decision rules, and edges if needed.

Round 8:
Perform adversarial critique:
- Which latent variables are not truly identifiable from the dataset?
- Which decision-tree rules are too weak or too dependent on a single measurement?
- Which edges are only correlations?
- Which observed variables were incorrectly treated as causes?
- Which latent variables are poorly measured?
- Which missingness patterns may bias definitions?
- Which edges create collider bias if adjusted for?
- Which variables should not be adjusted for when estimating effects on mortality?
- Which variables are mediators rather than confounders?
- Which latent definitions risk outcome leakage?

Only after these rounds, produce the final answer.

PHASE 8 - Required final answer representation

The final answer must be a structured, thesis-style technical report, not a loose narrative, a short summary, or an unordered list of findings.
Do not label the final answer as phases. The phases above are the work protocol; the final answer is the report.

Before Section 1, begin with a short opening synthesis of 2-4 paragraphs:
- State the clinically appropriate causal level for the dataset.
- State that observed labs, vitals, charted values, and tests are mainly manifestations or proxies of latent physiologic states, not primary causal variables.
- State the strongest high-level DAG pattern in one sentence, for example:
  background/case-mix latent acute physiology and organ dysfunction observed measurements / care intensity / measurement process
  mortality.
- State the main dataset limitations that constrain identifiability.

Your final answer must then contain exactly the following numbered top-level sections and order.

1. Dataset audit
Use compact Markdown tables.
Include:
- Variable name
- Variable type: background / observed time-varying / outcome / treatment / care-process / measurement-process / irrelevant-noisy
- Clinical meaning
- Unit or coding if available
- Notes

If the dataset has many variables, group the table into coherent blocks such as background variables, outcomes, interventions/care-process variables, time-varying vitals, laboratory variables, charted variables, and measurement-process variables.

2. Candidate latent variables
Use a Markdown table.
Include:
- Candidate latent variable name
- Clinical meaning
- Supporting observed variables
- Dataset support level: high / medium / low / absent
- Decision: keep / merge / optional / reject

- Rationale

This section must show that latent variables were considered before the DAG was built.

3. Final approved latent variables
Use a Markdown table.
Include:
- Latent variable name
- Clinical meaning
- Why it is latent rather than directly observed
- Observed indicators
- Suggested binary definition
- Missingness strategy
- Causal role: confounder / mediator / collider risk / treatment-effect modifier / outcome-proximal severity marker / proxy-label target
- Confidence: high / medium / low
- Rationale

This section must make clear which latent variables are actually allowed to appear as latent nodes in the DAG.

4. Rule-based binary decision trees
Provide one repeated subsection per approved latent variable.
For each tree include:
- Latent variable name
- Binary output name using the convention LAT_<LATENT_NAME>
- Clinical interpretation of label 1
- Input variables
- Domain indicators
- Score formula
- Binary threshold
- Missingness policy
- Pseudocode
- Compact Mermaid flowchart
- Clinical rationale
- False positive risks
- False negative risks
- Confidence level

Use this self-contained score-based tree pattern:
- Define several clinically meaningful Boolean domain indicators.
- Each domain indicator is based only on available observed measurements or metadata.
- Combine the indicators into a score: score = sum(domain indicators).
- Apply a threshold: if score >= THRESHOLD then LAT_<LATENT_NAME> = 1, otherwise LAT_<LATENT_NAME> = 0.

Do not define any tree using mortality, survival time, discharge disposition, post-outcome variables, or future measurements that would create leakage.

5. Observed-to-latent measurement model
Use a Markdown table.
Include:
- Observed variable
- Parent latent variable or variables
- Direction of abnormality
- Whether the measurement is specific or non-specific
- Whether it is used in a binary decision tree
- Whether it may also reflect treatment/care process or measurement intensity
- Notes

This section must treat observed variables primarily as measurements, proxies, or manifestations of latent states, unless the variable is truly an intervention or care-process variable.

6. Latent causal DAG edge list
Use a Markdown table.
Include:
- Source node
- Target node
- Edge type: background-to-latent / latent-to-latent / latent-to-observed / latent-to-mortality / background-to-mortality / treatment-or-care-process / measurement-process
- Clinical mechanism
- Confidence: high / medium / low
- Reasoning basis
- Decision: keep / optional / reject

Use sparse clinically meaningful edges, not a dense correlation graph.

7. Mortality mechanism summary
Explain the main causal paths to mortality in text.
Adapt the paths to the final DAG.
Examples of acceptable path templates:
- Chronic baseline risk → vulnerability/global severity → organ dysfunction → mortality.
- Acute insult → inflammation/sepsis burden → shock → renal/metabolic failure → mortality.
- Respiratory failure → hypoxemia/acidosis → global severity → mortality.
- Cardiac injury/strain → hemodynamic instability → global severity → mortality.

Do not describe mortality as being directly caused by raw measurements unless the variable is a true intervention or harmful exposure.

8. Missingness and irregular sampling analysis
Discuss:
- Which variables are likely MCAR, MAR, MNAR, or care-intensity-driven.

- Whether missingness and measurement frequency should be represented as separate nodes.
- Whether measurement frequency can act as a proxy for severity, clinician suspicion, or care intensity.
- How missingness should be handled when defining latent variables.
- How missingness enters or does not enter the binary decision trees.
- Whether using missingness creates bias, collider risk, or leakage risk.

9. Backdoor/confounding implications
For the purpose of estimating causal effects on mortality:
- Identify likely confounders.
- Identify likely mediators.
- Identify colliders to avoid conditioning on.
- Identify outcome-proximal severity markers that may be inappropriate adjustment variables for some treatment/exposure questions.
- Explain which variables are safe to adjust for depending on the treatment/exposure of interest.
- Explain whether the binary latent tags are safe adjustment variables or whether they may be descendants, mediators, colliders, or proxy outcomes.

10. Graph specification
Provide the final DAG in three formats:
A. Adjacency list.
B. Mermaid graph.
C. Python NetworkX edge list.

Use clear node prefixes:
- BG_ for background variables
- LAT_ for latent variables
- OBS_ for observed measurements
- TRT_ for interventions or care-process variables
- MISS_ for missingness/measurement-process variables
- OUT_ for mortality outcome

11. Decision-tree implementation specification
Provide a code-oriented specification for generating a latent tag table.
Include:
- Required input columns
- Derived summary features, such as first, worst, mean, slope, variability, count, and missingness
- One function per latent variable
- Expected binary output column names
- Handling of missing variables
- Unit assumptions
- Threshold table
- Validation checks

The specification must be concrete enough to implement without reinterpreting the clinical logic.

12. Rejected alternatives
List important variables, latent definitions, decision rules, or edges that were considered but rejected.
For each rejected item, give the reason, such as:
- Insufficient dataset support
- Leakage risk
- Weak measurement
- Causal ambiguity
- Collider or mediator risk
- Overly dense graph structure
- Not implementable from available variables

13. Final confidence and limitations
State:
- Which latent variables are strongest.
- Which latent variables are weakest.
- Which decision trees are most reliable.
- Which decision trees are most uncertain.
- Which DAG edges are most uncertain.
- What additional data would improve identifiability.
- Why the graph is a clinically plausible causal DAG rather than a proven ground-truth DAG.

Formatting requirements:
- Use clear Markdown headings matching the numbered sections above.
- Use Markdown tables for the dataset audit, candidate latent variables, approved latent variables, measurement model, and DAG edge list.
- Use fenced code blocks for pseudocode, Mermaid graphs, and NetworkX edge lists.
- Keep the report implementable: the reader should be able to turn the decision-tree section into latent-tag generation code.
- Be explicit when a variable is a measurement proxy, treatment/care-process variable, confounder, mediator, collider, or outcome-proximal severity marker.
- Do not reference any local files, images, prior generated reports, or external artifacts other than the official dataset description website specified at the top of this prompt.

PHASE 9 - Style and rigor requirements

Be rigorous and skeptical.

Every final latent variable must be selected before the DAG is constructed.
Every final latent variable must have a binary decision tree.
Every final decision tree must be implementable in code.
Every final edge must be justified.
Every latent variable must be clinically interpretable.
Every observed variable must be categorized.

```
Do not hallucinate variables not present in the dataset.
Do not overclaim certainty.
Do not present the graph as statistically proven from observational data alone.
Do not ignore informative missingness.
Do not treat labs/vitals as primary causal states.
Do not create a dense graph just because variables correlate.
Do not define latent variables after creating the DAG.
Do not construct the DAG before the latent variables and binary decision trees are stable.

The final output should be detailed enough to implement in code:
- latent tag construction,
- rule-based binary decision trees,
- latent_tags.csv generation,
- DAG construction,
- confounder selection,
- mortality-effect estimation,
- sensitivity analysis.

Begin now by opening DATASET-DESCRIPTION-LINK and auditing the dataset variables.
```

MIMIC-III ChatGPT Elicitation Output

Archived output corresponding to the general
latent-causal-model elicitation protocol

For **DATASET-NAME = MIMIC-III**, the clinically appropriate causal level is not the raw laboratory value, vital sign, medication row, or charted item. The causal level should be **latent physiology**: chronic vulnerability, acute inflammatory/infectious stress, shock, respiratory failure, renal dysfunction, neurologic dysfunction, metabolic derangement, hepatic/coagulation dysfunction, cardiac strain, and global acute severity.

The strongest high-level causal pattern is:

background/case-mix → latent acute physiology and organ dysfunction → observed measurements, care intensity, measurement intensity → mortality.

Observed measurements in DATASET-NAME are mostly **manifestations or proxies** of hidden patient states. For example, creatinine does not "cause" mortality in the causal model; renal dysfunction contributes to mortality and generates abnormal creatinine, BUN, urine output, potassium, and acid-base findings. DATASET-NAME's official documentation states that the database contains ICU demographics, vital signs, labs, medications, procedures, notes, imaging reports, and mortality, collected during routine care rather than by a fixed research protocol. This makes irregular sampling and informative missingness central to the causal representation. physionet.org +1

The main identifiability limitations are that many variables are dictionary-coded ITEMIDs, clinical measurements are ordered by clinicians, treatment decisions are confounded by severity, and chronic baseline state is only partially observed through demographics, ICD-9 codes, prior diagnoses, and admission context. The resulting DAG is therefore a **clinically plausible structural causal model**, not a proven ground-truth graph.

1. Dataset audit

DATASET-NAME is a relational ICU database. The public documentation describes 26 core tables linked by `SUBJECT_ID`, `HADM_ID`, and `ICUSTAY_ID`; schema metadata also lists 40 physical tables because `CHARTEVENTS` is partitioned. Public table-level documentation is complete, but the full clinical variable inventory must be enumerated locally from `D_ITEMS` and `D_LABITEMS`, because many measurements are ITEMID-coded rather than exposed as flat columns. physionet.org +1

1.1 Core identifiers, background, and outcomes

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
`SUBJECT_ID`	Identifier	Patient-level identifier	Integer	Links patient across admissions.
`HADM_ID`	Identifier	Hospital admission identifier	Integer	Links admission-level events.
`ICUSTAY_ID`	Identifier	ICU stay identifier	Integer	Links ICU-specific events.
`PATIENTS.GENDER`	Background	Recorded sex/gender	Categorical	Background cause of risk and disease distribution.
`PATIENTS.DOB`	Background	Date of birth	Shifted timestamp	Used to derive age; age >89 is deidentified as very old.
`PATIENTS.DOD`, `DOD_HOSP`, `DOD_SSN`	Outcome / survival metadata	Date of death	Shifted timestamp	`DOD` is null if alive at least 90 days post-discharge. lcp.mit.edu
`PATIENTS.EXPIRE_FLAG`	Outcome	Indicates patient died	Binary	Not allowed in latent-tag construction.
`ADMISSIONS.ADMITTIME`, `DISCHTIME`	Background / timing	Hospital admission and discharge	Timestamp	Used for cohort timing, not physiology.
`ADMISSIONS.DEATHTIME`	Outcome	Time of death	Timestamp	Outcome only; leakage if used for latent states.
`ADMISSIONS.ADMISSION_TYPE`	Background / care context	Emergency, elective, urgent, etc.	Categorical	Case-mix and confounding variable.
`ADMISSIONS.ADMISSION_LOCATION`	Background / care context	Origin before admission	Categorical	Proxy for illness pathway.
`ADMISSIONS.DISCHARGE_LOCATION`	Outcome-adjacent	Disposition	Categorical	Do not use for latent physiology because it is post-baseline.

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
`ADMISSIONS.INSURANCE`, `LANGUAGE`, `RELIGION`, `MARITAL_STATUS`, `ETHNICITY`	Background / social-context proxies	Sociodemographic and access/context variables	Categorical	Possible confounders for care intensity, measurement intensity, and outcomes.
`ADMISSIONS.DIAGNOSIS`	Background / admission-context proxy	Free-text admitting diagnosis	Text	Useful but noisy; avoid direct outcome leakage.
`ADMISSIONS.HOSPITAL_EXPIRE_FLAG`	Outcome	In-hospital mortality	Binary	Outcome only.
`ADMISSIONS.HAS_CHARTEVENTS_DATA`	Measurement-process	Whether charted observations exist	Binary	Observation availability indicator. lcp.mit.edu
`ICUSTAYS.FIRST_CAREUNIT`, `LAST_CAREUNIT`	Background / care context	ICU type	Categorical	Case-mix and care-process determinant.
`ICUSTAYS.INTIME`, `OUTTIME`, `LOS`	Timing / outcome-adjacent	ICU entry, exit, length of stay	Timestamp / days	`LOS` is outcome-adjacent; avoid for early latent labels. lcp.mit.edu

1.2 Time-varying observed measurements

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
`CHARTEVENTS`	Observed time-varying / measurement process	Bedside charted measurements, vitals, ventilator settings, nursing observations, GCS, RASS, code status, some labs	ITEMID-coded, `VALUE`, `VALUENUM`, `VALUEUOM`	Very large eve manifestation clinician docu
Heart rate	Observed time-varying	Cardiovascular stress, shock, fever, pain, arrhythmia	beats/min	Proxy for sho strain, global
Systolic/diastolic/MAP blood pressure	Observed time-varying	Perfusion pressure	mmHg	Proxy for sho
Respiratory rate	Observed time-varying	Respiratory distress / metabolic compensation	breaths/min	Proxy for resp metabolic de
Oxygen saturation	Observed time-varying	Peripheral oxygenation	%	Proxy for resp modified.
Temperature	Observed time-varying	Fever/hypothermia	°C or °F depending item	Proxy for infla normalization
Glasgow Coma Scale	Observed time-varying	Consciousness / neurologic function	3–15	Proxy for neu confounded b
Ventilator settings, FiO2, PEEP, ventilation status	Treatment / observed support	Respiratory support and oxygen delivery	Fraction, cmH2O, categorical	Care-process
`LABEVENTS`	Observed time-varying	Laboratory test results	ITEMID-coded, `VALUE`, `VALUENUM`, `VALUEUOM`, abnormal `FLAG`	Lab values are informative w
CBC: WBC, hemoglobin, platelets	Observed lab proxy	Inflammation, anemia, coagulation/hematologic dysfunction	K/μL, g/dL	Platelets are u dysfunction.
Chemistry: sodium, potassium, bicarbonate, chloride, BUN, creatinine, glucose	Observed lab proxy	Renal, metabolic, electrolyte, endocrine stress	mmol/L, mg/dL	Requires unit

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
Blood gas: pH, PaO2, PaCO2, base excess	Observed lab proxy	Respiratory and metabolic acid-base state	pH units, mmHg, mmol/L	Supports resp latent states.
Lactate	Observed lab proxy	Tissue hypoperfusion / cellular stress	mmol/L	Proxy for sho severity.
Bilirubin, AST, ALT, albumin	Observed lab proxy	Hepatic injury/synthetic function	mg/dL, U/L, g/dL	Hepatic dysfu measured.
INR, PT, PTT	Observed lab proxy	Coagulation dysfunction	Ratio / seconds	Careful: antic possible.
Troponin, CK-MB	Observed lab proxy	Cardiac injury/strain	ng/mL, µg/L, assay-dependent	Sparse and o

1.3 Interventions, care-process, and procedures

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
`INPUTEVENTS_CV`, `INPUTEVENTS_MV`	Treatment / care-process	Fluids, IV medications, infusions, nutrition	ITEMID-coded amounts/rates	Split by source system; fluid-input structure differs between CareVue and MetaVision. physionet.org +1
Vasopressor infusions	Treatment / care-process	Hemodynamic support	drug rate, dose units vary	Strong proxy for shock but also treatment variable.
IV fluids	Treatment / care-process	Resuscitation / volume management	mL, rates	Confounded by indication.
Antibiotics	Treatment / care-process	Suspected/treated infection	prescription rows	Proxy for clinician suspicion; not identical to infection.
`PRESCRIPTIONS`	Treatment / care-process	Ordered medications	drug, dose, route, dates	Use cautiously; time precision may vary.
`PROCEDUREEVENTS_MV`	Treatment / care-process	Timed procedures such as ventilation, dialysis, imaging	ITEMID-coded start/end	MetaVision only; includes procedure timing.
`PROCEDURES_ICD`, `CPTEVENTS`	Procedure / coding	Billing/procedure codes	ICD-9/CPT	Often post hoc and not always baseline-timed.
`OUTPUTEVENTS`	Observed time-varying / care-process	Urine, drains, fluid output	usually mL	Urine output is renal/perfusion proxy; output charting is care-process dependent.

1.4 Diagnoses, microbiology, notes, and dictionaries

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
`DIAGNOSES_ICD`	Background / phenotype metadata	ICD-9 diagnosis codes	ICD-9-CM	Useful for chronic burden, baseline comorbidity, and admission phenotype; no timestamps.
`D_ICD_DIAGNOSES`	Dictionary	Diagnosis code definitions	ICD-9 descriptions	Crosswalk for diagnosis concepts.
`MICROBIOLOGYEVENTS`	Observed proxy / care-process	Cultures, organisms, sensitivities	Specimen, organism, antibiotic interpretation	Negative cultures may be recorded by null organism name. physionet.org

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
`NOTEEVENTS`	Observed text / care-process	Notes, ECG reports, radiology reports, discharge summaries	Free text	Rich but high leakage risk if discharge summaries are used.
`D_ITEMS`	Dictionary	Non-lab charted item definitions	ITEMID dictionary	Required for variable extraction.
`D_LABITEMS`	Dictionary	Lab item definitions, LOINC where available	ITEMID dictionary	Required for lab extraction.
`SERVICES`, `TRANSFERS`	Background / care-process	Service line and location transitions	Categorical + timestamps	ICU type, service changes, and transfer patterns affect measurement and outcomes.
`CALLOUT`	Outcome-adjacent care-process	Readiness for transfer/discharge	Categorical/timestamps	Avoid for early mortality/physiology labels.

2. Candidate latent variables

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Rationale
Chronic baseline burden	Pre-ICU vulnerability from age, comorbidity, chronic organ disease, malignancy, frailty proxies	Age, ICD-9 diagnoses, admission type, ICU type, prior admissions if derived	High	Keep	Strong confou measurement mortality.
Acute inflammation / infection burden	Dysregulated host response or suspected infection	Temperature, WBC, lactate, cultures, antibiotics, microbiology	Medium-high	Keep	Central ICU la is partly unob emphasizes ir dysfunction ra PMC
Global acute severity	Multi-domain physiologic instability	Vitals, GCS, urine output, ventilation, labs	High	Keep	Severity score represent org physiology ra MSD Manuals +1
Circulatory shock	Inadequate perfusion / hemodynamic instability	MAP/SBP, lactate, vasopressors, urine output, pH/base deficit	High	Keep	Strong morta vasopressors must be sepa
Respiratory failure	Hypoxemia, ventilatory failure, need for support	SpO2, PaO2/FiO2, FiO2, PEEP, ventilation, RR, PaCO2/pH	High	Keep	Strongly supp blood gases, ARDS thresho based severity
Renal dysfunction	AKI or clinically important renal failure	Creatinine, BUN, urine output, potassium, bicarbonate, dialysis	High	Keep	KDIGO suppo output criteria outputs. MSD
Hepatic dysfunction	Liver injury or impaired clearance/synthetic function	Bilirubin, AST, ALT, albumin, INR	Medium	Merge with coagulation	Hepatic and c are clinically r measured sep windows.
Coagulation / hematologic dysfunction	Platelet and clotting-system derangement	Platelets, INR, PT, PTT, hemoglobin	Medium	Merge with hepatic	SOFA uses pla organ domair implementab
Neurologic dysfunction	Depressed consciousness or acute brain dysfunction	GCS, RASS, pupil response,	Medium-high	Keep	GCS is widely sedation/intu interpretation

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Rationale
		sedation/intubation indicators			
Metabolic / acid-base-electrolyte derangement	Acidosis/alkalosis, lactate, electrolyte, glucose instability	pH, bicarbonate, lactate, base excess, Na, K, glucose	High	Keep	Cross-organ s mortality path
Cardiac injury / strain	Myocardial injury, arrhythmia burden, cardiac stress	Troponin, CK-MB, HR, BP, ECG report, care unit	Medium-low	Keep, low confidence	Clinically imp sparse/selecti weak latent p
Volume status / fluid balance burden	Overload or depletion	Inputs, outputs, weight, edema notes, CVP	Low-medium	Reject as standalone	Measurement source system dependent.
Nutritional burden	Malnutrition/catabolism	Albumin, weight, nutrition inputs	Low	Reject	Albumin is nc inflammation,
Treatment intensity	Clinician response and organ support	Vasopressors, ventilation, dialysis, fluids, antibiotics, monitoring frequency	High	Do not treat as biological latent	Represent as primary laten
Missingness / observation intensity	Clinician-driven measurement process	Lab counts, chart counts, missing indicators	High	Separate measurement-process node	Informative m collapsed into

3. Final approved latent variables

le	Clinical meaning	Why latent	Observed indicators	Suggested binary definition	Missingness str
`_BURDEN`	Baseline vulnerability before ICU physiology	Comorbidity/frailty is not directly measured	Age, ICD-9 comorbidity groups, chronic organ disease, malignancy, admission context	Score ≥2 among age, comorbidity, chronic organ disease, malignancy/immunosuppression, nonelective high-risk admission	Missing ICD → unknown/0 with not infer absent from missing co strongly
`ATION_SEPSIS`	Suspected infection/inflammatory burden	True infection and dysregulated host response are hidden	Temp, WBC, lactate, cultures, antibiotics, microbiology	Score ≥2, with at least one infection-evidence or inflammatory-evidence domain	Missing cultures not equal no inf include measure flag
`SEVERITY`	Multi-system acute physiologic severity	Severity is a construct across organs	Vitals, GCS, ventilation, urine output, lactate, labs	≥3 abnormal organ domains or ≥2 severe domains	Calculate from a domains; require minimum obser domains or mar uncertain
	Circulatory failure / tissue hypoperfusion	Perfusion adequacy is not directly measured	MAP/SBP, vasopressors, lactate, urine output, acidosis	Score ≥2 or vasopressor + hypoperfusion evidence	If BP missing bu vasopressor pre not mark missin normal
`TORY_FAILURE`	Hypoxemic or ventilatory failure	Lung gas exchange state is latent	SpO2, PaO2/FiO2, FiO2, PEEP, ventilation, RR, PaCO2/pH	Score ≥2 or mechanical ventilation with oxygenation/gas-exchange abnormality	Missing ABG allo SpO2/FiO2 fallb available
`YSFUNCTION`	AKI or clinically important renal failure	Kidney function is inferred from filtration/output	Creatinine, BUN, urine output, K, bicarbonate, dialysis	Score ≥2 or dialysis/procedure evidence	Require time wi missing urine ou normal unless o table available
`_COAG_DYSFUNC`	Liver/coagulation/hematologic organ dysfunction	Synthetic/clearance state is not directly observed	Bilirubin, platelets, INR/PT/PTT, AST/ALT, albumin	Score ≥2	Missing INR/bili common; do no normal without

le	Clinical meaning	Why latent	Observed indicators	Suggested binary definition	Missingness stra
GIC_DYSFUNCTI	Depressed consciousness/acute neurologic dysfunction	Neurologic state is partly obscured by sedation/intubation	GCS, RASS, pupil response, charted mental status	GCS abnormal plus another neuro/sedation-context indicator, or very low GCS alone	Missing GCS not sedation/intuba required
IC_DERANGEMEN	Acid-base, lactate, electrolyte, glucose instability	Systemic metabolic homeostasis is latent	pH, bicarbonate, base excess, lactate, Na, K, glucose	Score ≥2	If blood gas mis on chemistry; in count flag
_STRAIN	Myocardial injury/strain or clinically important arrhythmia burden	Cardiac injury is selectively tested and incompletely observed	Troponin, CK-MB, HR extremes, arrhythmia charting, ECG reports	Score ≥2, with biomarker or rhythm evidence required	Missing troponi unmeasured, no low-confidence

4. Rule-based binary decision trees

All rules use an early analysis window, preferably **first 24 or first 48 ICU hours**, with no mortality, discharge location, survival time, ICU length of stay, or post-outcome data. Use `first`, `worst`, `min`, `max`, `mean`, `slope`, `count`, and `missing` summaries computed only within the chosen window.

4.1 `LAT_CHRONIC_BURDEN`

Binary output: `LAT_CHRONIC_BURDEN`
Label 1 means: high pre-ICU vulnerability or chronic disease burden.

Input variables: age, ICD-9 diagnosis groups, admission type, ICU type, prior admission count if derived before index ICU stay.

Domain indicators

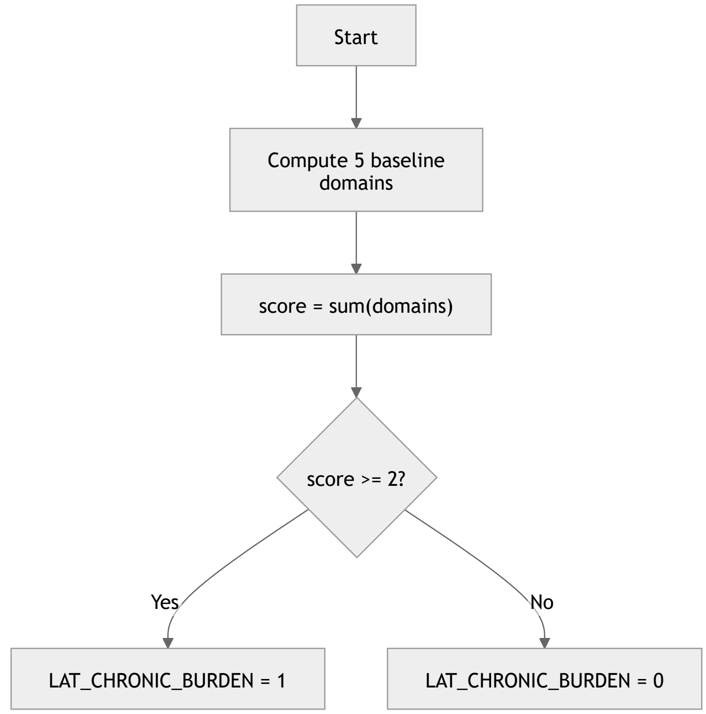
Indicator	Rule
Age burden	age ≥75, or deidentified age category >89
Comorbidity burden	Charlson/Elixhauser count ≥2, or equivalent ICD-derived comorbidity score high
Chronic organ disease	ICD evidence of chronic heart, lung, kidney, liver, neurologic disease
Cancer/immunosuppression	ICD evidence of metastatic cancer, hematologic malignancy, transplant, immunosuppression
High-risk admission context	nonelective admission plus medical ICU or emergency admission source

Score formula: `score = sum(indicators)`
Threshold: `LAT_CHRONIC_BURDEN = 1 if score >= 2 else 0`

Missingness policy: missing ICD codes are treated as "not documented," not proof of absence. Add `MISS_ICD_COMORBIDITY_UNKNOWN = 1` if diagnosis coding is unavailable.

Python

```python
def tag_chronic_burden(x):
    score = 0
    score += int(x.age >= 75 or x.age_is_deidentified_old)
    score += int(x.comorbidity_count >= 2 or x.elixhauser_score >= 5)
    score += int(x.has_chronic_organ_disease)
    score += int(x.has_malignancy_or_immunosuppression)
    score += int(x.admission_type in ["EMERGENCY", "URGENT"] and x.first_careunit in ["MICU", "SICU", "CSRU"])
    return int(score >= 2)
```



Clinical rationale: chronic vulnerability modifies the effect of acute insults and mortality risk.
False positives: age alone may overstate frailty.
False negatives: undercoded comorbidities.
Confidence: high.

4.2 `LAT_INFLAMMATION_SEPSIS`

Binary output: `LAT_INFLAMMATION_SEPSIS`
Label 1 means: evidence of systemic inflammation, suspected infection, or sepsis burden.

Input variables: temperature, WBC, lactate, microbiology cultures, antibiotics, admission diagnosis/ICD if within baseline definition.

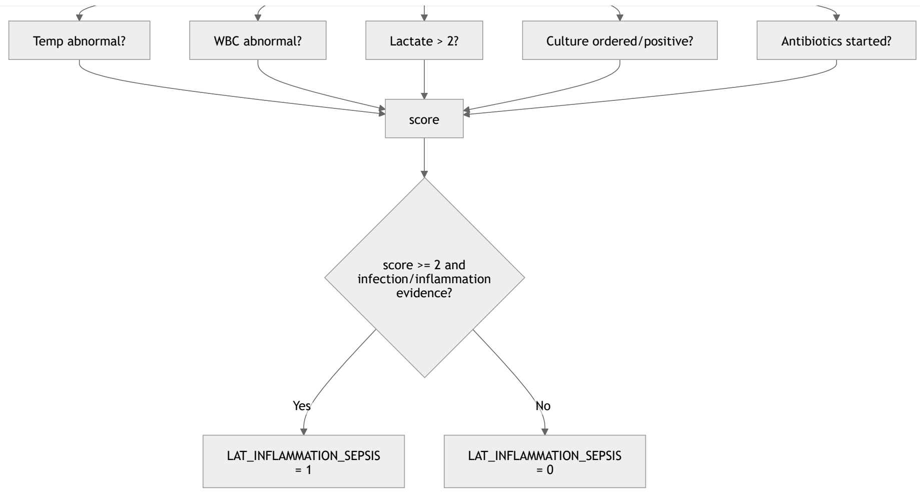
Domain indicators

Indicator	Rule
Temperature dysregulation	temp_max ≥38.3°C or temp_min <36°C
WBC abnormality	WBC_max >12 K/µL or WBC_min <4 K/µL
Hypoperfusion/inflammatory stress	lactate_max >2 mmol/L
Infection testing evidence	blood/urine/respiratory culture ordered or positive culture
Anti-infective treatment evidence	systemic antibiotic prescription started in early window

Score formula: `score = sum(indicators)`
Threshold: `LAT_INFLAMMATION_SEPSIS = 1 if score >= 2 and (culture_or_antibiotic or temp_wbc_abnormal) else 0`

Python

```python
def tag_inflammation_sepsis(x):
    temp = x.temp_max_c >= 38.3 or x.temp_min_c < 36
    wbc = x.wbc_max > 12 or x.wbc_min < 4
    lactate = x.lactate_max > 2
    culture = x.culture_ordered or x.culture_positive
    antibiotics = x.systemic_antibiotic_started
    score = sum(map(int, [temp, wbc, lactate, culture, antibiotics]))
    return int(score >= 2 and (culture or antibiotics or temp or wbc))
```



Clinical rationale: infection and inflammation generate fever, leukocyte abnormalities, lactate elevation, clinician testing, and antibiotics.
False positives: postoperative inflammation, trauma, pancreatitis, prophylactic antibiotics.
False negatives: culture-negative sepsis, immunosuppressed patients without fever/WBC response.
Confidence: medium-high.

4.3 `LAT_GLOBAL_SEVERITY`

Binary output: `LAT_GLOBAL_SEVERITY`
Label 1 means: multi-domain acute physiologic severity.

Input variables: MAP/SBP, HR, RR, SpO2, PaO2/FiO2, ventilation, GCS, creatinine, urine output, bilirubin, platelets, lactate, pH, bicarbonate, temperature.

Domain indicators

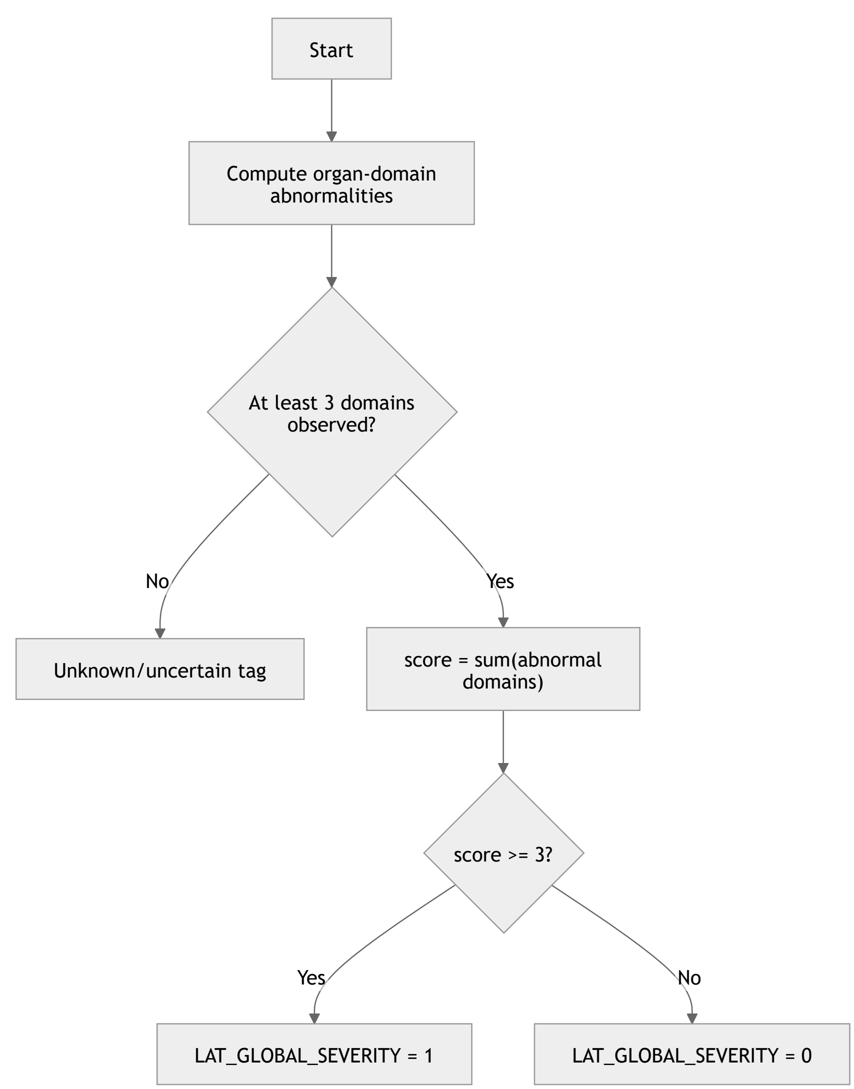
Indicator	Rule
Circulatory domain	MAP_min <70 or SBP_min ≤100 or vasopressor present
Respiratory domain	SpO2_min <92, PaO2/FiO2 ≤300, or mechanical ventilation
Neurologic domain	GCS_min <15 or RASS ≤-3 / ≥2
Renal domain	creatinine_max ≥2 or urine_output_24h <500 mL
Metabolic domain	lactate_max >2, pH_min <7.30, or bicarbonate_min <18
Inflammatory domain	temp abnormal or WBC abnormal
Hepatic/coag domain	bilirubin ≥2, platelets <100, or INR ≥1.5

Score formula: `score = sum(domains)`
Threshold: `LAT_GLOBAL_SEVERITY = 1 if score >= 3 else 0`

Python Run

```python
def tag_global_severity(x):
    domains = [
        x.map_min < 70 or x.sbp_min <= 100 or x.vasopressor_present,
        x.spo2_min < 92 or x.pao2_fio2_min <= 300 or x.mechanical_ventilation,
        x.gcs_min < 15 or x.rass_min <= -3 or x.rass_max >= 2,
        x.creatinine_max >= 2 or x.urine_output_24h < 500,
        x.lactate_max > 2 or x.ph_min < 7.30 or x.bicarbonate_min < 18,
        x.temp_max_c >= 38.3 or x.temp_min_c < 36 or x.wbc_max > 12 or x.wbc_min < 4,
```

```
        x.bilirubin_max >= 2 or x.platelets_min < 100 or x.inr_max >= 1.5,
    ]
    observed_domains = [d for d in domains if d is not None]
    if len(observed_domains) < 3:
        return None
    return int(sum(map(int, observed_domains)) >= 3)
```



Clinical rationale: operationally resembles SOFA/OASIS-style multi-organ severity without using mortality. SOFA explicitly uses respiratory, coagulation, liver, cardiovascular, neurologic, and renal domains. MSD Manuals

False positives: transient abnormalities or treatment-induced values.

False negatives: sparse labs or early unmeasured organ failure.

Confidence: high.

4.4 `LAT_SHOCK`

Binary output: `LAT_SHOCK`

Label 1 means: evidence of circulatory shock or clinically meaningful hypoperfusion.

Input variables: MAP, SBP, HR, lactate, vasopressors, urine output, pH/base excess.

Domain indicators

Indicator	Rule
Hypotension	MAP_min <65, MAP_min <70 sustained, or SBP_min ≤90

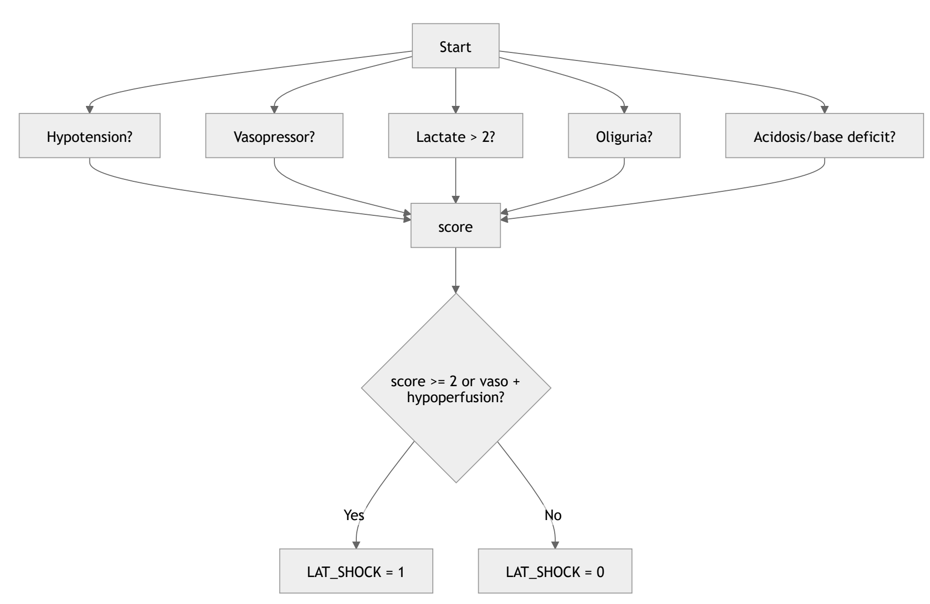
Indicator	Rule
Vasopressor support	norepinephrine, epinephrine, dopamine, phenylephrine, vasopressin, or dobutamine infusion
Tissue hypoperfusion	lactate_max >2 mmol/L
Renal perfusion consequence	urine_output_6h <0.5 mL/kg/h or urine_output_24h <500 mL
Acid-base consequence	pH_min <7.30 or base_excess_min ≤-5

Score formula: `score = sum(indicators)`

Threshold: `LAT_SHOCK = 1 if score >= 2 else 0`; also positive if vasopressor support plus lactate/acidosis.

Python

```python
def tag_shock(x):
    hypotension = x.map_min < 65 or x.sbp_min <= 90 or x.map_sustained_lt70
    vaso = x.vasopressor_present
    hypoperfusion = x.lactate_max > 2
    oliguria = x.urine_output_ml_per_kg_hr_6h < 0.5 or x.urine_output_24h < 500
    acidosis = x.ph_min < 7.30 or x.base_excess_min <= -5
    score = sum(map(int, [hypotension, vaso, hypoperfusion, oliguria, acidosis]))
    return int(score >= 2 or (vaso and (hypoperfusion or acidosis or hypotension)))
```



Clinical rationale: Sepsis-3 septic shock uses vasopressor-requiring hypotension and lactate elevation; SOFA cardiovascular scoring uses MAP and vasopressor support. PMC +1

False positives: vasopressors after anesthesia or cardiac surgery.

False negatives: shock treated before ICU arrival or missing lactate.

Confidence: high.

4.5 `LAT_RESPIRATORY_FAILURE`

Binary output: `LAT_RESPIRATORY_FAILURE`

Label 1 means: clinically meaningful hypoxemic or ventilatory failure.

Input variables: SpO2, PaO2, FiO2, PaCO2, pH, respiratory rate, mechanical ventilation, PEEP.

Domain indicators

Indicator	Rule
Hypoxemia	SpO2_min <90–92%

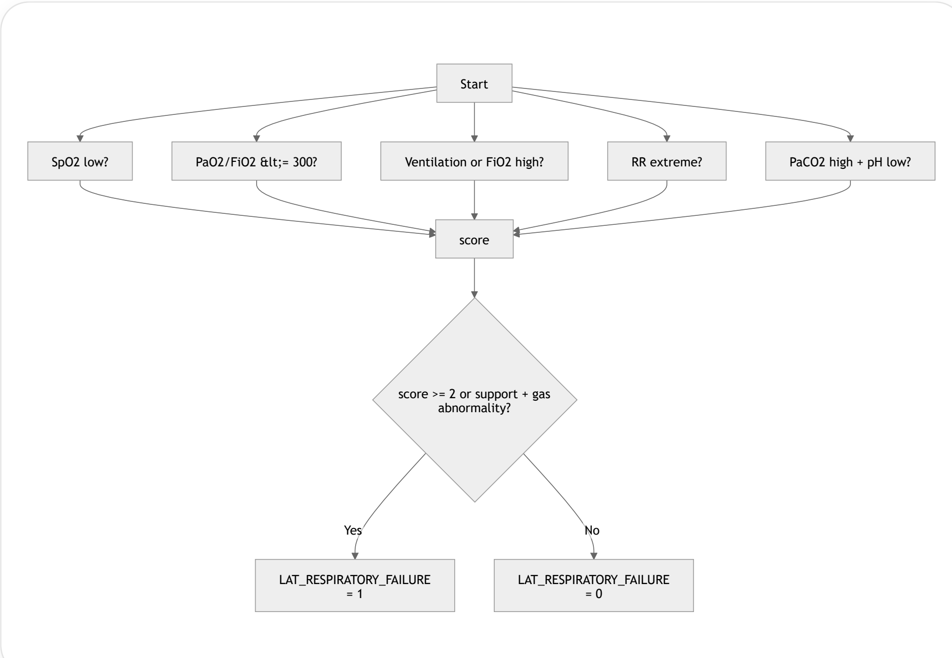
Indicator	Rule
Oxygenation failure	PaO2/FiO2_min ≤300
Respiratory support	invasive ventilation, noninvasive ventilation, or FiO2 ≥0.5
Work of breathing	RR_max ≥30 or RR_min ≤8
Ventilatory failure	PaCO2_max ≥50 with pH_min <7.30

Score formula: `score = sum(indicators)`

Threshold: `LAT_RESPIRATORY_FAILURE = 1 if score >= 2 else 0`

Python

```python
def tag_respiratory_failure(x):
    hypoxemia = x.spo2_min < 92
    oxygenation = x.pao2_fio2_min <= 300
    support = x.mechanical_ventilation or x.noninvasive_ventilation or x.fio2_max >= 0.5
    rr_abnormal = x.resp_rate_max >= 30 or x.resp_rate_min <= 8
    hypercapnia = x.paco2_max >= 50 and x.ph_min < 7.30
    score = sum(map(int, [hypoxemia, oxygenation, support, rr_abnormal, hypercapnia]))
    return int(score >= 2 or (support and (oxygenation or hypoxemia or hypercapnia)))
```



Clinical rationale: Berlin ARDS oxygenation categories use PaO2/FiO2 thresholds ≤300, ≤200, and ≤100 with PEEP/CPAP support. MSD Manuals

False positives: oxygen given prophylactically after surgery.

False negatives: missing ABG, pulse oximetry artifact, oxygenation corrected by support.

Confidence: high.

4.6 `LAT_RENAL_DYSFUNCTION`

Binary output: `LAT_RENAL_DYSFUNCTION`

Label 1 means: AKI or clinically meaningful renal dysfunction.

Input variables: creatinine, BUN, urine output, potassium, bicarbonate/pH, dialysis procedure.

Domain indicators

Indicator	Rule
Creatinine elevation	creatinine_max ≥2.0 mg/dL, or rise ≥0.3 mg/dL if baseline available
Azotemia	BUN_max ≥40 mg/dL

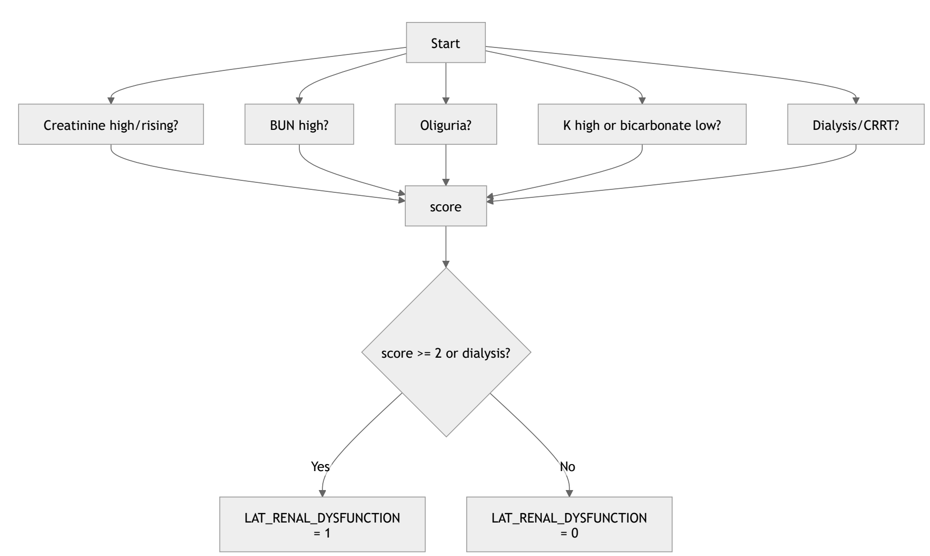
Indicator	Rule
Oliguria	urine_output_6h <0.5 mL/kg/h or urine_output_24h <500 mL
Renal metabolic consequence	potassium_max ≥5.5 or bicarbonate_min <18
Renal replacement therapy	dialysis/CRRT procedure or dialysis-related chart item

Score formula: `score = sum(indicators)`

Threshold: `LAT_RENAL_DYSFUNCTION = 1 if score >= 2 or dialysis_present else 0`

Python

```python
def tag_renal_dysfunction(x):
    creat = x.creatinine_max >= 2.0 or x.creatinine_delta_from_baseline >= 0.3
    bun = x.bun_max >= 40
    oliguria = x.urine_output_ml_per_kg_hr_6h < 0.5 or x.urine_output_24h < 500
    metabolic = x.potassium_max >= 5.5 or x.bicarbonate_min < 18
    dialysis = x.dialysis_or_crrt_present
    score = sum(map(int, [creat, bun, oliguria, metabolic, dialysis]))
    return int(score >= 2 or dialysis)
```



Clinical rationale: KDIGO AKI staging uses creatinine rise and urine-output decline; severe stages include very high creatinine, oliguria/anuria, and renal replacement therapy. MSD Manuals

False positives: chronic kidney disease without acute injury.

False negatives: no baseline creatinine, uncharted urine output.

Confidence: high.

4.7 `LAT_HEPATIC_COAG_DYSFUNCTION`

Binary output: `LAT_HEPATIC_COAG_DYSFUNCTION`

Label 1 means: hepatic, coagulation, or hematologic organ dysfunction.

Input variables: bilirubin, AST, ALT, platelets, INR, PT, PTT, albumin.

Domain indicators

Indicator	Rule
Cholestatic/clearance dysfunction	bilirubin_max ≥2.0 mg/dL
Hepatocellular injury	AST_max or ALT_max ≥3× upper limit of normal, or ≥120 U/L fallback
Thrombocytopenia	platelets_min <100 K/μL

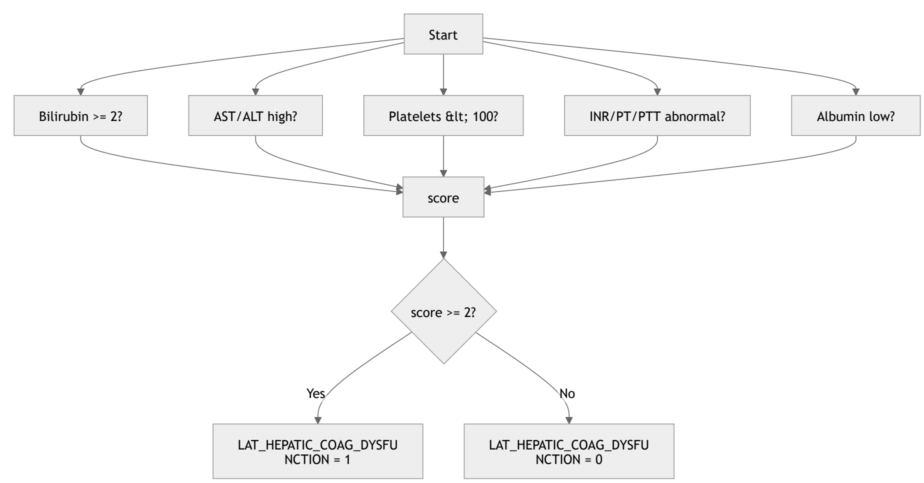
Indicator	Rule
Coagulopathy	INR_max ≥1.5, PT prolonged, or PTT prolonged
Synthetic/chronic liver signal	albumin_min <2.5 g/dL, only as weak indicator

Score formula: `score = sum(indicators)`

Threshold: `LAT_HEPATIC_COAG_DYSFUNCTION = 1 if score >= 2 else 0`

Python Run

```python
def tag_hepatic_coag_dysfunction(x):
    bilirubin = x.bilirubin_max >= 2.0
    transaminitis = x.ast_max >= 120 or x.alt_max >= 120
    platelets = x.platelets_min < 100
    coag = x.inr_max >= 1.5 or x.pt_max_prolonged or x.ptt_max_prolonged
    albumin = x.albumin_min < 2.5
    score = sum(map(int, [bilirubin, transaminitis, platelets, coag, albumin]))
    return int(score >= 2)
```



Clinical rationale: SOFA uses bilirubin for liver dysfunction and platelets for coagulation dysfunction.

MSD Manuals

False positives: anticoagulation, chronic liver disease, dilutional thrombocytopenia.
False negatives: labs not ordered in early window.
Confidence: medium.

4.8 `LAT_NEUROLOGIC_DYSFUNCTION`

Binary output: `LAT_NEUROLOGIC_DYSFUNCTION`
Label 1 means: depressed consciousness or acute neurologic dysfunction.

Input variables: GCS total/components, RASS, mental-status charting, pupil response, sedation/intubation indicators.

Domain indicators

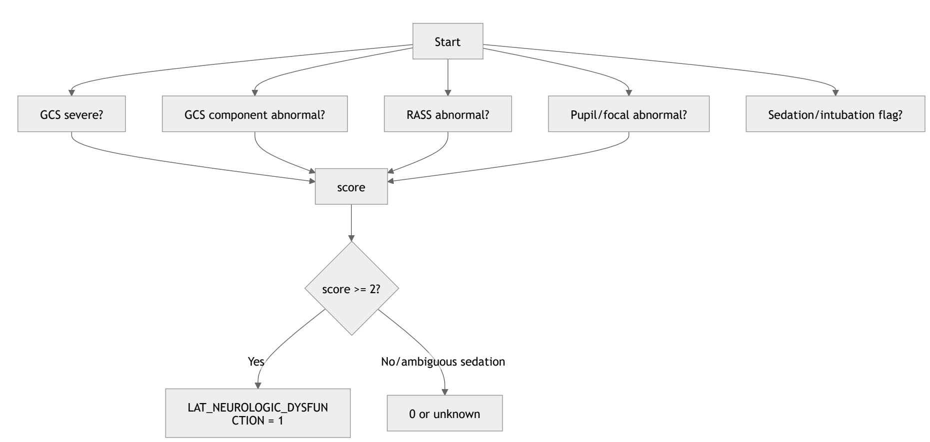
Indicator	Rule
GCS depression	GCS_min ≤13; severe if GCS_min ≤8
Motor/verbal/eye abnormality	abnormal GCS component score
RASS abnormality	RASS_min ≤-3 or RASS_max ≥2
Pupillary/neurologic abnormality	abnormal pupil response or focal neurologic charting

Indicator	Rule
Sedation/intubation context	sedative infusion or intubation present; used as confound flag, not positive evidence alone

Score formula: severe GCS counts 2; otherwise score is sum of neuro indicators.
Threshold: `LAT_NEUROLOGIC_DYSFUNCTION = 1 if score >= 2 else 0`

Python

```python
def tag_neurologic_dysfunction(x):
    score = 0
    if x.gcs_min <= 8:
        score += 2
    elif x.gcs_min <= 13:
        score += 1
    score += int(x.gcs_component_abnormal)
    score += int(x.rass_min <= -3 or x.rass_max >= 2)
    score += int(x.pupil_or_focal_neuro_abnormal)
    # sedation flag is not positive by itself; it marks ambiguity
    if x.sedation_or_intubation and score == 1:
        return None
    return int(score >= 2)
```



Clinical rationale: neurologic SOFA is based on GCS, but ICU sedation can mask neurologic status.
False positives: sedatives, postoperative anesthesia, intubation limiting verbal response.
False negatives: missing GCS or neuro exams.
Confidence: medium-high.

4.9 `LAT_METABOLIC_DERANGEMENT`

Binary output: `LAT_METABOLIC_DERANGEMENT`
Label 1 means: clinically meaningful acid-base, lactate, glucose, or electrolyte instability.

Input variables: pH, bicarbonate, base excess, lactate, sodium, potassium, glucose, anion gap if derived.

Domain indicators

Indicator	Rule
Acidemia/alkalemia	pH_min <7.30 or pH_max >7.55
Bicarbonate/base disturbance	bicarbonate_min <18, bicarbonate_max >35, or base_excess_min ≤-5
Lactate stress	lactate_max >2 mmol/L
Potassium disorder	potassium_min <3.0 or potassium_max ≥5.5
Sodium disorder	sodium_min <130 or sodium_max >150

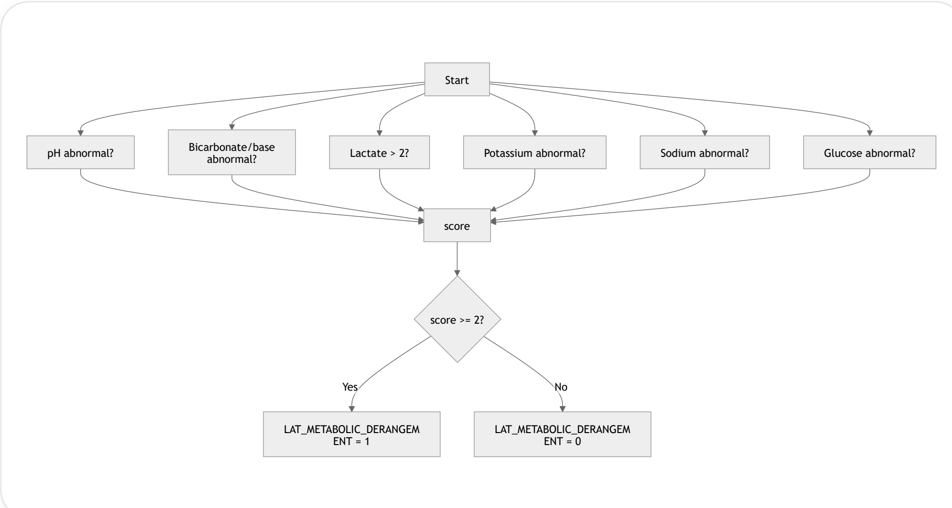
Indicator	Rule
Glucose disorder	glucose_min <70 or glucose_max >250 mg/dL

Score formula: `score = sum(indicators)`

Threshold: `LAT_METABOLIC_DERANGEMENT = 1 if score >= 2 else 0`

Python

```
def tag_metabolic_derangement(x):
    ph = x.ph_min < 7.30 or x.ph_max > 7.55
    bicarb = x.bicarbonate_min < 18 or x.bicarbonate_max > 35 or x.base_excess_min <= -5
    lactate = x.lactate_max > 2
    potassium = x.potassium_min < 3.0 or x.potassium_max >= 5.5
    sodium = x.sodium_min < 130 or x.sodium_max > 150
    glucose = x.glucose_min < 70 or x.glucose_max > 250
    score = sum(map(int, [ph, bicarb, lactate, potassium, sodium, glucose]))
    return int(score >= 2)
```



Clinical rationale: metabolic derangement is a downstream mediator of shock, renal dysfunction, respiratory failure, and sepsis.
False positives: chronic renal failure, diabetic ketoacidosis, treatment-induced electrolyte shifts.
False negatives: missing ABG or chemistry panels.
Confidence: high.

4.10 `LAT_CARDIAC_STRAIN`

Binary output: `LAT_CARDIAC_STRAIN`
Label 1 means: evidence of myocardial injury, strain, or major rhythm stress.

Input variables: troponin, CK-MB, heart rate, arrhythmia charting, ECG report metadata, vasopressor/cardiac ICU context as weak contextual signal.

Domain indicators

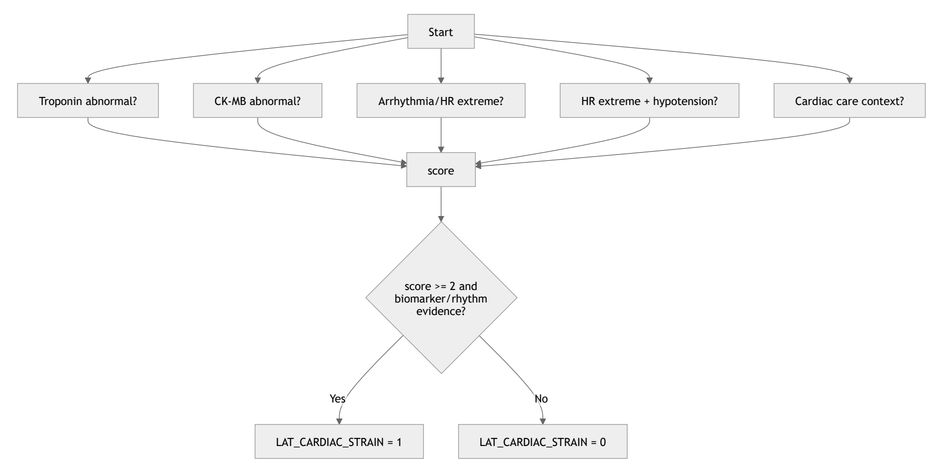
Indicator	Rule
Myocardial biomarker	troponin above local assay upper reference limit, or troponin qualitatively abnormal
CK-MB injury signal	CK-MB above reference range
Rhythm instability	arrhythmia charting, atrial/ventricular tachyarrhythmia, severe bradycardia
Hemodynamic cardiac stress	HR_max >130 or HR_min <40 with hypotension
Cardiac-care context	CSRU/CCU/cardiac surgery context, weak indicator only

Score formula: biomarker or rhythm evidence required; `score = sum(indicators)`

Threshold: `LAT_CARDIAC_STRAIN = 1 if score >= 2 and (biomarker or rhythm) else 0`

Python

```python
def tag_cardiac_strain(x):
    biomarker = x.troponin_above_uln or x.troponin_flag_abnormal
    ckmb = x.ckmb_above_uln
    rhythm = x.arrhythmia_documented or x.hr_max > 150 or x.hr_min < 40
    hemodynamic = (x.hr_max > 130 or x.hr_min < 40) and (x.map_min < 70 or x.sbp_min <= 100)
    context = x.first_careunit in ["CSRU", "CCU"] or x.cardiac_surgery_context
    score = sum(map(int, [biomarker, ckmb, rhythm, hemodynamic, context]))
    return int(score >= 2 and (biomarker or rhythm))
```



Clinical rationale: cardiac injury can precipitate shock and global severity, but measurement is selective.
False positives: demand ischemia, renal failure elevating troponin, postoperative cardiac surgery.
False negatives: troponin not ordered.
Confidence: low-medium.

5. Observed-to-latent measurement model

served variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care-process / measurement-intensity influence?	No
e	`LAT_CHRONIC_BURDEN` / background	Older → higher vulnerability	Nonspecific	Yes	No	De dei spe
/gender	Background	Context-dependent	Nonspecific	No	Possible	Co not
nicity/insurance/langua	Background/social context	Context-dependent	Nonspecific	No	Yes	Ma me
nission type/location	Background	Emergency/urgent → higher acute risk	Nonspecific	Yes	Yes	Co ind
type	Background/care context	Unit-specific risk	Nonspecific	Yes	Yes	Ca me
-9 diagnoses	`LAT_CHRONIC_BURDEN` , admission phenotype	Chronic disease codes increase burden	Medium	Yes	Coding process	No out
art rate	`LAT_SHOCK` , `LAT_INFLAMMATION_SEPSIS` , `LAT_CARDIAC_STRAIN` , `LAT_GLOBAL_SEVERITY`	High or very low abnormal	Nonspecific	Yes	Yes	Pai sho

served variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care-process / measurement-intensity influence?	No
P/SBP	`LAT_SHOCK` , `LAT_GLOBAL_SEVERITY`	Low abnormal; very high may severity	Medium	Yes	Strongly treatment-modified	Vas cha
piratory rate	`LAT_RESPIRATORY_FAILURE` , `LAT_METABOLIC_DERANGEMENT` , `LAT_INFLAMMATION_SEPSIS`	High/low abnormal	Nonspecific	Yes	Yes	Co anc
)2	`LAT_RESPIRATORY_FAILURE`	Low abnormal	Medium	Yes	Oxygen therapy modifies	Ser
)2/FiO2	`LAT_RESPIRATORY_FAILURE`	Lower worse	Specific	Yes	Ventilator/FiO2 dependent	Rec FiC
:O2/pH	`LAT_RESPIRATORY_FAILURE` , `LAT_METABOLIC_DERANGEMENT`	Hypercapnia with low pH worse	Medium	Yes	ABG ordering informative	Str
nperature	`LAT_INFLAMMATION_SEPSIS` , `LAT_GLOBAL_SEVERITY`	Fever/hypothermia abnormal	Nonspecific	Yes	Frequent vitals	Hy ICL
C	`LAT_INFLAMMATION_SEPSIS`	High/low abnormal	Nonspecific	Yes	Lab ordering	Ste cor
tate	`LAT_SHOCK` , `LAT_INFLAMMATION_SEPSIS` , `LAT_METABOLIC_DERANGEMENT` , `LAT_GLOBAL_SEVERITY`	Higher worse	Medium	Yes	Highly order-dependent	Str ma
atinine	`LAT_RENAL_DYSFUNCTION`	Higher/rising worse	Medium	Yes	Lab ordering	Chı cor
N	`LAT_RENAL_DYSFUNCTION` , `LAT_CHRONIC_BURDEN`	Higher worse	Nonspecific	Yes	Lab ordering	Als ble
ne output	`LAT_RENAL_DYSFUNCTION` , `LAT_SHOCK`	Lower worse	Medium	Yes	Charting and catheter use	Ou inf
assium	`LAT_RENAL_DYSFUNCTION` , `LAT_METABOLIC_DERANGEMENT`	High/low abnormal	Medium	Yes	Treatment-modified	Dia aff
arbonate/base excess	`LAT_METABOLIC_DERANGEMENT` , `LAT_RENAL_DYSFUNCTION` , `LAT_SHOCK`	Low or extreme abnormal	Medium	Yes	Lab/ABG ordering	Aci
lium	`LAT_METABOLIC_DERANGEMENT`	Low/high abnormal	Nonspecific	Yes	Treatment/fluid modified	Flu
cose	`LAT_METABOLIC_DERANGEMENT` , `LAT_INFLAMMATION_SEPSIS`	Low/high abnormal	Nonspecific	Yes	Frequent if diabetic/insulin	Me inf
:elets	`LAT_HEPATIC_COAG_DYSFUNCTION` , `LAT_INFLAMMATION_SEPSIS`	Lower worse	Medium	Yes	Lab ordering	Sep dil
/PT/PTT	`LAT_HEPATIC_COAG_DYSFUNCTION`	Higher/prolonged worse	Medium	Yes	Anticoagulants/procedures	Str cor
ubin	`LAT_HEPATIC_COAG_DYSFUNCTION`	Higher worse	Medium	Yes	Selectively ordered	SO
/ALT	`LAT_HEPATIC_COAG_DYSFUNCTION` , `LAT_SHOCK`	Higher worse	Nonspecific	Yes	Selectively ordered	Sho
umin	`LAT_CHRONIC_BURDEN` , `LAT_HEPATIC_COAG_DYSFUNCTION`	Lower worse	Nonspecific	Weak	Selectively ordered	Chı infl ma
S	`LAT_NEUROLOGIC_DYSFUNCTION` , `LAT_GLOBAL_SEVERITY`	Lower worse	Medium	Yes	Sedation/intubation	Ne

served variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care-process / measurement-intensity influence?	No
S	LAT_NEUROLOGIC_DYSFUNCTION , care process	Extreme low/high abnormal	Medium	Yes	Sedation target	No
ponin/CK-MB	LAT_CARDIAC_STRAIN , LAT_RENAL_DYSFUNCTION	Higher worse	Medium	Yes	Selectively ordered	Spa cor
opressor administration	TRT_VASOPRESSORS , proxy for LAT_SHOCK	Present → shock severity proxy	Specific for treatment, not biology	Yes	Assignment by clinician	Tre me
chanical ventilation	TRT_MECH_VENT , proxy for LAT_RESPIRATORY_FAILURE	Present → severe respiratory/airway support	Medium	Yes	Clinical decision	Car tha
lysis/CRRT	TRT_DIALYSIS , proxy for LAT_RENAL_DYSFUNCTION	Present → severe renal/metabolic failure	High	Yes	Clinical decision	Tre ma
ibiotics/cultures	TRT_ANTIBIOTICS , proxy for LAT_INFLAMMATION_SEPSIS	Present/ordered → suspected infection	Medium	Yes	Clinician suspicion	No infe
/vital counts	MISS_MEASUREMENT_INTENSITY	More frequent often higher severity	Nonspecific	Limited	Directly	Info bio

6. Latent causal DAG edge list

Source node	Target node	Edge type	Clinical mechanism	Confidence	Reasoning basi
BG_AGE	LAT_CHRONIC_BURDEN	background-to-latent	Aging increases chronic vulnerability and frailty burden	High	Clinical plausibility
BG_SEX	LAT_CHRONIC_BURDEN	background-to-latent	Sex differences in comorbidity and physiology	Medium	Case-mix
BG_ETHNICITY_INSURANCE_LANGUAGE	MISS_MEASUREMENT_INTENSITY	measurement-process	Social/access factors may affect care and documentation	Medium	Informative observation literature
BG_ADMISSION_CONTEXT	LAT_INFLAMMATION_SEPSIS	background-to-latent	Emergency/medical admissions more likely acute infection	Medium	Case-mix
BG_ADMISSION_CONTEXT	LAT_GLOBAL_SEVERITY	background-to-latent	Admission source/type captures acute presentation	Medium	Case-mix
BG_ICU_UNIT	LAT_GLOBAL_SEVERITY	background-to-latent	ICU type reflects illness mix	Medium	Dataset suppor
BG_ICU_UNIT	MISS_MEASUREMENT_INTENSITY	measurement-process	Units differ in monitoring and charting	High	Dataset source/care process
LAT_CHRONIC_BURDEN	LAT_GLOBAL_SEVERITY	latent-to-latent	Vulnerability worsens physiologic response to acute illness	High	Clinical mechanism
LAT_CHRONIC_BURDEN	OUT_MORTALITY	latent-to-mortality	Frailty/comorbidity directly affects survival reserve	High	Clinical mechanism
LAT_INFLAMMATION_SEPSIS	LAT_GLOBAL_SEVERITY	latent-to-latent	Systemic inflammation drives multi-organ instability	High	Sepsis/organ dysfunction
LAT_INFLAMMATION_SEPSIS	LAT_SHOCK	latent-to-latent	Vasodilation, capillary leak, myocardial depression	High	Septic shock mechanism

Source node	Target node	Edge type	Clinical mechanism	Confidence	Reasoning basi
LAT_INFLAMMATION_SEPSIS	LAT_HEPATIC_COAG_DYSFUNCTION	latent-to-latent	Sepsis-associated thrombocytopenia/coagulopathy/liver injury	Medium	Clinical mechanism
LAT_GLOBAL_SEVERITY	LAT_RESPIRATORY_FAILURE	latent-to-latent	Severe illness can lead to hypoxemia/ventilatory failure	Medium	ICU physiology
LAT_GLOBAL_SEVERITY	LAT_NEUROLOGIC_DYSFUNCTION	latent-to-latent	Encephalopathy, sedation need, hypoperfusion	Medium	ICU physiology
LAT_GLOBAL_SEVERITY	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Multi-organ severity disrupts homeostasis	High	Clinical mechanism
LAT_CARDIAC_STRAIN	LAT_SHOCK	latent-to-latent	Pump failure or arrhythmia can cause hypoperfusion	Medium	Clinical mechanism
LAT_SHOCK	LAT_RENAL_DYSFUNCTION	latent-to-latent	Renal hypoperfusion and ischemic tubular injury	High	Clinical mechanism
LAT_SHOCK	LAT_HEPATIC_COAG_DYSFUNCTION	latent-to-latent	Shock liver, coagulopathy, DIC risk	Medium	Clinical mechanism
LAT_SHOCK	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Lactate/acidosis from hypoperfusion	High	Clinical mechanism
LAT_RESPIRATORY_FAILURE	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hypercapnia/acidosis and hypoxemia	High	Clinical mechanism
LAT_RENAL_DYSFUNCTION	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hyperkalemia, acidosis, uremia	High	KDIGO/metabo mechanism
LAT_SHOCK	OUT_MORTALITY	latent-to-mortality	Inadequate perfusion causes organ failure/death	High	Clinical mechanism
LAT_RESPIRATORY_FAILURE	OUT_MORTALITY	latent-to-mortality	Hypoxemia/hypercapnia and ventilatory failure	High	Clinical mechanism
LAT_RENAL_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	AKI reflects and worsens multi-organ failure	High	Clinical mechanism
LAT_HEPATIC_COAG_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	Bleeding, thrombosis, liver failure, severe systemic illness	Medium	SOFA organ domains
LAT_NEUROLOGIC_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	Coma/encephalopathy/brain injury indicates reduced survival	High	SOFA/OASIS domains
LAT_METABOLIC_DERANGEMENT	OUT_MORTALITY	latent-to-mortality	Severe acidosis/electrolyte/glucose derangement can be fatal	High	Clinical mechanism
LAT_CARDIAC_STRAIN	OUT_MORTALITY	latent-to-mortality	Myocardial injury/arrhythmia increases mortality	Medium	Clinical mechanism
LAT_*	OBS_*	latent-to-observed	Latent states generate labs/vitals/charted values	High	Measurement model
LAT_GLOBAL_SEVERITY	MISS_MEASUREMENT_INTENSITY	measurement-process	Sicker patients receive more tests and monitoring	High	Informative observation literature
LAT_INFLAMMATION_SEPSIS	TRT_ANTIBIOTICS	treatment/care-process	Suspected infection causes antibiotic ordering	High	Clinical care process
LAT_SHOCK	TRT_VASOPRESSORS	treatment/care-process	Shock causes vasopressor treatment	High	Clinical care process

Source node	Target node	Edge type	Clinical mechanism	Confidence	Reasoning basi
LAT_RESPIRATORY_FAILURE	TRT_MECH_VENT	treatment/care-process	Respiratory failure causes ventilatory support	High	Clinical care process
LAT_RENAL_DYSFUNCTION	TRT_DIALYSIS	treatment/care-process	Severe renal/metabolic failure causes dialysis	High	Clinical care process
TRT_*	OBS_*	treatment-to-observed	Treatments change observed vitals/labs	Medium	Treatment response
MISS_MEASUREMENT_INTENSITY	OBS_AVAILABILITY	measurement-process	Ordering/charting determines whether measurements exist	High	EHR process

7. Mortality mechanism summary

Mortality in this DAG is caused by **latent clinical states**, not raw measurements. The main mortality pathways are:

1. **Chronic vulnerability pathway:**
   `BG_AGE / BG_ADMISSION_CONTEXT → LAT_CHRONIC_BURDEN → LAT_GLOBAL_SEVERITY → organ dysfunction → OUT_MORTALITY`.
2. **Sepsis/shock pathway:**
   `LAT_INFLAMMATION_SEPSIS → LAT_SHOCK → LAT_RENAL_DYSFUNCTION / LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY`.
3. **Respiratory failure pathway:**
   `LAT_RESPIRATORY_FAILURE → hypoxemia/hypercapnia/acidosis represented by LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY`.
4. **Renal-metabolic pathway:**
   `LAT_RENAL_DYSFUNCTION → LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY`.
5. **Cardiac-hemodynamic pathway:**
   `LAT_CARDIAC_STRAIN → LAT_SHOCK → LAT_GLOBAL_SEVERITY / LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY`.

The observed values are children of these states. For example, lactate is an observed proxy of shock/metabolic stress; it is not modeled as the root cause of death. Similarly, low PaO2/FiO2 indicates respiratory failure, and creatinine indicates renal dysfunction.

8. Missingness and irregular sampling analysis

DATASET-NAME is not sampled by a fixed research protocol. It is routine-care EHR data: nurses chart vitals, clinicians order labs, medications are recorded when administered or prescribed, and some procedures are recorded only in specific source systems. The official documentation states that the database was populated from routine hospital care and includes time-stamped bedside measurements, medications, fluid balances, labs, and reports. physionet.org

Missingness is therefore rarely pure MCAR. The most plausible mechanisms are:

Variable class	Likely missingness mechanism	Interpretation
Hourly vitals	MAR/care-intensity-driven	Usually frequent in ICU, but frequency still reflects acuity and unit practice.
ABGs, lactate, troponin, coagulation labs	MNAR / informative observation	Ordered when clinicians suspect hypoxemia, shock, cardiac injury, bleeding, or severe illness.
Microbiology cultures	Clinician-suspicion-driven	Absence of culture does not prove absence of infection.
Urine output	Care-process dependent	Requires charting and often catheterization.
Ventilator settings	Treatment-dependent	Missing may mean not ventilated or simply unrecorded depending extraction logic.

Variable class	Likely missingness mechanism	Interpretation
ICD diagnoses	Coding process	No timestamp; missing code is not equivalent to absence.
Notes/reports	Care-process and documentation-driven	Highly informative but can contain leakage if discharge summaries are used.

Informative presence and informative observation are well-described in routine health data: the presence, timing, and frequency of observations can themselves carry information about health state, and the literature specifically notes that more severe conditions often lead to more intense monitoring. OUP Academic

For latent labels:

- Do **not** treat missing high-cost/selective labs as normal.
- Use missingness indicators and observation counts as separate features.
- Allow "unknown" or low-confidence tags when too few domains are observed.
- Use measurement frequency only as a weak auxiliary signal in `LAT_GLOBAL_SEVERITY` ; do not let missingness dominate biological definitions.
- Represent `MISS_MEASUREMENT_INTENSITY` as a separate node influenced by severity, ICU unit, clinician suspicion, and care processes.

Bias risks:

- Conditioning on measurement availability can create collider bias because severity and care unit both affect whether labs are measured.
- Using discharge summaries or post-outcome notes can create leakage.
- Using treatment presence as if it were pure disease severity can induce confounding by indication.

9. Backdoor/confounding implications

For causal effects on mortality, adjustment depends on the exposure.

Likely confounders

Exposure type	Candidate confounders
Baseline exposure, e.g. admission type or ICU unit	Age, sex, ethnicity/social-context proxies, chronic burden, admission source
Early treatment, e.g. vasopressors	Chronic burden, admission context, initial shock severity, sepsis burden, cardiac strain
Ventilation strategy	Chronic lung disease, admission type, respiratory failure severity, neurologic dysfunction, surgical context
Antibiotics	Infection suspicion, temperature/WBC, cultures, chronic immunosuppression, admission diagnosis
Dialysis	Baseline kidney disease, renal dysfunction severity, shock, metabolic derangement

Likely mediators

Variable	Why mediator
`LAT_SHOCK`	Mediates sepsis/cardiac injury effects on mortality.
`LAT_RENAL_DYSFUNCTION`	Mediates shock/sepsis effects on mortality.
`LAT_METABOLIC_DERANGEMENT`	Mediates shock, renal, and respiratory effects.
`LAT_GLOBAL_SEVERITY`	Often summarizes downstream acute physiology; inappropriate adjustment for total effects of upstream exposures.
Treatments	May mediate effects of latent severity on later outcomes, while also being confounded by severity.

Colliders and variables to avoid conditioning on

Variable	Collider risk
`MISS_MEASUREMENT_INTENSITY`	Caused by latent severity, clinician suspicion, ICU unit, and social/care factors.
Lab availability	Conditioning on ordered labs selects on clinician suspicion and severity.
ICU length of stay	Affected by survival, severity, and discharge readiness.
Discharge location	Downstream of outcome trajectory.
`LAT_GLOBAL_SEVERITY`	Can be a collider/proxy outcome if estimating effects of upstream disease states or treatments.
Treatment receipt	Collider/confounder-by-indication if conditioned on without longitudinal causal design.

For estimating the **total effect** of an upstream exposure such as sepsis burden on mortality, do not adjust for downstream `LAT_SHOCK`, `LAT_RENAL_DYSFUNCTION`, `LAT_METABOLIC_DERANGEMENT`, or `LAT_GLOBAL_SEVERITY`. For estimating a **controlled direct effect**, adjustment may be appropriate only with a clearly specified target trial and temporal ordering.

For ICU time-varying treatments, standard regression adjustment is often biased when prior treatment affects later confounders; ICU causal inference reviews emphasize target-trial framing and G-methods for treatment-affected time-varying confounding. Nature

Binary latent tags are therefore not automatically safe adjustment variables. They may be:

- confounders when measured before exposure,
- mediators when downstream of exposure,
- colliders when defined using measurement intensity or treatment decisions,
- proxy outcomes when they are outcome-proximal severity markers.

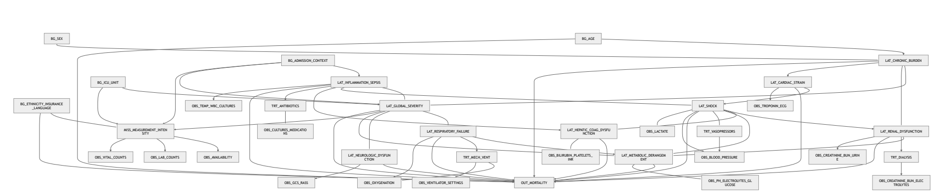
10. Graph specification

A. Adjacency list

```
BG_AGE:
  - LAT_CHRONIC_BURDEN
  - OUT_MORTALITY

BG_SEX:
  - LAT_CHRONIC_BURDEN

BG_ETHNICITY_INSURANCE_LANGUAGE:
  - MISS_MEASUREMENT_INTENSITY
  - OUT_MORTALITY

BG_ADMISSION_CONTEXT:
  - LAT_INFLAMMATION_SEPSIS
  - LAT_GLOBAL_SEVERITY
  - MISS_MEASUREMENT_INTENSITY

BG_ICU_UNIT:
  - LAT_GLOBAL_SEVERITY
  - MISS_MEASUREMENT_INTENSITY

LAT_CHRONIC_BURDEN:
  - LAT_GLOBAL_SEVERITY
  - LAT_RENAL_DYSFUNCTION
  - LAT_CARDIAC_STRAIN
  - OUT_MORTALITY

LAT_INFLAMMATION_SEPSIS:
  - LAT_GLOBAL_SEVERITY
  - LAT_SHOCK
  - LAT_HEPATIC_COAG_DYSFUNCTION
  - TRT_ANTIBIOTICS
  - OUT_MORTALITY
```

```
LAT_GLOBAL_SEVERITY:
  - LAT_RESPIRATORY_FAILURE
  - LAT_NEUROLOGIC_DYSFUNCTION
  - LAT_METABOLIC_DERANGEMENT
  - MISS_MEASUREMENT_INTENSITY
  - OUT_MORTALITY

LAT_CARDIAC_STRAIN:
  - LAT_SHOCK
  - OUT_MORTALITY

LAT_SHOCK:
  - LAT_RENAL_DYSFUNCTION
  - LAT_HEPATIC_COAG_DYSFUNCTION
  - LAT_METABOLIC_DERANGEMENT
  - TRT_VASOPRESSORS
  - OUT_MORTALITY

LAT_RESPIRATORY_FAILURE:
  - LAT_METABOLIC_DERANGEMENT
  - TRT_MECH_VENT
  - OUT_MORTALITY

LAT_RENAL_DYSFUNCTION:
  - LAT_METABOLIC_DERANGEMENT
  - TRT_DIALYSIS
  - OUT_MORTALITY

LAT_HEPATIC_COAG_DYSFUNCTION:
  - OUT_MORTALITY

LAT_NEUROLOGIC_DYSFUNCTION:
  - OUT_MORTALITY

LAT_METABOLIC_DERANGEMENT:
  - OUT_MORTALITY

TRT_ANTIBIOTICS:
  - OBS_CULTURES_MEDICATIONS

TRT_VASOPRESSORS:
  - OBS_BLOOD_PRESSURE
  - OBS_LACTATE
  - OUT_MORTALITY

TRT_MECH_VENT:
  - OBS_OXYGENATION
  - OBS_VENTILATOR_SETTINGS
  - OUT_MORTALITY

TRT_DIALYSIS:
  - OBS_CREATININE_BUN_ELECTROLYTES
  - OUT_MORTALITY

MISS_MEASUREMENT_INTENSITY:
  - OBS_AVAILABILITY
  - OBS_LAB_COUNTS
  - OBS_VITAL_COUNTS
```

B. Mermaid graph



C. Python NetworkX edge list

Python

```python
edges = [
    ("BG_AGE", "LAT_CHRONIC_BURDEN"),
    ("BG_AGE", "OUT_MORTALITY"),
    ("BG_SEX", "LAT_CHRONIC_BURDEN"),
    ("BG_ETHNICITY_INSURANCE_LANGUAGE", "MISS_MEASUREMENT_INTENSITY"),
    ("BG_ETHNICITY_INSURANCE_LANGUAGE", "OUT_MORTALITY"),
    ("BG_ADMISSION_CONTEXT", "LAT_INFLAMMATION_SEPSIS"),
    ("BG_ADMISSION_CONTEXT", "LAT_GLOBAL_SEVERITY"),
    ("BG_ADMISSION_CONTEXT", "MISS_MEASUREMENT_INTENSITY"),
    ("BG_ICU_UNIT", "LAT_GLOBAL_SEVERITY"),
    ("BG_ICU_UNIT", "MISS_MEASUREMENT_INTENSITY"),

    ("LAT_CHRONIC_BURDEN", "LAT_GLOBAL_SEVERITY"),
    ("LAT_CHRONIC_BURDEN", "LAT_RENAL_DYSFUNCTION"),
    ("LAT_CHRONIC_BURDEN", "LAT_CARDIAC_STRAIN"),
    ("LAT_CHRONIC_BURDEN", "OUT_MORTALITY"),

    ("LAT_INFLAMMATION_SEPSIS", "LAT_GLOBAL_SEVERITY"),
    ("LAT_INFLAMMATION_SEPSIS", "LAT_SHOCK"),
    ("LAT_INFLAMMATION_SEPSIS", "LAT_HEPATIC_COAG_DYSFUNCTION"),
    ("LAT_INFLAMMATION_SEPSIS", "TRT_ANTIBIOTICS"),
    ("LAT_INFLAMMATION_SEPSIS", "OUT_MORTALITY"),

    ("LAT_GLOBAL_SEVERITY", "LAT_RESPIRATORY_FAILURE"),
    ("LAT_GLOBAL_SEVERITY", "LAT_NEUROLOGIC_DYSFUNCTION"),
    ("LAT_GLOBAL_SEVERITY", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_GLOBAL_SEVERITY", "MISS_MEASUREMENT_INTENSITY"),
    ("LAT_GLOBAL_SEVERITY", "OUT_MORTALITY"),

    ("LAT_CARDIAC_STRAIN", "LAT_SHOCK"),
    ("LAT_CARDIAC_STRAIN", "OUT_MORTALITY"),

    ("LAT_SHOCK", "LAT_RENAL_DYSFUNCTION"),
    ("LAT_SHOCK", "LAT_HEPATIC_COAG_DYSFUNCTION"),
    ("LAT_SHOCK", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_SHOCK", "TRT_VASOPRESSORS"),
    ("LAT_SHOCK", "OUT_MORTALITY"),

    ("LAT_RESPIRATORY_FAILURE", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_RESPIRATORY_FAILURE", "TRT_MECH_VENT"),
    ("LAT_RESPIRATORY_FAILURE", "OUT_MORTALITY"),

    ("LAT_RENAL_DYSFUNCTION", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_RENAL_DYSFUNCTION", "TRT_DIALYSIS"),
    ("LAT_RENAL_DYSFUNCTION", "OUT_MORTALITY"),

    ("LAT_HEPATIC_COAG_DYSFUNCTION", "OUT_MORTALITY"),
    ("LAT_NEUROLOGIC_DYSFUNCTION", "OUT_MORTALITY"),
    ("LAT_METABOLIC_DERANGEMENT", "OUT_MORTALITY"),

    ("LAT_SHOCK", "OBS_BLOOD_PRESSURE"),
    ("LAT_SHOCK", "OBS_LACTATE"),
    ("LAT_RESPIRATORY_FAILURE", "OBS_OXYGENATION"),
    ("LAT_RENAL_DYSFUNCTION", "OBS_CREATININE_BUN_URINE"),
    ("LAT_HEPATIC_COAG_DYSFUNCTION", "OBS_BILIRUBIN_PLATELETS_INR"),
    ("LAT_NEUROLOGIC_DYSFUNCTION", "OBS_GCS_RASS"),
    ("LAT_METABOLIC_DERANGEMENT", "OBS_PH_ELECTROLYTES_GLUCOSE"),
    ("LAT_INFLAMMATION_SEPSIS", "OBS_TEMP_WBC_CULTURES"),
    ("LAT_CARDIAC_STRAIN", "OBS_TROPONIN_ECG"),

    ("TRT_ANTIBIOTICS", "OBS_CULTURES_MEDICATIONS"),
    ("TRT_VASOPRESSORS", "OBS_BLOOD_PRESSURE"),
    ("TRT_MECH_VENT", "OBS_OXYGENATION"),
    ("TRT_MECH_VENT", "OBS_VENTILATOR_SETTINGS"),
    ("TRT_DIALYSIS", "OBS_CREATININE_BUN_ELECTROLYTES"),

    ("MISS_MEASUREMENT_INTENSITY", "OBS_AVAILABILITY"),
    ("MISS_MEASUREMENT_INTENSITY", "OBS_LAB_COUNTS"),
    ("MISS_MEASUREMENT_INTENSITY", "OBS_VITAL_COUNTS"),
]
```

11. Decision-tree implementation specification

11.1 Required input columns

At minimum, generate one analytic row per `ICUSTAY_ID` and early window.

Column group	Required columns / features
Identifiers	`subject_id`, `hadm_id`, `icustay_id`, `window_start`, `window_end`
Background	`age`, `gender`, `ethnicity`, `insurance`, `language`, `admission_type`, `admission_location`, `first_careunit`
ICD-derived	`comorbidity_count`, `elixhauser_score`, chronic organ disease flags, malignancy flag
Vitals	HR, SBP, DBP, MAP, RR, SpO2, temperature
Neuro	GCS total/components, RASS, pupil response if available
Respiratory	PaO2, FiO2, PaO2/FiO2, PaCO2, pH, PEEP, ventilation status
Renal/output	creatinine, BUN, urine output, dialysis/CRRT flag
Metabolic	bicarbonate, base excess, lactate, sodium, potassium, glucose
Hepatic/coag	bilirubin, AST, ALT, albumin, platelets, INR, PT, PTT
Infection	WBC, cultures ordered/positive, antibiotic started
Cardiac	troponin, CK-MB, arrhythmia flag, ECG-report-derived flag if used
Treatments	vasopressor present, mechanical ventilation, dialysis, antibiotics, fluids
Missingness	count per variable, missing flag per variable, time since last observation

11.2 Derived summary features

For each numeric time-varying variable within the early window:

```
first_value
last_value
min_value
max_value
mean_value
slope_per_hour
std_or_variability
count_observed
missing_indicator
time_to_first_measurement
```

For irregular data:

```
- Use timestamps, not row order.
- Convert units before thresholds.
- Compute worst values clinically:
    BP/SpO2/GCS/pH/bicarbonate: lower can be worse.
    HR/RR/temp/glucose/sodium/potassium: both extremes can be worse.
    creatinine/BUN/lactate/bilirubin/INR: higher worse.
- Compute PaO2/FiO2 only when paired within a clinically reasonable interval.
```

11.3 Expected output columns

```
LAT_CHRONIC_BURDEN
LAT_INFLAMMATION_SEPSIS
LAT_GLOBAL_SEVERITY
LAT_SHOCK
LAT_RESPIRATORY_FAILURE
LAT_RENAL_DYSFUNCTION
LAT_HEPATIC_COAG_DYSFUNCTION
LAT_NEUROLOGIC_DYSFUNCTION
LAT_METABOLIC_DERANGEMENT
LAT_CARDIAC_STRAIN

Optional uncertainty flags:
```

```
LAT_*_UNKNOWN
LAT_*_LOW_CONFIDENCE
MISS_MEASUREMENT_INTENSITY_SCORE
```

11.4 Threshold table

Feature	Threshold
Age burden	≥75 years
MAP low	<65 mmHg, or sustained <70 mmHg
SBP low	≤90 mmHg; ≤100 as weaker qSOFA-style threshold
Lactate high	>2 mmol/L
SpO2 low	<92%; severe <90%
PaO2/FiO2 abnormal	≤300
RR high	≥30; sepsis screening weaker threshold ≥22
GCS abnormal	<15; moderate ≤13; severe ≤8
Creatinine high	≥2.0 mg/dL or rise ≥0.3 mg/dL
Urine output low	<0.5 mL/kg/h over 6h or <500 mL/day
BUN high	≥40 mg/dL
Potassium abnormal	<3.0 or ≥5.5 mmol/L
Sodium abnormal	<130 or >150 mmol/L
Bicarbonate low	<18 mmol/L
pH abnormal	<7.30 or >7.55
WBC abnormal	<4 or >12 K/μL
Temperature abnormal	<36°C or ≥38.3°C
Bilirubin high	≥2 mg/dL
Platelets low	<100 K/μL
INR high	≥1.5
Glucose abnormal	<70 or >250 mg/dL

11.5 Validation checks

Python

Run

```python
required_outputs = [
    "LAT_CHRONIC_BURDEN",
    "LAT_INFLAMMATION_SEPSIS",
    "LAT_GLOBAL_SEVERITY",
    "LAT_SHOCK",
    "LAT_RESPIRATORY_FAILURE",
    "LAT_RENAL_DYSFUNCTION",
    "LAT_HEPATIC_COAG_DYSFUNCTION",
    "LAT_NEUROLOGIC_DYSFUNCTION",
    "LAT_METABOLIC_DERANGEMENT",
    "LAT_CARDIAC_STRAIN",
]

for col in required_outputs:
    assert col in latent_tags.columns
    assert set(latent_tags[col].dropna().unique()).issubset({0, 1})
```

```
for forbidden in [
    "deathtime", "hospital_expire_flag", "expire_flag",
    "discharge_location", "los", "survival_time"
]:
    assert forbidden not in features_used_for_latent_tags

assert all(feature_time <= window_end for feature_time in extracted_event_times)
assert units_normalized is True
assert itemid_mapping_version_logged is True
assert missingness_flags_created is True
```

12. Rejected alternatives

Rejected item	Reason
Raw lab-to-lab causal graph	Labs are mostly manifestations of physiology, not primary causal mechanisms.
Direct edge from creatinine to mortality	Creatinine is a renal dysfunction proxy; renal dysfunction causes mortality risk.
Direct edge from lactate to mortality as primary cause	Lactate is mainly a hypoperfusion/metabolic stress marker.
Treating antibiotics as infection itself	Antibiotics are clinician response and suspected-infection proxy.
Treating vasopressors as shock itself	Vasopressors are treatment; shock is latent hemodynamic failure.
Using discharge location in latent tags	Post-baseline and outcome-adjacent leakage.
Using ICU length of stay in severity tags	Affected by survival, complications, and discharge practices.
Using hospital mortality to tune thresholds	Outcome leakage.
Fluid-balance burden as approved latent	Source-system differences and strong treatment dependence reduce identifiability.
Nutritional burden as approved latent	Albumin/weight/nutrition inputs are nonspecific and incompletely measured.
Separate hepatic and coagulation latents	Weak early support and overlapping SOFA-style organ dysfunction; merged.
Measurement frequency as biological severity	It is an observation process, not a physiologic state.
Conditioning on lab availability in causal mortality models	Can open collider paths through clinician suspicion and severity.
Dense edges among all organ dysfunctions	Overfits clinical correlations and obscures causal mechanisms.
Bidirectional acute-state feedback in a single static DAG	Real ICU physiology is dynamic; this static DAG must remain acyclic.

13. Final confidence and limitations

The strongest latent variables are:

Strongest variable	Why
`LAT_SHOCK`	Multiple strong indicators: BP, vasopressors, lactate, urine output, acidosis.
`LAT_RESPIRATORY_FAILURE`	Supported by SpO2, ABG, FiO2, PEEP, ventilation.
`LAT_RENAL_DYSFUNCTION`	Supported by creatinine, BUN, urine output, dialysis; KDIGO-compatible.
`LAT_GLOBAL_SEVERITY`	Multi-domain construct aligned with SOFA/OASIS-style severity.

Strongest variable	Why
`LAT_METABOLIC_DERANGEMENT`	Broad lab support and clear clinical mechanisms.

The weakest latent variables are:

Weakest variable	Why
`LAT_CARDIAC_STRAIN`	Troponin/ECG testing is selective; assay differences and renal confounding.
`LAT_HEPATIC_COAG_DYSFUNCTION`	Bilirubin/INR may be missing; anticoagulation and chronic liver disease confound.
`LAT_INFLAMMATION_SEPSIS`	Infection is partly hidden; cultures/antibiotics reflect clinician suspicion.
`LAT_NEUROLOGIC_DYSFUNCTION`	GCS/RASS are confounded by sedation and intubation.

The most reliable decision trees are `LAT_SHOCK` , `LAT_RESPIRATORY_FAILURE` , `LAT_RENAL_DYSFUNCTION` , `LAT_METABOLIC_DERANGEMENT` , and `LAT_GLOBAL_SEVERITY` . The most uncertain are `LAT_CARDIAC_STRAIN` , `LAT_HEPATIC_COAG_DYSFUNCTION` , and `LAT_INFLAMMATION_SEPSIS` .

The most uncertain DAG edges are:

Edge	Uncertainty
`LAT_CARDIAC_STRAIN → LAT_SHOCK`	True in cardiogenic shock, but cardiac biomarkers are selectively measured.
`LAT_GLOBAL_SEVERITY → LAT_RESPIRATORY_FAILURE`	Respiratory failure may also precede global severity. Static DAG simplifies dynamics.
`LAT_INFLAMMATION_SEPSIS → LAT_HEPATIC_COAG_DYSFUNCTION`	Plausible but not always identifiable without infection confirmation.
`BG_ETHNICITY_INSURANCE_LANGUAGE → OUT_MORTALITY`	May reflect social, structural, access, and measurement pathways; causal interpretation requires care.

Additional data that would improve identifiability:

- pre-ICU baseline labs,
- clinician-entered problem onset times,
- medication indication,
- reliable sedation depth and timing,
- radiology-derived respiratory findings,
- validated infection adjudication,
- echocardiography/ECG structured findings,
- baseline functional status and frailty,
- external validation dataset.

This graph is clinically plausible because it respects the ICU data-generating process: background risk and acute insults generate latent physiologic states; latent states generate measurements, treatments, missingness patterns, and mortality. It is not a statistically proven ground-truth DAG. It should be treated as a transparent, clinically motivated structural hypothesis for latent-tag generation, causal adjustment planning, sensitivity analysis, and mortality modeling.

Sources

PhysioNet 2012 ChatGPT Elicitation Output

Archived output corresponding to the general
latent-causal-model elicitation protocol

For DATASET-NAME, the clinically appropriate causal level is **latent acute physiology and organ dysfunction**, not individual charted values. The 48-hour measurements are noisy, irregularly sampled observations of ICU physiology: blood pressure, gases, labs, urine output, neurologic scores, and ventilation status are better treated as **manifestations of hidden clinical states** than as the primary causal variables.

The strongest high-level graph is: **background/case-mix → chronic vulnerability and acute latent physiologic stress → organ dysfunction and global severity → observed measurements, care intensity, measurement intensity, and mortality.** This matches the dataset design: adult ICU stays, first 48 hours of irregular clinical observations, and in-hospital mortality as the prediction outcome. The official description states that DATASET-NAME contains 12,000 ICU stays, adult patients from cardiac, medical, surgical, and trauma ICUs, first-48-hour observations, six admission descriptors, and up to 37 time-varying variables.

physionet.org

The main identifiability limitation is that DATASET-NAME lacks diagnoses, medications, vasopressors, antibiotics, procedures, microbiology, comorbidity codes, admission diagnosis, and clinician intent. Therefore, latent states such as sepsis, shock, and cardiac injury can be approximated only as **proxy labels**, not confirmed clinical diagnoses. Missingness is also informative: EHR measurements are generated by care, values are measured at irregular intervals, and missingness may depend on expected abnormality or clinician suspicion rather than being random. JMIR Medical Inf...

1. Dataset audit

1.1 Collection design and global rules

Item	Dataset fact	Causal relevance
Population	Adult ICU stays from cardiac, medical, surgical, and trauma ICUs	Strong case-mix confounding; ICU type is not just a label but a proxy for disease spectrum and care pathway
Observation window	First 48 hours after ICU admission	Latent labels should represent first-48h physiology unless explicitly restricted to early windows
Records	12,000 ICU stays; set A has 4,000 training records, other sets were test/validation	Outcome labels are not universally available
Inclusion rule	ICU stays under 48 hours excluded	Selection/collider risk: early death or rapid discharge cases are structurally absent
Missing coding	Valid values are non-negative; `-1` means missing or unknown	Must convert `-1` to missing before feature extraction
Sampling	Time-stamped measurements, regular or irregular; variables may be observed once, repeatedly, or not at all	Measurement frequency is a separate observation process
Outliers	Official notes mention occasional outliers, especially with multiple sensors/methods	Robust summaries needed: first, worst, median, count, and physiologic filters

The official description specifies the first-48-hour window, six admission descriptors, 37 time-series variables, non-negative valid values, `-1` missingness, irregular measurement, and occasional outliers. physionet.org

1.2 Background/static variables

Variable	Type	Clinical meaning	Unit/coding	Notes
`RecordID`	Irrelevant-noisy / identifier	Unique ICU stay ID	Integer	Do not use as predictor or causal variable
`Age`	Background	Age-related frailty and mortality vulnerability	Years	Strong confounder for mortality and severity
`Gender`	Background	Sex/gender marker	0 female, 1 male	Possible demographic confounder; weak physiologic specificity
`Height`	Background / measurement proxy	Body size	cm	Used with weight for BMI; missing often
`Weight` admission	Background / baseline body size	Admission weight	kg	Also appears as time-varying variable; baseline weight differs from fluid-balance weight
`ICUType`	Background / care-unit case mix	ICU service type	1 CCU, 2 CSRU, 3 MICU, 4 SICU	Strong proxy for admission reason, care pathway, and monitoring

1.3 Time-varying vitals, respiratory variables, and charted care variables

Variable	Type	Clinical meaning	Unit/coding	Notes
`HR`	Observed time-varying	Heart rate	bpm	Proxy for shock, inflammation, cardiac strain, global severity
`RespRate`	Observed time-varying	Respiratory rate	breaths/min	Proxy for respiratory failure, acidosis, inflammatory stress
`Temp`	Observed time-varying	Temperature	°C	Proxy for inflammation, infection, thermoregulatory failure
`SysABP`	Observed time-varying	Invasive systolic arterial pressure	mmHg	Proxy for circulatory state; invasive line implies monitoring intensity
`DiasABP`	Observed time-varying	Invasive diastolic arterial pressure	mmHg	Proxy for circulatory state
`MAP`	Observed time-varying	Invasive mean arterial pressure	mmHg	Key shock/hemodynamic proxy
`NISysABP`	Observed time-varying	Non-invasive systolic pressure	mmHg	Same physiologic construct as systolic BP, different measurement process
`NIDiasABP`	Observed time-varying	Non-invasive diastolic pressure	mmHg	Same physiologic construct as diastolic BP, different measurement process
`NIMAP`	Observed time-varying	Non-invasive mean arterial pressure	mmHg	Hemodynamic proxy; less invasive measurement
`SaO2`	Observed time-varying	Oxygen saturation	%	Proxy for hypoxemia; affected by oxygen therapy
`PaO2`	Observed time-varying	Arterial oxygen partial pressure	mmHg	ABG-derived proxy for respiratory failure; ordering is informative
`PaCO2`	Observed time-varying	Arterial carbon dioxide partial pressure	mmHg	Proxy for ventilation failure and acid-base disturbance
`FiO2`	Care-process / observed time-varying	Fraction of inspired oxygen	0–1	Treatment setting and respiratory severity proxy
`MechVent`	Treatment / care-process / proxy	Mechanical ventilation status	0 false, 1 true	True intervention/care process and proxy for respiratory/global severity
`GCS`	Observed time-varying	Glasgow Coma Scale	3–15	Neurologic dysfunction proxy; sedation and ventilation confounding possible
`Urine`	Observed time-varying / charted process	Urine output	mL	Renal perfusion/function proxy; charting intensity matters
`Weight` time series	Observed time-varying / care-process proxy	Serial weight	kg	Often used for fluid balance; noisy and measurement-process dependent

1.4 Laboratory variables

Variable	Type	Clinical meaning	Unit/coding	Notes
`Albumin`	Observed lab proxy	Nutrition, inflammation, hepatic synthetic reserve, chronic illness	g/dL	Non-specific; useful for chronic vulnerability and hepatic/inflammatory burden

Variable	Type	Clinical meaning	Unit/coding	Notes
`ALP`	Observed lab proxy	Cholestasis/hepatobiliary injury	IU/L	Hepatic dysfunction proxy
`ALT`	Observed lab proxy	Hepatocellular injury	IU/L	Hepatic injury proxy
`AST`	Observed lab proxy	Hepatocellular or muscle injury	IU/L	Hepatic/cardiac/muscle injury proxy
`Bilirubin`	Observed lab proxy	Cholestasis/liver dysfunction	mg/dL	SOFA hepatic component uses bilirubin
`BUN`	Observed lab proxy	Renal dysfunction, catabolism, hypovolemia	mg/dL	Renal/perfusion proxy
`Creatinine`	Observed lab proxy	Renal filtration	mg/dL	Renal dysfunction proxy; baseline unavailable
`Glucose`	Observed lab proxy	Metabolic stress, diabetes, nutrition/insulin effects	mg/dL	Metabolic derangement proxy
`HCO3`	Observed lab proxy	Metabolic acid-base status	mmol/L	Metabolic/renal/respiratory compensation proxy
`HCT`	Observed lab proxy	Hematocrit, anemia/hemoconcentration	%	Hematologic status and fluid balance proxy
`K`	Observed lab proxy	Potassium	mEq/L	Renal/metabolic derangement proxy
`Lactate`	Observed lab proxy	Tissue hypoperfusion, mitochondrial stress, sepsis/shock severity	mmol/L	Strong shock/metabolic/global severity proxy; ordering is informative
`Mg`	Observed lab proxy	Magnesium	mmol/L	Metabolic/electrolyte proxy
`Na`	Observed lab proxy	Sodium	mEq/L	Fluid/electrolyte/neurologic risk proxy
`pH`	Observed lab proxy	Arterial acid-base status	0–14	Metabolic/respiratory derangement proxy
`Platelets`	Observed lab proxy	Coagulation/hematologic dysfunction	cells/nL	SOFA coagulation component uses platelets
`WBC`	Observed lab proxy	Leukocyte response	cells/nL	Inflammation/infection/stress proxy
`Cholesterol`	Observed lab proxy / weak relevance	Lipid/nutritional/chronic marker	mg/dL	Sparse/low acute causal specificity
`TropI`	Observed lab proxy	Troponin-I myocardial injury marker	µg/L	Cardiac injury proxy; test ordering is highly informative
`TropT`	Observed lab proxy	Troponin-T myocardial injury marker	µg/L	Cardiac injury proxy; assay-specific threshold uncertainty

1.5 Outcome-related descriptors

Variable	Type	Clinical meaning	Unit/coding	Notes
`SAPS-I score`	Outcome-related severity descriptor	ICU severity score	Score	Training outcome file; avoid for latent-label construction to prevent dataset split dependence

Variable	Type	Clinical meaning	Unit/coding	Notes
`SOFA score`	Outcome-related severity descriptor	Organ failure score	Score	Training outcome file; useful for validation only, not for proxy-label generation
`Length of stay`	Outcome/post-outcome	Hospital length of stay	Days	Do not use for latent labels; post-baseline outcome process
`Survival`	Outcome/post-outcome	Days from ICU admission to death, or `-1` if survivor	Days	Do not use; direct outcome leakage
`In-hospital death`	Outcome	Mortality label	0 survivor, 1 died	Primary outcome node `OUT_InHospitalMortality`

SAPS-I, SOFA, length of stay, survival, and in-hospital death are stored separately as outcome-related descriptors, and only set A outcomes were available to challenge participants. physionet.org

2. Candidate latent variables

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Ra
Chronic baseline vulnerability	Frailty/comorbidity/body reserve/case-mix risk	Age, ICUType, BMI, Albumin, admission Weight	Medium-low	Keep	Im no co
Acute global severity	Multi-domain acute physiologic instability	BP, HR, GCS, lactate, pH, HCO3, PaO2/FiO2, creatinine, urine, bilirubin, platelets	High	Keep	Cl ge
Circulatory shock / hemodynamic instability	Hypoperfusion and inadequate circulatory support	MAP, SysABP/NISysABP, lactate, HR, urine, pH/HCO3	High	Keep	Str he pr
Respiratory failure / pulmonary dysfunction	Hypoxemia, ventilatory failure, oxygen/ventilator dependence	PaO2, FiO2, SaO2, RespRate, PaCO2, pH, MechVent	High	Keep	Str do
Renal dysfunction / kidney injury	Impaired filtration or urine output	Creatinine, BUN, Urine, K, HCO3	High	Keep	Str do
Hepatic dysfunction / liver injury	Cholestasis/hepatocellular injury/synthetic reserve	Bilirubin, AST, ALT, ALP, Albumin	Medium	Keep	Bili fur IN
Coagulation/hematologic dysfunction	Thrombocytopenia, anemia, marrow/stress response	Platelets, HCT, WBC	Medium	Keep	Pla co
Inflammation / sepsis-like burden	Systemic host response, possible infection	Temp, WBC, HR, RespRate, PaCO2, Lactate, platelets	Medium-low	Keep, rename as inflammatory/sepsis-like burden	No cul so
Neurologic dysfunction	Depressed consciousness or brain dysfunction	GCS, metabolic derangements, hypoxemia/hypercapnia	Medium	Keep	GC an like
Cardiac injury/strain	Myocardial injury or ischemic/cardiac strain	TropI, TropT, HR, BP, lactate, ICUType	Medium-low	Keep as weakly measured	Tro me ass
Metabolic/acid-base/electrolyte derangement	Acidosis/alkalosis, lactate, electrolytes, glucose	pH, HCO3, Lactate, Na, K, Mg, Glucose, PaCO2	High	Keep	Bro oft fail

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Ra
Fluid balance / volume-status burden	Fluid overload or depletion	Weight time series, urine, Na, BP	Low	Reject / merge	We flui
Treatment/care intensity	Intensity of clinical support	MechVent, FiO2, measurement counts, ABG/lactate/troponin ordering	High as process, not physiology	Model separately as care/measurement process	No des clir
Confirmed sepsis	Infection plus dysregulated host response	Temp, WBC, lactate, organ dysfunction	Absent for confirmation	Reject as confirmed diagnosis	No sus tim
Vasopressor-dependent shock	Shock requiring vasopressors	MAP, lactate only	Absent	Reject / approximate as shock	Vas un
ARDS	Acute respiratory distress syndrome	PaO2/FiO2, MechVent	Low-medium	Reject as formal ARDS	No tim
Chronic kidney disease	Baseline renal disease	Creatinine, BUN	Low	Reject as separate variable	No cre

SOFA motivates respiratory, coagulation, hepatic, cardiovascular, neurologic, and renal organ dysfunction domains; the table version uses PaO2/FiO2, platelets, bilirubin, MAP/vasopressors, GCS, creatinine, and urine output. MSD Manuals Sepsis-3 motivates separating infection/inflammation from organ dysfunction because sepsis is defined as life-threatening organ dysfunction from a dysregulated host response, but DATASET-NAME lacks infection-confirming variables. jamanetwork.com

3. Final approved latent variables

Latent variable	Clinical meaning	Why latent	Observed indicators	Suggested bi definition
`LAT_CHRONIC_BASELINE_RISK`	Baseline vulnerability before ICU physiology	True comorbid burden/frailty is not observed	Age, ICUType, BMI, Albumin	Score ≥2 fron habitus, low a high-risk ICU
`LAT_GLOBAL_SEVERITY`	Multi-system acute illness severity	Severity is a latent synthesis of multiple organ states	Hemodynamics, respiratory, neuro, renal, hepatic/coag, metabolic, cardiac domains	Score ≥3 dom abnormal, or critical abnorı
`LAT_SHOCK`	Circulatory failure/hypoperfusion	Shock is not equal to any single BP or lactate value	MAP/SBP, lactate, HR, urine, pH/HCO3	Score ≥2 sho
`LAT_RESPIRATORY_FAILURE`	Hypoxemia/ventilatory failure or ventilator dependence	Oxygenation state depends on disease and treatment settings	PaO2/FiO2, SaO2, RespRate, PaCO2, pH, MechVent	Score ≥2 resp indicators
`LAT_RENAL_DYSFUNCTION`	Reduced filtration/oliguria or renal-metabolic consequences	Renal function state is not fully observed without baseline	Creatinine, BUN, Urine, K, HCO3	Score ≥2 rena
`LAT_HEPATIC_DYSFUNCTION`	Liver injury/cholestasis/synthetic reserve impairment	No direct liver diagnosis or INR	Bilirubin, AST, ALT, ALP, Albumin	Score ≥2 hep indicators, wit weighted stro
`LAT_COAG_HEME_DYSFUNCTION`	Thrombocytopenia/anemia/hematologic stress	Coagulation system not directly measured	Platelets, HCT, WBC	Weighted sco

Latent variable	Clinical meaning	Why latent	Observed indicators	Suggested bi definition
`LAT_INFLAMMATION_SEPSIS_BURDEN`	Systemic inflammatory or sepsis-like host response	Infection source and antibiotics unavailable	Temp, WBC, HR, RR/PaCO2, lactate, organ dysfunction proxies	Score ≥3, or S score ≥2 plus lactate/organ
`LAT_NEUROLOGIC_DYSFUNCTION`	Depressed consciousness/brain dysfunction	GCS is observed but confounded by sedation and ventilation	GCS, hypoxemia/hypercapnia, Na/glucose extremes, pH	Weighted sco severe GCS w
`LAT_CARDIAC_INJURY_STRAIN`	Myocardial injury/ischemia/strain	True cardiac injury requires clinical context	TropI, TropT, HR, hypotension, lactate, ICUType	Weighted sco troponin abn weighted
`LAT_METABOLIC_DERANGEMENT`	Acid-base, lactate, electrolyte, glucose instability	Metabolic state is multidimensional	pH, HCO3, lactate, Na, K, Mg, glucose, PaCO2	Score ≥2 met indicators

These latent variables are **proxy-label targets**, not ground-truth diagnoses. Causal representation learning explicitly motivates discovering high-level causal variables from low-level observations, rather than assuming raw observations are already the causal variables. research-explor...

4. Rule-based binary decision trees

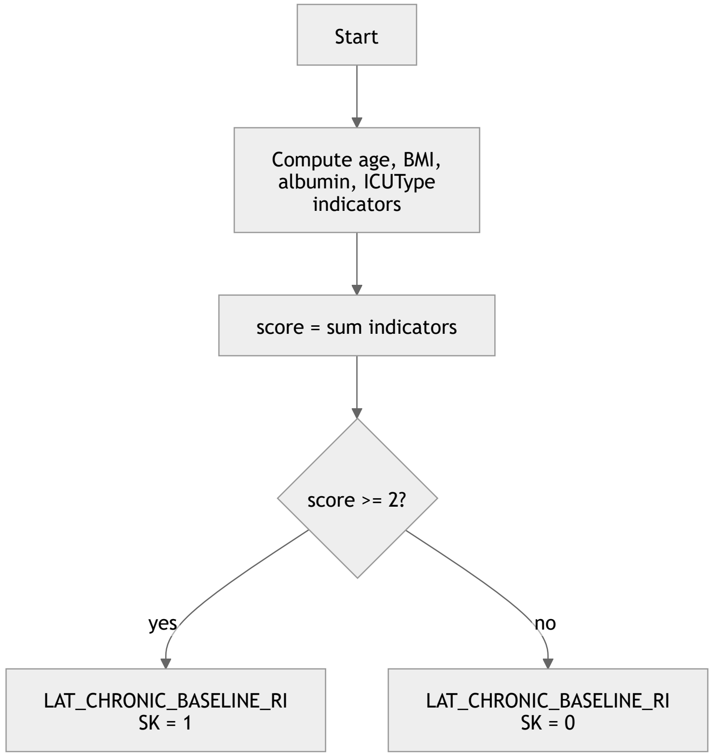
All trees use first-48h summaries unless an implementation chooses an earlier landmark. `min` , `max` , `first` , `last` , `delta` , `count` , and `missing` refer to values after converting `-1` to missing and applying physiologic sanity filters.

4.1 Chronic baseline vulnerability

Latent variable: `LAT_CHRONIC_BASELINE_RISK`
Clinical interpretation of label 1: evidence of high baseline vulnerability or limited physiologic reserve before or at ICU admission.

Component	Rule
Input variables	`Age` , `Gender` , `Height` , admission `Weight` , `ICUType` , `Albumin`
Domain indicators	`AGE_RISK = Age >= 75` ; `BODY_RISK = BMI < 18.5 OR BMI >= 40` ; `LOW_ALBUMIN = Albumin_min < 3.0` ; `CASEMIX_RISK = ICUType in {1,3}`
Score	`score = AGE_RISK + BODY_RISK + LOW_ALBUMIN + CASEMIX_RISK`
Threshold	`LAT_CHRONIC_BASELINE_RISK = 1 if score >= 2 else 0`
Missingness	Missing height/weight means `BODY_RISK = 0` ; missing albumin means `LOW_ALBUMIN = 0` ; store `available_components`
Confidence	Medium-low

```
def tag_chronic_baseline_risk(x):
    age_risk = x.Age >= 75
    bmi = x.Weight_first / (x.Height / 100)**2 if x.Height and x.Weight_first else None
    body_risk = bmi is not None and (bmi < 18.5 or bmi >= 40)
    low_albumin = present(x.Albumin_min) and x.Albumin_min < 3.0
    casemix_risk = x.ICUType in [1, 3]
    score = sum([age_risk, body_risk, low_albumin, casemix_risk])
    return int(score >= 2)
```



Clinical rationale: age, poor nutritional reserve, extreme body habitus, and medical/coronary ICU case-mix can proxy chronic vulnerability.
False positives: hypoalbuminemia from acute inflammation; ICUType reflects care setting, not disease.
False negatives: younger patients with severe comorbidity not captured by the dataset.

4.2 Global severity of illness

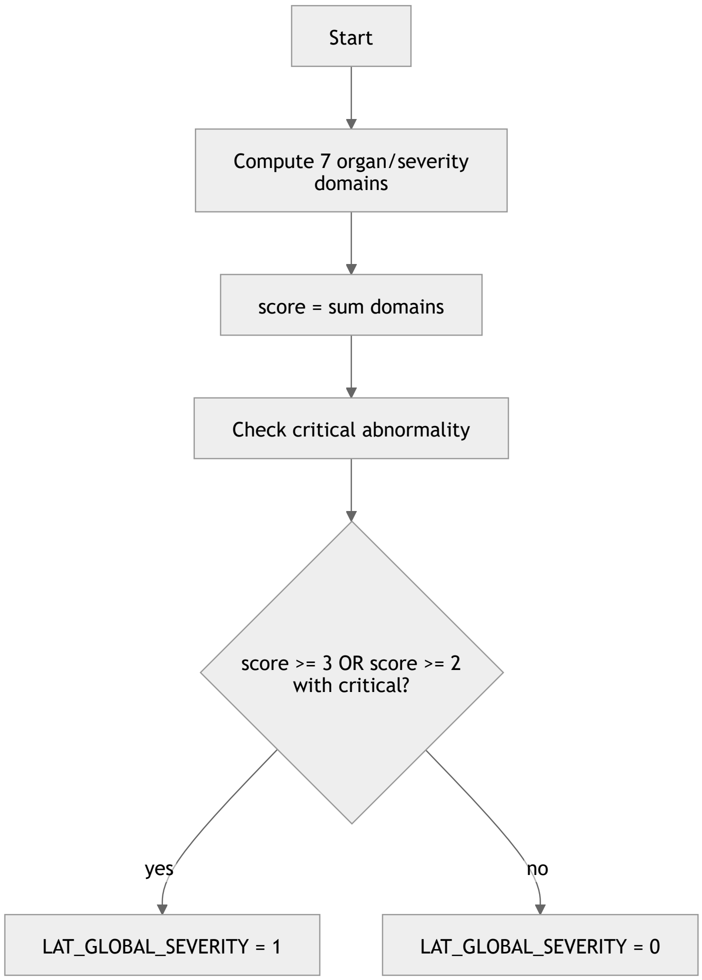
Latent variable: `LAT_GLOBAL_SEVERITY`

Clinical interpretation of label 1: evidence of multi-domain acute physiologic severity during the first 48 hours.

Component	Rule
Input variables	BP, HR, GCS, respiratory variables, renal variables, liver/coag labs, lactate, acid-base, troponin
Domain indicators	`HEMO = MAP_min < 70 OR SysABP_min <= 100 OR NIMAP_min < 70` ; `RESP = MechVent_max == 1 OR PF_min < 300 OR SaO2_min < 92 OR RespRate_max >= 22` ; `NEURO = GCS_min < 15` ; `RENAL = Creatinine_max >= 2.0 OR BUN_max >= 40 OR Urine_24h_min < 500` ; `HEP_COAG = Bilirubin_max >= 2.0 OR Platelets_min < 100` ; `METAB = Lactate_max > 2 OR pH_min < 7.30 OR HCO3_min < 18` ; `CARDIAC = TropI_abn OR TropT_abn OR HR_max >= 130`
Score	`score = sum(domain indicators)`
Threshold	`1 if score >= 3 OR (score >= 2 AND CRITICAL == 1)`
Critical	`Lactate_max >= 4 OR pH_min < 7.20 OR MechVent_max == 1 OR GCS_min <= 8 OR MAP_min < 60`
Missingness	Missing domains score 0; require at least two available domains for positive label unless a critical rule is present
Confidence	High

```
def tag_global_severity(x):
    hemo = any_present_true([
        lt(x.MAP_min, 70), lt(x.NIMAP_min, 70),
        le(x.SysABP_min, 100), le(x.NISysABP_min, 100)
    ])
```

```
    resp = any_present_true([
        eq(x.MechVent_max, 1), lt(x.PF_min, 300),
        lt(x.SaO2_min, 92), ge(x.RespRate_max, 22)
    ])
    neuro = present(x.GCS_min) and x.GCS_min < 15
    renal = any_present_true([ge(x.Creatinine_max, 2.0), ge(x.BUN_max, 40), lt(x.Urine_24h_min, 500)])
    hep_coag = any_present_true([ge(x.Bilirubin_max, 2.0), lt(x.Platelets_min, 100)])
    metab = any_present_true([gt(x.Lactate_max, 2), lt(x.pH_min, 7.30), lt(x.HCO3_min, 18)])
    cardiac = any_present_true([x.TropI_abn, x.TropT_abn, ge(x.HR_max, 130)])
    score = sum([hemo, resp, neuro, renal, hep_coag, metab, cardiac])
    critical = any_present_true([ge(x.Lactate_max, 4), lt(x.pH_min, 7.20), eq(x.MechVent_max, 1), le(x.GCS_min,
    return int(score >= 3 or (score >= 2 and critical))
```



Clinical rationale: SOFA-like organ dysfunction is a clinically accepted severity construct using respiratory, coagulation, hepatic, cardiovascular, neurologic, and renal domains. MSD Manuals
False positives: post-operative ventilation, transient lab artifacts.
False negatives: severe disease in unmeasured domains or under-sampled patients.

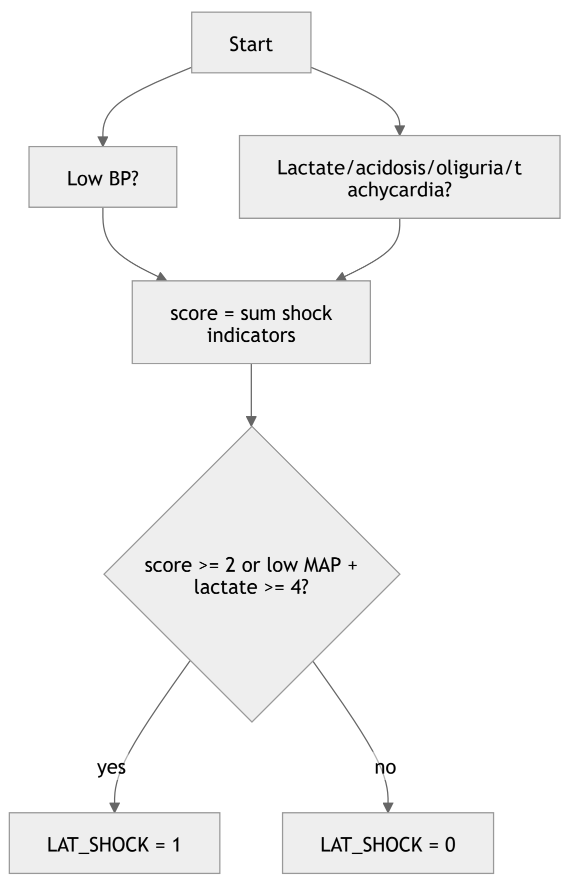
4.3 Circulatory shock / hemodynamic instability

Latent variable: `LAT_SHOCK`
Clinical interpretation of label 1: evidence of clinically meaningful hypotension or tissue hypoperfusion.

Component	Rule
Input variables	`MAP`, `NIMAP`, `SysABP`, `NISysABP`, `HR`, `Lactate`, `Urine`, `pH`, `HCO3`
Domain indicators	`LOW_MAP = MAP_min < 70 OR NIMAP_min < 70`; `LOW_SBP = SysABP_min <= 90 OR NISysABP_min <= 90`; `HYPOPERFUSION = Lactate_max > 2`; `TACHY = HR_max >= 110`; `OLIGURIA = Urine_24h_min < 500`; `ACIDOSIS = pH_min < 7.30 OR HCO3_min < 18`

Component	Rule
Score	`score = sum(indicators)`
Threshold	`1 if score >= 2`, or `1 if LOW_MAP and Lactate_max >= 4`
Missingness	Lactate missing does not rule out shock; urine missing does not imply oliguria
Confidence	High

```
def tag_shock(x):
    low_map = any_present_true([lt(x.MAP_min, 70), lt(x.NIMAP_min, 70)])
    low_sbp = any_present_true([le(x.SysABP_min, 90), le(x.NISysABP_min, 90)])
    hypoperfusion = present(x.Lactate_max) and x.Lactate_max > 2
    tachy = present(x.HR_max) and x.HR_max >= 110
    oliguria = present(x.Urine_24h_min) and x.Urine_24h_min < 500
    acidosis = any_present_true([lt(x.pH_min, 7.30), lt(x.HCO3_min, 18)])
    score = sum([low_map, low_sbp, hypoperfusion, tachy, oliguria, acidosis])
    severe_pair = low_map and present(x.Lactate_max) and x.Lactate_max >= 4
    return int(score >= 2 or severe_pair)
```



Clinical rationale: cardiovascular SOFA includes MAP <70, and lactate/acidosis/oliguria add hypoperfusion evidence. MSD Manuals

False positives: isolated low BP during sedation, measurement artifact.

False negatives: vasopressor-supported shock with normalized BP because vasopressors are unavailable.

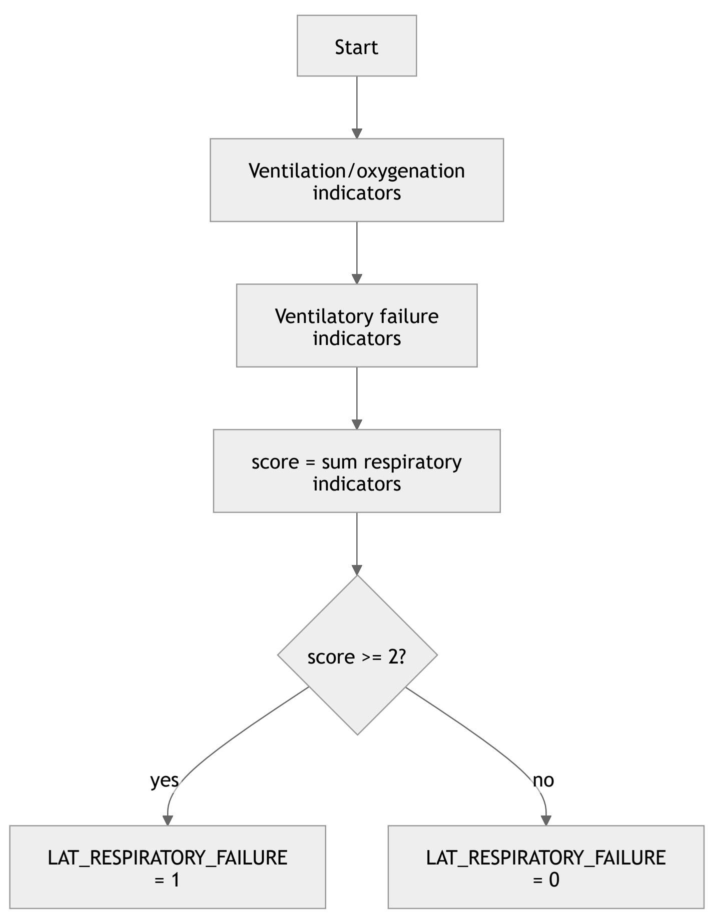
4.4 Respiratory failure / pulmonary dysfunction

Latent variable: `LAT_RESPIRATORY_FAILURE`

Clinical interpretation of label 1: evidence of hypoxemia, ventilatory failure, or mechanical ventilatory support.

Component	Rule
Input variables	`PaO2` , `FiO2` , `SaO2` , `RespRate` , `PaCO2` , `pH` , `MechVent`
Domain indicators	`VENT = MechVent_max == 1` ; `LOW_PF = PF_min < 300` ; `HYPOXEMIA = SaO2_min < 92 OR PaO2_min < 60` ; `TACHYPNEA = RespRate_max >= 22 OR RespRate_min < 8` ; `VENTILATORY_FAILURE = PaCO2_max >= 50 OR (PaCO2_max >= 45 AND pH_min < 7.30)`
Score	`score = sum(indicators)`
Threshold	`1 if score >= 2` , or `1 if VENT and LOW_PF`
Missingness	Compute PaO2/FiO2 only when FiO2 and PaO2 are near-paired; missing ABG not abnormal
Confidence	High

```
def tag_respiratory_failure(x):
    vent = x.MechVent_max == 1
    low_pf = present(x.PF_min) and x.PF_min < 300
    hypoxemia = any_present_true([lt(x.SaO2_min, 92), lt(x.PaO2_min, 60)])
    tachypnea = any_present_true([ge(x.RespRate_max, 22), lt(x.RespRate_min, 8)])
    ventilatory_failure = any_present_true([
        ge(x.PaCO2_max, 50),
        (present(x.PaCO2_max) and x.PaCO2_max >= 45 and present(x.pH_min) and x.pH_min < 7.30)
    ])
    score = sum([vent, low_pf, hypoxemia, tachypnea, ventilatory_failure])
    return int(score >= 2 or (vent and low_pf))
```



Clinical rationale: PaO2/FiO2 thresholds are core SOFA respiratory markers, and mechanical ventilation is both treatment and severity proxy. MSD Manuals

False positives: elective post-operative ventilation.
False negatives: oxygen therapy without recorded FiO2 or ABG.

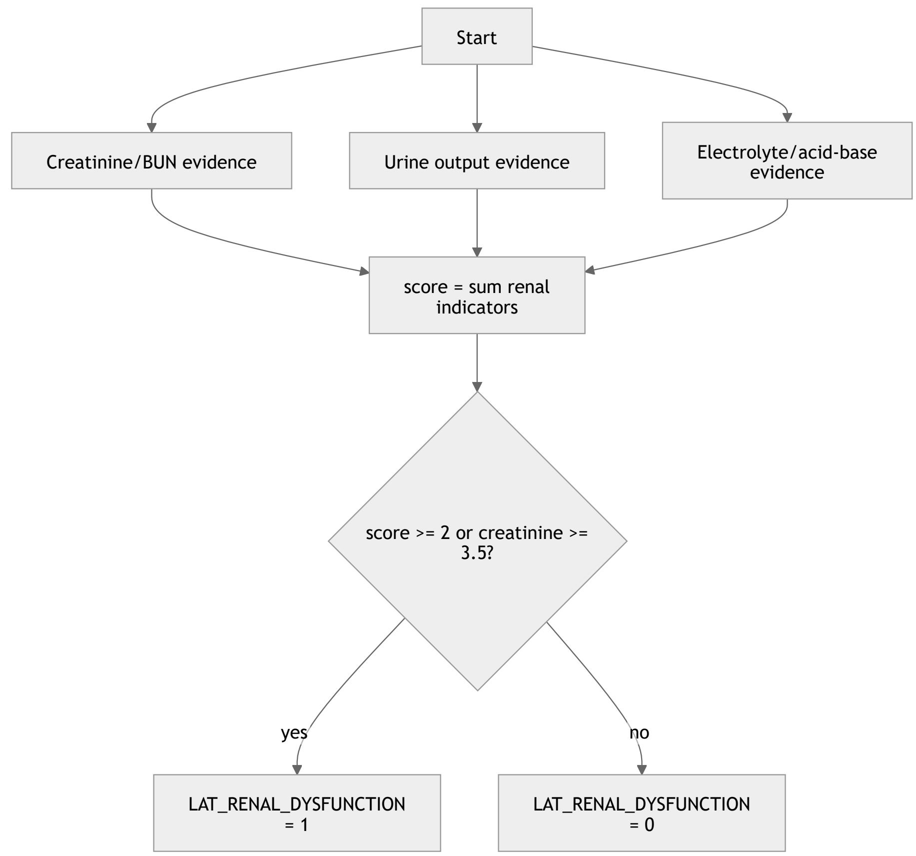
4.5 Renal dysfunction

Latent variable: `LAT_RENAL_DYSFUNCTION`

Clinical interpretation of label 1: evidence of renal filtration impairment, oliguria, or renal-metabolic consequences.

Component	Rule
Input variables	`Creatinine`, `BUN`, `Urine`, `K`, `HCO3`
Domain indicators	`CREAT_HIGH = Creatinine_max >= 2.0`; `CREAT_RISE = Creatinine_max - Creatinine_first >= 0.3`; `BUN_HIGH = BUN_max >= 40`; `OLIGURIA = Urine_24h_min < 500`; `RENAL_METAB = K_max >= 5.5 OR HCO3_min < 18`
Score	`score = sum(indicators)`
Threshold	`1 if score >= 2`, or `1 if Creatinine_max >= 3.5`
Missingness	Baseline creatinine unavailable; use within-window rise only if two creatinine values exist
Confidence	High

```
def tag_renal_dysfunction(x):
    creat_high = present(x.Creatinine_max) and x.Creatinine_max >= 2.0
    creat_rise = present(x.Creatinine_first) and present(x.Creatinine_max) and (x.Creatinine_max - x.Creatinine_
    bun_high = present(x.BUN_max) and x.BUN_max >= 40
    oliguria = present(x.Urine_24h_min) and x.Urine_24h_min < 500
    renal_metab = any_present_true([ge(x.K_max, 5.5), lt(x.HCO3_min, 18)])
    score = sum([creat_high, creat_rise, bun_high, oliguria, renal_metab])
    return int(score >= 2 or ge(x.Creatinine_max, 3.5))
```



Clinical rationale: SOFA renal scoring uses creatinine and urine output, both available here. MSD Manuals
False positives: chronic kidney disease misclassified as acute dysfunction.
False negatives: early AKI without creatinine rise or uncharted urine.

4.6 Hepatic dysfunction

Latent variable: `LAT_HEPATIC_DYSFUNCTION`

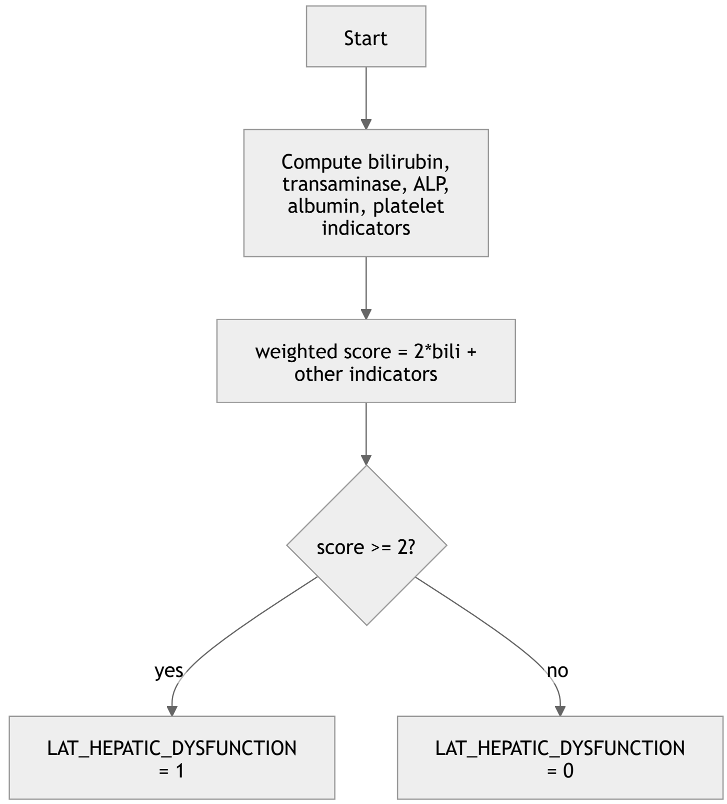
Clinical interpretation of label 1: evidence of cholestatic or hepatocellular injury or poor synthetic/nutritional hepatic reserve.

Component	Rule
Input variables	`Bilirubin` , `AST` , `ALT` , `ALP` , `Albumin` , `Platelets`
Domain indicators	`BILI = Bilirubin_max >= 2.0` ; `TRANSAM = AST_max >= 200 OR ALT_max >= 200` ; `CHOL = ALP_max >= 250` ; `LOW_ALB = Albumin_min < 2.5` ; `PORTAL_OR_SEVERE = Platelets_min < 100`
Score	`score = 2*BILI + TRANSAM + CHOL + LOW_ALB + PORTAL_OR_SEVERE`
Threshold	`1 if score >= 2`
Missingness	Missing liver panel elements score 0 and should produce a low-evidence flag
Confidence	Medium

```
def tag_hepatic_dysfunction(x):
    bili = present(x.Bilirubin_max) and x.Bilirubin_max >= 2.0
    transam = any_present_true([ge(x.AST_max, 200), ge(x.ALT_max, 200)])
    chol = present(x.ALP_max) and x.ALP_max >= 250
    low_alb = present(x.Albumin_min) and x.Albumin_min < 2.5
    portal_or_severe = present(x.Platelets_min) and x.Platelets_min < 100
```

Python

Run



Clinical rationale: bilirubin is the SOFA hepatic marker; AST/ALT/ALP/albumin add injury and reserve context.

MSD Manuals

False positives: chronic liver disease, hemolysis, nutrition-related hypoalbuminemia.

False negatives: coagulopathy without bilirubin elevation because INR/PT are absent.

4.7 Coagulation or hematologic dysfunction

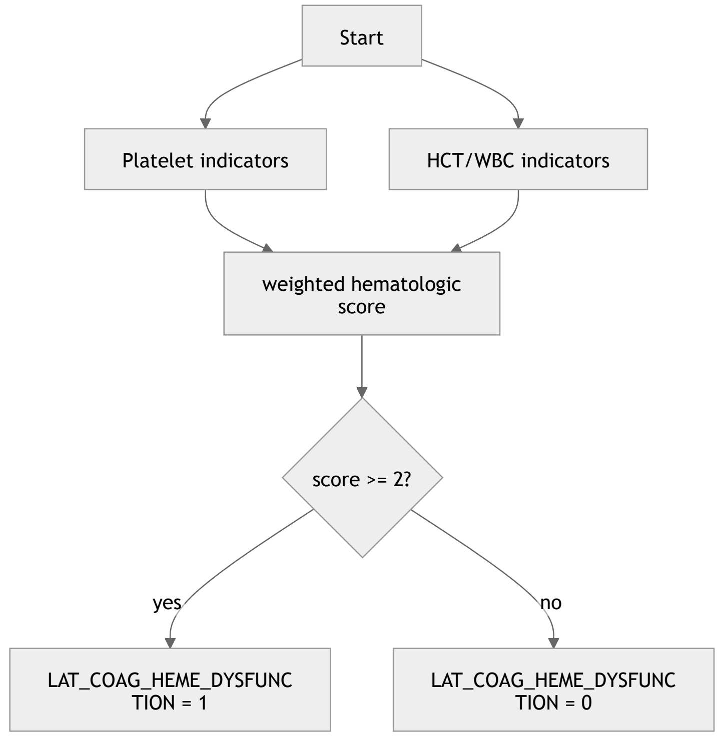
Latent variable: `LAT_COAG_HEME_DYSFUNCTION`

Clinical interpretation of label 1: evidence of thrombocytopenia, anemia, or hematologic stress.

Component	Rule
Input variables	`Platelets` , `HCT` , `WBC`
Domain indicators	`SEVERE_PLT = Platelets_min < 100` ; `MILD_PLT = Platelets_min < 150` ; `HCT_ABN = HCT_min < 25 OR HCT_max > 55` ; `WBC_EXTREME = WBC_min < 4 OR WBC_max > 20` ; `PLT_DROP = Platelets_first - Platelets_min >= 50`
Score	`score = 2*SEVERE_PLT + MILD_PLT + HCT_ABN + WBC_EXTREME + PLT_DROP`
Threshold	`1 if score >= 2`
Missingness	Missing CBC components score 0; do not infer normality
Confidence	Medium

```
def tag_coag_heme_dysfunction(x):
    severe_plt = present(x.Platelets_min) and x.Platelets_min < 100
    mild_plt = present(x.Platelets_min) and x.Platelets_min < 150
    hct_abn = any_present_true([lt(x.HCT_min, 25), gt(x.HCT_max, 55)])
    wbc_extreme = any_present_true([lt(x.WBC_min, 4), gt(x.WBC_max, 20)])
```

```
    plt_drop = present(x.Platelets_first) and present(x.Platelets_min) and (x.Platelets_first - x.Platelets_min
    score = 2*int(severe_plt) + sum([mild_plt, hct_abn, wbc_extreme, plt_drop])
    return int(score >= 2)
```



Clinical rationale: SOFA coagulation uses platelet count, while HCT and WBC add hematologic context.

MSD Manuals

False positives: chronic thrombocytopenia, dilutional anemia.

False negatives: coagulopathy without thrombocytopenia because INR/PT/PTT unavailable.

4.8 Inflammatory / sepsis-like burden

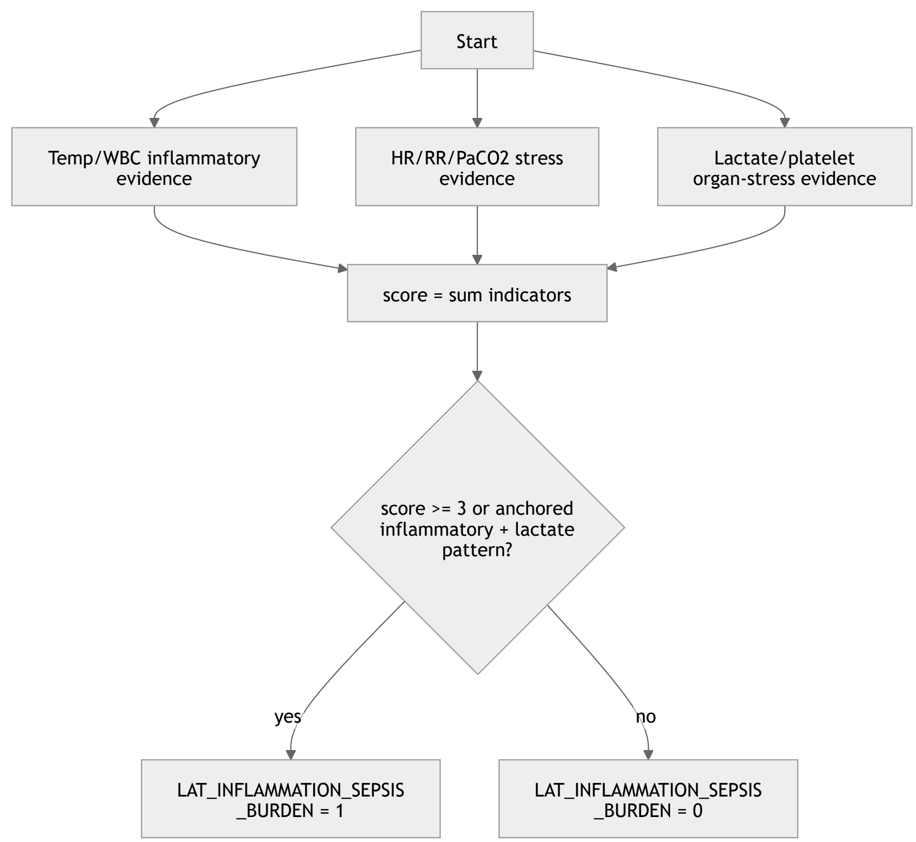
Latent variable: `LAT_INFLAMMATION_SEPSIS_BURDEN`

Clinical interpretation of label 1: systemic inflammatory response with possible infection-related organ stress, not confirmed sepsis.

Component	Rule
Input variables	`Temp`, `WBC`, `HR`, `RespRate`, `PaCO2`, `Lactate`, `Platelets`
Domain indicators	`TEMP_ABN = Temp_max > 38.3 OR Temp_min < 36`; `WBC_ABN = WBC_max > 12 OR WBC_min < 4`; `TACHY = HR_max > 90`; `RESP_STRESS = RespRate_max > 20 OR PaCO2_min < 32`; `LACTATE_STRESS = Lactate_max > 2`; `PLT_LOW = Platelets_min < 150`
Score	`score = sum(indicators)`
Threshold	`1 if score >= 3`, or `1 if TEMP_ABN + WBC_ABN >= 1 and LACTATE_STRESS and score >= 2`
Missingness	Missing WBC/temp should not be treated as absence of inflammation
Confidence	Medium-low

```
def tag_inflammation_sepsis_burden(x):
    temp_abn = any_present_true([gt(x.Temp_max, 38.3), lt(x.Temp_min, 36)])
    wbc_abn = any_present_true([gt(x.WBC_max, 12), lt(x.WBC_min, 4)])
    tachy = present(x.HR_max) and x.HR_max > 90
    resp_stress = any_present_true([gt(x.RespRate_max, 20), lt(x.PaCO2_min, 32)])
```

```
    lactate_stress = present(x.Lactate_max) and x.Lactate_max > 2
    plt_low = present(x.Platelets_min) and x.Platelets_min < 150
    score = sum([temp_abn, wbc_abn, tachy, resp_stress, lactate_stress, plt_low])
    anchored = (temp_abn or wbc_abn) and lactate_stress and score >= 2
    return int(score >= 3 or anchored)
```



Clinical rationale: Sepsis-3 links dysregulated host response to organ dysfunction, but this dataset lacks infection confirmation, so the label is intentionally “sepsis-like burden.” jamanetwork.com

False positives: trauma, surgery, pancreatitis, cardiogenic shock.

False negatives: immunosuppressed or elderly patients without fever or leukocytosis.

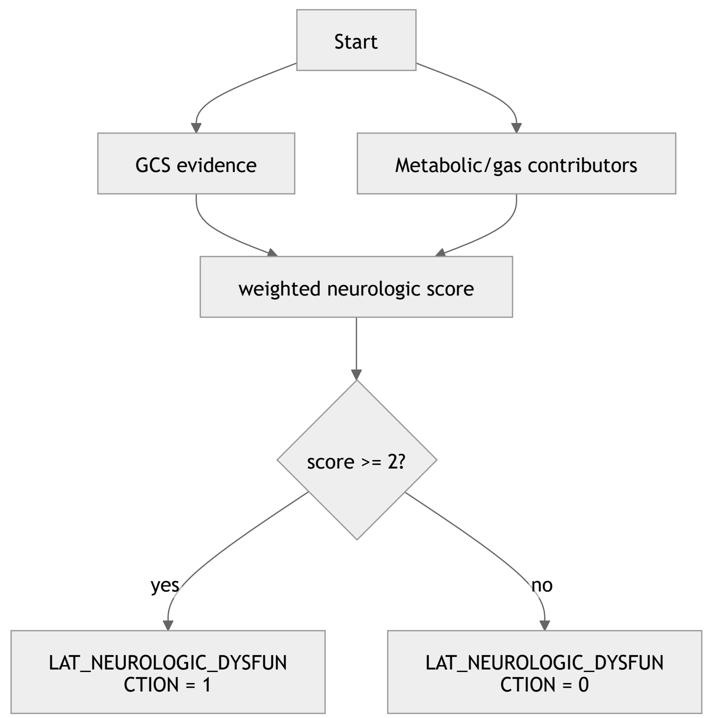
4.9 Neurologic dysfunction

Latent variable: `LAT_NEUROLOGIC_DYSFUNCTION`

Clinical interpretation of label 1: depressed consciousness or neurologic dysfunction, acknowledging sedation ambiguity.

Component	Rule
Input variables	`GCS` , `Na` , `Glucose` , `PaCO2` , `SaO2` , `pH` , `MechVent`
Domain indicators	`GCS_MILD = GCS_min < 15` ; `GCS_SEVERE = GCS_min <= 12` ; `METAB_NEURO = Na_min < 130 OR Na_max > 150 OR Glucose_min < 70 OR Glucose_max > 300` ; `GAS_NEURO = PaCO2_max >= 50 OR SaO2_min < 90 OR pH_min < 7.25` ; `VENT_CONTEXT = MechVent_max == 1`
Score	`score = 2*GCS_SEVERE + GCS_MILD + METAB_NEURO + GAS_NEURO`
Threshold	`1 if score >= 2`
Missingness	Missing GCS scores 0; ventilation is stored as ambiguity/context flag, not a score point
Confidence	Medium

```
def tag_neurologic_dysfunction(x):
    gcs_mild = present(x.GCS_min) and x.GCS_min < 15
    gcs_severe = present(x.GCS_min) and x.GCS_min <= 12
    metab_neuro = any_present_true([lt(x.Na_min, 130), gt(x.Na_max, 150), lt(x.Glucose_min, 70), gt(x.Glucose_ma
    gas_neuro = any_present_true([ge(x.PaCO2_max, 50), lt(x.SaO2_min, 90), lt(x.pH_min, 7.25)])
    score = 2*int(gcs_severe) + sum([gcs_mild, metab_neuro, gas_neuro])
    return int(score >= 2)
```



Clinical rationale: SOFA neurologic scoring uses GCS, but GCS can be affected by sedation, ventilation, and metabolic disturbance. MSD Manuals

False positives: sedated or intubated patients.

False negatives: neurologic injury without recorded GCS.

4.10 Cardiac injury or cardiac strain

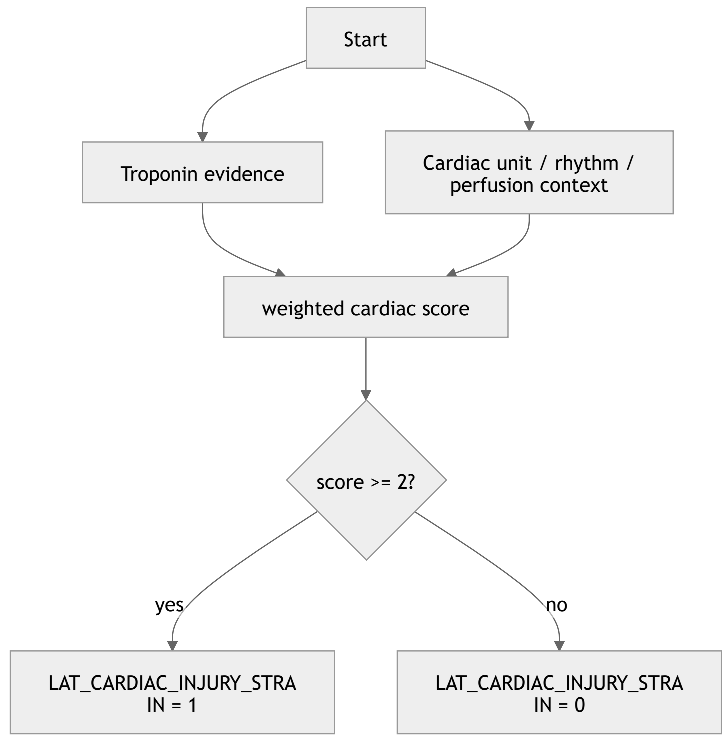
Latent variable: `LAT_CARDIAC_INJURY_STRAIN`

Clinical interpretation of label 1: evidence of myocardial injury, ischemia, or severe cardiac strain.

Component	Rule
Input variables	`TropI`, `TropT`, `HR`, `MAP`, `SysABP`, `Lactate`, `ICUType`
Domain indicators	`TROP_ABN = TropI_max > 0.1 OR TropT_max > 0.01`; `CARDIAC_UNIT = ICUType in {1,2}`; `ARRHYTHMIC_STRESS = HR_max >= 130 OR HR_min < 50`; `HEMO_STRAIN = MAP_min < 70 OR SysABP_min <= 90`; `PERFUSION_STRESS = Lactate_max > 2`
Score	`score = 2*TROP_ABN + CARDIAC_UNIT + ARRHYTHMIC_STRESS + HEMO_STRAIN + PERFUSION_STRESS`
Threshold	`1 if score >= 2`
Missingness	Missing troponin does not imply no cardiac injury; output `troponin_measured` flag
Confidence	Medium-low

```
def tag_cardiac_injury_strain(x):
    trop_abn = any_present_true([gt(x.TropI_max, 0.1), gt(x.TropT_max, 0.01)])
    cardiac_unit = x.ICUType in [1, 2]
```

```
    arrhythmic_stress = any_present_true([ge(x.HR_max, 130), lt(x.HR_min, 50)])
    hemo_strain = any_present_true([lt(x.MAP_min, 70), le(x.SysABP_min, 90)])
    perfusion_stress = present(x.Lactate_max) and x.Lactate_max > 2
    score = 2*int(trop_abn) + sum([cardiac_unit, arrhythmic_stress, hemo_strain, perfusion_stress])
    return int(score >= 2)
```



Clinical rationale: troponin is specific when measured, while HR, hypotension, and lactate capture strain and hypoperfusion context.
False positives: demand ischemia, renal failure-related troponin elevation, post-operative cardiac monitoring.
False negatives: unmeasured troponin or assay-specific thresholds.

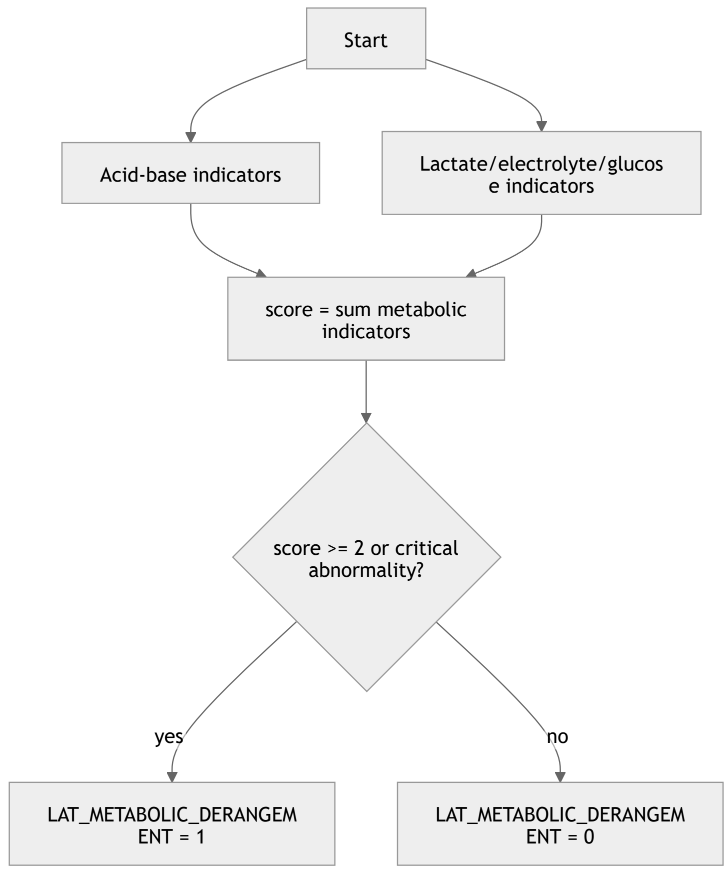
4.11 Metabolic, acid-base, and electrolyte derangement

Latent variable: `LAT_METABOLIC_DERANGEMENT`

Clinical interpretation of label 1: clinically meaningful acid-base, lactate, electrolyte, or glucose derangement.

Component	Rule
Input variables	`pH` , `HCO3` , `Lactate` , `Na` , `K` , `Mg` , `Glucose` , `PaCO2`
Domain indicators	`PH_ABN = pH_min < 7.30 OR pH_max > 7.50` ; `BICARB_ABN = HCO3_min < 18 OR HCO3_max > 32` ; `LACTATE_ABN = Lactate_max > 2` ; `ELECTROLYTE_ABN = Na_min < 130 OR Na_max > 150 OR K_min < 3.0 OR K_max > 5.5 OR Mg_min < 0.6 OR Mg_max > 1.2` ; `GLUCOSE_ABN = Glucose_min < 70 OR Glucose_max > 250` ; `VENT_COMP = PaCO2_min < 32 OR PaCO2_max > 50`
Score	`score = sum(indicators)`
Threshold	`1 if score >= 2` , or `1 if pH_min < 7.20 OR Lactate_max >= 4 OR K_max >= 6.0`
Missingness	Missing ABG/electrolytes score 0; store evidence counts
Confidence	High

```
def tag_metabolic_derangement(x):
    ph_abn = any_present_true([lt(x.pH_min, 7.30), gt(x.pH_max, 7.50)])
    bicarb_abn = any_present_true([lt(x.HCO3_min, 18), gt(x.HCO3_max, 32)])
    lactate_abn = present(x.Lactate_max) and x.Lactate_max > 2
    electrolyte_abn = any_present_true([
        lt(x.Na_min, 130), gt(x.Na_max, 150),
        lt(x.K_min, 3.0), gt(x.K_max, 5.5),
        lt(x.Mg_min, 0.6), gt(x.Mg_max, 1.2)
    ])
    glucose_abn = any_present_true([lt(x.Glucose_min, 70), gt(x.Glucose_max, 250)])
    vent_comp = any_present_true([lt(x.PaCO2_min, 32), gt(x.PaCO2_max, 50)])
    score = sum([ph_abn, bicarb_abn, lactate_abn, electrolyte_abn, glucose_abn, vent_comp])
    critical = any_present_true([lt(x.pH_min, 7.20), ge(x.Lactate_max, 4), ge(x.K_max, 6.0)])
    return int(score >= 2 or critical)
```



Clinical rationale: metabolic derangement is a common mediator from shock, renal failure, respiratory failure, and sepsis-like inflammation to mortality.
False positives: transient treatment-related electrolyte or glucose changes.
False negatives: missing ABG or labs in less-monitored patients.

5. Observed-to-latent measurement model

Observed variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care/measurer process influer
`Age`	`LAT_CHRONIC_BASELINE_RISK`; direct BG role	Higher worse	Non-specific	Yes	No
`Gender`	BG only	Context-dependent	Non-specific	No	No
`Height`	`LAT_CHRONIC_BASELINE_RISK` via BMI	Extreme BMI worse	Non-specific	Yes	Missingness possible

Observed variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care/measurer process influer
Admission `Weight`	`LAT_CHRONIC_BASELINE_RISK` via BMI	Extreme BMI worse	Non-specific	Yes	Measurement e possible
`ICUType`	`LAT_CHRONIC_BASELINE_RISK` , `LAT_CARDIAC_INJURY_STRAIN` ; BG case-mix	Medical/cardiac units imply different risk	Non-specific	Yes	Yes
`Albumin`	Chronic risk, hepatic, inflammation	Low worse	Non-specific	Yes	Lab ordering
`ALP`	Hepatic dysfunction	High worse	Medium	Yes	Lab ordering
`ALT`	Hepatic dysfunction	High worse	Medium	Yes	Lab ordering
`AST`	Hepatic, cardiac/muscle injury	High worse	Low-medium	Yes	Lab ordering
`Bilirubin`	Hepatic dysfunction, global severity	High worse	Medium-high	Yes	Lab ordering
`BUN`	Renal dysfunction, shock/dehydration	High worse	Medium	Yes	Lab ordering
`Cholesterol`	Chronic/nutritional state	Low or high context-dependent	Low	No	Lab ordering
`Creatinine`	Renal dysfunction	High/rising worse	High for renal filtration	Yes	Lab ordering
`DiasABP`	Shock	Low worse	Medium	No direct, BP family	Invasive monito
`SysABP`	Shock/global severity	Low worse	Medium	Yes	Invasive monito
`MAP`	Shock/global severity	Low worse	High	Yes	Invasive monito
`NIDiasABP`	Shock	Low worse	Medium	No direct, BP family	Non-invasive frequency
`NISysABP`	Shock/global severity	Low worse	Medium	Yes	Non-invasive frequency
`NIMAP`	Shock/global severity	Low worse	High	Yes	Non-invasive frequency
`FiO2`	Respiratory failure; treatment intensity	High oxygen requirement worse	Medium	Yes via PF	Strong care-pro
`GCS`	Neurologic dysfunction/global severity	Low worse	Medium	Yes	Sedation/ventil
`Glucose`	Metabolic derangement	Low or high worse	Medium	Yes	Treatment/nutr
`HCO3`	Metabolic, renal, shock	Low/high abnormal	Medium	Yes	Lab/ABG order
`HCT`	Coag/heme, fluid status	Low/high abnormal	Medium-low	Yes	Transfusion/flui effects unavaila

Observed variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care/measurer process influer
HR	Shock, inflammation, cardiac strain, global severity	High or very low worse	Non-specific	Yes	Continuous monitoring
K	Renal/metabolic derangement	Low/high worse	Medium	Yes	Lab ordering/treatr
Lactate	Shock, metabolic, inflammation, global severity	High worse	High for hypoperfusion/stress	Yes	Ordering highly informative
Mg	Metabolic derangement	Low/high worse	Medium-low	Yes	Lab ordering/treatr
MechVent	Respiratory failure, global severity; care process	1 worse as severity proxy	Medium	Yes	True intervention/ca process
Na	Metabolic/neurologic derangement	Low/high worse	Medium	Yes	Lab ordering/treatr
PaCO2	Respiratory failure, metabolic compensation	Low/high abnormal	Medium	Yes	ABG ordering
PaO2	Respiratory failure	Low worse, but depends on FiO2	Medium	Yes	ABG/oxygen therapy
pH	Metabolic/respiratory derangement/global severity	Low or high worse	High	Yes	ABG ordering
Platelets	Coag/heme, inflammation, hepatic/global severity	Low worse	High for platelet domain	Yes	Lab ordering
RespRate	Respiratory failure, inflammation, metabolic acidosis	High or very low worse	Medium	Yes	Charting freque
SaO2	Respiratory failure	Low worse	Medium	Yes	Oxygen therap
Temp	Inflammation/sepsis-like burden	High/low worse	Medium	Yes	Measurement frequency
TropI	Cardiac injury/strain	High worse	High when measured	Yes	Ordering highly informative
TropT	Cardiac injury/strain	High worse	High when measured	Yes	Ordering highly informative
Urine	Renal dysfunction, shock, care/charting	Low worse	Medium-high	Yes	Foley/charting intensity
WBC	Inflammation, coag/heme stress	Low/high worse	Medium	Yes	Lab ordering
Serial Weight	Fluid balance/care process	Rapid gain/loss possibly worse	Low	No main tree	Measurement frequency

6. Latent causal DAG edge list

Source node	Target node	Edge type	Clinical mechanism	Confidence	Re ba
BG_Age	LAT_CHRONIC_BASELINE_RISK	background-to-latent	Older age increases frailty/comorbidity vulnerability	High	Cli

Source node	Target node	Edge type	Clinical mechanism	Confidence	Re ba
BG_HeightWeightBMI	LAT_CHRONIC_BASELINE_RISK	background-to-latent	Extreme body habitus affects physiologic reserve	Medium	Cli
BG_ICUType	LAT_CHRONIC_BASELINE_RISK	background-to-latent	ICU unit captures case-mix and admission pathway	Medium	Da
BG_ICUType	LAT_CARDIAC_INJURY_STRAIN	background-to-latent	CCU/CSRU case-mix increases cardiac injury likelihood	Medium	Da
LAT_CHRONIC_BASELINE_RISK	LAT_GLOBAL_SEVERITY	latent-to-latent	Low reserve worsens physiologic response to acute illness	Medium	Cli
LAT_CHRONIC_BASELINE_RISK	OUT_InHospitalMortality	latent-to-mortality	Baseline vulnerability may affect mortality independent of measured physiology	Medium	Cli
LAT_INFLAMMATION_SEPSIS_BURDEN	LAT_GLOBAL_SEVERITY	latent-to-latent	Systemic inflammation can drive multi-organ dysfunction	Medium	Se dy lite
LAT_INFLAMMATION_SEPSIS_BURDEN	LAT_SHOCK	latent-to-latent	Distributive shock from inflammatory vasodilation/capillary leak	Medium	Se ph
LAT_INFLAMMATION_SEPSIS_BURDEN	LAT_COAG_HEME_DYSFUNCTION	latent-to-latent	Inflammation can lower platelets and cause hematologic stress	Medium	Cli
LAT_CARDIAC_INJURY_STRAIN	LAT_SHOCK	latent-to-latent	Pump failure/ischemia can cause hypotension and hypoperfusion	Medium	Cli
LAT_RESPIRATORY_FAILURE	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hypercapnia/hypoxemia contributes to acidosis and lactate stress	Medium	Cli
LAT_SHOCK	LAT_RENAL_DYSFUNCTION	latent-to-latent	Renal hypoperfusion causes oliguria and kidney injury	High	Cli
LAT_SHOCK	LAT_HEPATIC_DYSFUNCTION	latent-to-latent	Hypoperfusion can cause ischemic/cholestatic liver injury	Medium	Cli
LAT_SHOCK	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hypoperfusion raises lactate and acidosis	High	Cli
LAT_RENAL_DYSFUNCTION	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Kidney failure causes electrolyte and acid-base derangements	High	Cli
LAT_HEPATIC_DYSFUNCTION	LAT_COAG_HEME_DYSFUNCTION	latent-to-latent	Liver disease can contribute to thrombocytopenia/coagulopathy	Low	INI we ob
LAT_GLOBAL_SEVERITY	OUT_InHospitalMortality	latent-to-mortality	Multi-organ severity increases death risk	High	ICU lite
LAT_SHOCK	OUT_InHospitalMortality	latent-to-mortality	Hypoperfusion and circulatory collapse increase mortality	High	Cli
LAT_RESPIRATORY_FAILURE	OUT_InHospitalMortality	latent-to-mortality	Hypoxemia/ventilatory failure can be fatal	High	Cli
LAT_RENAL_DYSFUNCTION	OUT_InHospitalMortality	latent-to-mortality	Kidney failure is organ dysfunction and severity marker	Medium-high	SO
LAT_METABOLIC_DERANGEMENT	OUT_InHospitalMortality	latent-to-mortality	Severe acidosis/electrolyte derangement can precipitate death	High	Cli

Source node	Target node	Edge type	Clinical mechanism	Confidence	Re ba
LAT_NEUROLOGIC_DYSFUNCTION	OUT_InHospitalMortality	latent-to-mortality	Depressed consciousness reflects severe brain/systemic dysfunction	Medium	SO
LAT_CARDIAC_INJURY_STRAIN	OUT_InHospitalMortality	latent-to-mortality	Myocardial injury/strain may cause or mark fatal physiology	Medium	Cli
LAT_SHOCK	OBS_Hemodynamics	latent-to-observed	Shock manifests as low BP/tachycardia	High	Me mc
LAT_RESPIRATORY_FAILURE	OBS_RespiratoryGasExchange	latent-to-observed	Respiratory failure manifests as low PF ratio, hypoxemia, hypercapnia	High	Me mc
LAT_RENAL_DYSFUNCTION	OBS_RenalLabsUrine	latent-to-observed	Renal dysfunction manifests as creatinine/BUN/urine/K/HCO3 changes	High	Me mc
LAT_HEPATIC_DYSFUNCTION	OBS_LiverLabs	latent-to-observed	Liver dysfunction manifests as bilirubin/transaminase/albumin abnormalities	Medium-high	Me mc
LAT_COAG_HEME_DYSFUNCTION	OBS_CBCPlatelets	latent-to-observed	Hematologic dysfunction manifests as platelet/HCT/WBC abnormalities	High	Me mc
LAT_INFLAMMATION_SEPSIS_BURDEN	OBS_TempWBCInflam	latent-to-observed	Inflammation manifests as fever/hypothermia and WBC abnormalities	Medium	Me mc
LAT_NEUROLOGIC_DYSFUNCTION	OBS_GCS	latent-to-observed	Neurologic dysfunction manifests as lower GCS	Medium-high	Me mc
LAT_CARDIAC_INJURY_STRAIN	OBS_TroponinHR	latent-to-observed	Cardiac injury manifests as troponin and rhythm/HR abnormalities	Medium	Me mc
LAT_METABOLIC_DERANGEMENT	OBS_MetabolicLabsABG	latent-to-observed	Metabolic derangement manifests as pH/HCO3/lactate/electrolyte/glucose abnormalities	High	Me mc
LAT_RESPIRATORY_FAILURE	TRT_MechanicalVentilation	treatment/care-process	Respiratory failure triggers ventilation	High	Ca
LAT_GLOBAL_SEVERITY	TRT_MechanicalVentilation	treatment/care-process	Overall severity increases probability of ventilation	Medium	Co by
TRT_MechanicalVentilation	OBS_RespiratoryGasExchange	treatment-or-care-process	Ventilation and FiO2 affect measured gases/oxygenation	High	Cli me
LAT_GLOBAL_SEVERITY	MISS_MeasurementIntensity	measurement-process	Sicker patients are measured more often	High	EH mi lite
LAT_SHOCK	MISS_LactateABGOrdering	measurement-process	Suspected hypoperfusion triggers lactate/ABG testing	High	Cli wo
LAT_CARDIAC_INJURY_STRAIN	MISS_TroponinOrdering	measurement-process	Suspected cardiac injury triggers troponin measurement	High	Cli wo
BG_ICUType	MISS_MeasurementIntensity	measurement-process	ICUs differ in monitoring/lab routines	Medium	Da wo
MISS_MeasurementIntensity	OBS_AvailabilityCounts	measurement-process	Ordering/charting determines missingness and counts	High	EH

DAG-based causal inference requires making assumptions explicit; confounders close back-door paths, mediators lie on causal paths, and conditioning on colliders can open biasing paths. ora.ox.ac.uk

7. Mortality mechanism summary

The main mortality mechanism is not “low MAP causes death” or “high creatinine causes death” as isolated measurements. Rather, **latent severity states** cause both abnormal observations and mortality.

The dominant causal paths are:

1. `BG_Age / BG_ICUType / BG_HeightWeightBMI → LAT_CHRONIC_BASELINE_RISK → LAT_GLOBAL_SEVERITY → OUT_InHospitalMortality`.
2. `LAT_INFLAMMATION_SEPSIS_BURDEN → LAT_SHOCK → LAT_RENAL_DYSFUNCTION → LAT_METABOLIC_DERANGEMENT → OUT_InHospitalMortality`.
3. `LAT_RESPIRATORY_FAILURE → TRT_MechanicalVentilation → OBS_RespiratoryGasExchange`, while `LAT_RESPIRATORY_FAILURE → LAT_METABOLIC_DERANGEMENT → OUT_InHospitalMortality`.
4. `LAT_CARDIAC_INJURY_STRAIN → LAT_SHOCK → LAT_GLOBAL_SEVERITY → OUT_InHospitalMortality`.
5. `LAT_SHOCK → OBS_MAP/SBP/Lactate/Urine/pH` and `LAT_SHOCK → OUT_InHospitalMortality`; the observations are downstream manifestations, not the underlying causal state itself.
6. `LAT_GLOBAL_SEVERITY → MISS_MeasurementIntensity → OBS_AvailabilityCounts`; the act of being measured more often is a consequence of illness and clinical concern, not a randomly assigned exposure.

8. Missingness and irregular sampling analysis

DATASET-NAME should not be treated as MCAR. The official description states that variables may be observed once, repeatedly, or not at all, with irregular measurement intervals and occasional outliers. physionet.org EHR missingness can arise because measurements are generated by clinical care, and measurement frequency can depend on how abnormal a value is expected to be; missingness can therefore be informative and may be MNAR. JMIR Medical Inf...

Variable family	Likely missingness mechanism	Why	Modeling recommendation
Routine vitals: HR, BP, Temp, RespRate	MAR / care-intensity-driven	More frequent in sicker or monitored patients	Use values plus counts/frequency
Invasive BP: `SysABP`, `DiasABP`, `MAP`	MNAR / care-process	Arterial lines placed in unstable patients	Separate invasive-monitoring indicator
ABGs: `PaO2`, `PaCO2`, `pH`, `SaO2`, `FiO2`	MNAR / suspicion-driven	Ordered for respiratory/metabolic concern	Do not impute normal; include ABG-ordering sensitivity
Lactate	MNAR / suspicion-driven	Ordered when shock/sepsis/hypoperfusion suspected	Missing lactate is not normal; count/order flag useful
Troponin	MNAR / suspicion-driven	Ordered for suspected myocardial injury	Missing troponin should not mean no injury
Liver labs	MAR/MNAR	Ordered based on illness, medication, liver concern	Missing panel reduces confidence
Urine	Care-process / charting-dependent	Foley and hourly charting more likely in severe illness	Use only when available; include urine-count flag
GCS	MAR/MNAR	Sedation/intubation and neuro concern affect measurement	Ventilation context flag needed
Serial weight	Care-process-driven	Often related to fluid management workflow	Avoid primary latent definition

For binary trees, missingness should generally **not count as abnormal**. However, missingness indicators and measurement counts should be preserved in a separate feature table because they may proxy clinician suspicion and care intensity. Using missingness directly in severity labels risks encoding local practice patterns and collider bias; using it as a sensitivity feature is safer.

9. Backdoor/confounding implications

For causal effects on mortality, safe adjustment depends on the exposure.

Exposure question	Likely confounders	Likely mediators to avoid for total effect	Collider risks	Recommendation
Effect of `TRT_MechanicalVentilation` on mortality	Age, ICUType, chronic baseline risk, pre-ventilation respiratory failure/global severity	Post-ventilation gases, later global severity	Conditioning on measurement intensity or only ventilated patients	Need time-indexed de do not naively adjust f downstream oxygenat
Effect of high lactate as exposure	Chronic risk, ICUType, shock/sepsis/cardiac state	Metabolic derangement, renal dysfunction if downstream	Lactate ordering is a collider of clinician suspicion and severity	Prefer modeling `LAT_S` not lactate as causal exposure
Effect of renal dysfunction on mortality	Age, chronic risk, ICUType, shock/sepsis	Metabolic derangement may mediate renal effect	Lab ordering and complete cases	Adjust for pre-renal confounders, not downstream acidosis i effect desired
Effect of inflammation/sepsis-like burden on mortality	Age, chronic risk, ICUType	Shock, organ dysfunction, global severity	Measuring only patients with lactate/WBC can induce selection	Do not adjust for glob severity when estimati total inflammatory bu effect
Effect of ICUType on mortality	Age, baseline risk/case mix	Acute severity may mediate case-mix effect	Conditioning on 48h survivors/stays is selection	Interpret cautiously; IC is not randomized
Effect of global severity on mortality	Age, chronic risk, ICUType	Organ-specific downstream variables if defined later	Measurement intensity	Severity is outcome-pr useful for prediction, n always valid adjustmen

Binary latent tags are not automatically safe adjustment variables. `LAT_GLOBAL_SEVERITY`, `LAT_SHOCK`, `LAT_METABOLIC_DERANGEMENT`, and organ dysfunction tags are often **mediators or outcome-proximal proxy outcomes**. They may be appropriate for risk adjustment or prediction, but adjusting for them can block part of the causal pathway for treatment or exposure effects. DAG literature specifically warns against treating mediators like confounders and against conditioning on colliders. ora.ox.ac.uk

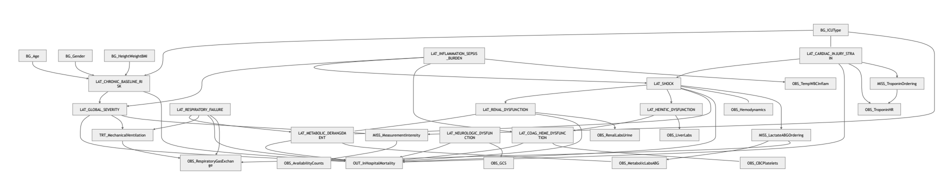
10. Graph specification

A. Adjacency list

```
BG_Age:
  -> LAT_CHRONIC_BASELINE_RISK

BG_Gender:
  -> LAT_CHRONIC_BASELINE_RISK

BG_HeightWeightBMI:
  -> LAT_CHRONIC_BASELINE_RISK

BG_ICUType:
  -> LAT_CHRONIC_BASELINE_RISK
  -> LAT_CARDIAC_INJURY_STRAIN
  -> MISS_MeasurementIntensity

LAT_CHRONIC_BASELINE_RISK:
  -> LAT_GLOBAL_SEVERITY
  -> OUT_InHospitalMortality

LAT_INFLAMMATION_SEPSIS_BURDEN:
  -> LAT_GLOBAL_SEVERITY
  -> LAT_SHOCK
  -> LAT_COAG_HEME_DYSFUNCTION
  -> OBS_TempWBCInflam

LAT_CARDIAC_INJURY_STRAIN:
  -> LAT_SHOCK
  -> OUT_InHospitalMortality
```

```
  -> OBS_TroponinHR
  -> MISS_TroponinOrdering

LAT_RESPIRATORY_FAILURE:
  -> LAT_METABOLIC_DERANGEMENT
  -> TRT_MechanicalVentilation
  -> OUT_InHospitalMortality
  -> OBS_RespiratoryGasExchange

LAT_SHOCK:
  -> LAT_RENAL_DYSFUNCTION
  -> LAT_HEPATIC_DYSFUNCTION
  -> LAT_METABOLIC_DERANGEMENT
  -> OUT_InHospitalMortality
  -> OBS_Hemodynamics
  -> MISS_LactateABGOrdering

LAT_RENAL_DYSFUNCTION:
  -> LAT_METABOLIC_DERANGEMENT
  -> OUT_InHospitalMortality
  -> OBS_RenalLabsUrine

LAT_HEPATIC_DYSFUNCTION:
  -> LAT_COAG_HEME_DYSFUNCTION
  -> OBS_LiverLabs

LAT_COAG_HEME_DYSFUNCTION:
  -> OUT_InHospitalMortality
  -> OBS_CBCPlatelets

LAT_NEUROLOGIC_DYSFUNCTION:
  -> OUT_InHospitalMortality
  -> OBS_GCS

LAT_METABOLIC_DERANGEMENT:
  -> OUT_InHospitalMortality
  -> OBS_MetabolicLabsABG

LAT_GLOBAL_SEVERITY:
  -> OUT_InHospitalMortality
  -> TRT_MechanicalVentilation
  -> MISS_MeasurementIntensity

TRT_MechanicalVentilation:
  -> OBS_RespiratoryGasExchange

MISS_MeasurementIntensity:
  -> OBS_AvailabilityCounts

MISS_LactateABGOrdering:
  -> OBS_MetabolicLabsABG
  -> OBS_RespiratoryGasExchange

MISS_TroponinOrdering:
  -> OBS_TroponinHR
```

B. Mermaid graph



C. Python NetworkX edge list

```
edges = [
    ("BG_Age", "LAT_CHRONIC_BASELINE_RISK"),
    ("BG_Gender", "LAT_CHRONIC_BASELINE_RISK"),
```

```
    ("BG_HeightWeightBMI", "LAT_CHRONIC_BASELINE_RISK"),
    ("BG_ICUType", "LAT_CHRONIC_BASELINE_RISK"),
    ("BG_ICUType", "LAT_CARDIAC_INJURY_STRAIN"),
    ("BG_ICUType", "MISS_MeasurementIntensity"),

    ("LAT_CHRONIC_BASELINE_RISK", "LAT_GLOBAL_SEVERITY"),
    ("LAT_CHRONIC_BASELINE_RISK", "OUT_InHospitalMortality"),

    ("LAT_INFLAMMATION_SEPSIS_BURDEN", "LAT_GLOBAL_SEVERITY"),
    ("LAT_INFLAMMATION_SEPSIS_BURDEN", "LAT_SHOCK"),
    ("LAT_INFLAMMATION_SEPSIS_BURDEN", "LAT_COAG_HEME_DYSFUNCTION"),
    ("LAT_INFLAMMATION_SEPSIS_BURDEN", "OBS_TempWBCInflam"),

    ("LAT_CARDIAC_INJURY_STRAIN", "LAT_SHOCK"),
    ("LAT_CARDIAC_INJURY_STRAIN", "OUT_InHospitalMortality"),
    ("LAT_CARDIAC_INJURY_STRAIN", "OBS_TroponinHR"),
    ("LAT_CARDIAC_INJURY_STRAIN", "MISS_TroponinOrdering"),

    ("LAT_RESPIRATORY_FAILURE", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_RESPIRATORY_FAILURE", "TRT_MechanicalVentilation"),
    ("LAT_RESPIRATORY_FAILURE", "OUT_InHospitalMortality"),
    ("LAT_RESPIRATORY_FAILURE", "OBS_RespiratoryGasExchange"),

    ("LAT_SHOCK", "LAT_RENAL_DYSFUNCTION"),
    ("LAT_SHOCK", "LAT_HEPATIC_DYSFUNCTION"),
    ("LAT_SHOCK", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_SHOCK", "OUT_InHospitalMortality"),
    ("LAT_SHOCK", "OBS_Hemodynamics"),
    ("LAT_SHOCK", "MISS_LactateABGOrdering"),

    ("LAT_RENAL_DYSFUNCTION", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_RENAL_DYSFUNCTION", "OUT_InHospitalMortality"),
    ("LAT_RENAL_DYSFUNCTION", "OBS_RenalLabsUrine"),

    ("LAT_HEPATIC_DYSFUNCTION", "LAT_COAG_HEME_DYSFUNCTION"),
    ("LAT_HEPATIC_DYSFUNCTION", "OBS_LiverLabs"),

    ("LAT_COAG_HEME_DYSFUNCTION", "OUT_InHospitalMortality"),
    ("LAT_COAG_HEME_DYSFUNCTION", "OBS_CBCPlatelets"),

    ("LAT_NEUROLOGIC_DYSFUNCTION", "OUT_InHospitalMortality"),
    ("LAT_NEUROLOGIC_DYSFUNCTION", "OBS_GCS"),

    ("LAT_METABOLIC_DERANGEMENT", "OUT_InHospitalMortality"),
    ("LAT_METABOLIC_DERANGEMENT", "OBS_MetabolicLabsABG"),

    ("LAT_GLOBAL_SEVERITY", "OUT_InHospitalMortality"),
    ("LAT_GLOBAL_SEVERITY", "TRT_MechanicalVentilation"),
    ("LAT_GLOBAL_SEVERITY", "MISS_MeasurementIntensity"),

    ("TRT_MechanicalVentilation", "OBS_RespiratoryGasExchange"),
    ("MISS_MeasurementIntensity", "OBS_AvailabilityCounts"),
    ("MISS_LactateABGOrdering", "OBS_MetabolicLabsABG"),
    ("MISS_LactateABGOrdering", "OBS_RespiratoryGasExchange"),
    ("MISS_TroponinOrdering", "OBS_TroponinHR"),
]

import networkx as nx
G = nx.DiGraph()
G.add_edges_from(edges)
assert nx.is_directed_acyclic_graph(G)
```

11. Decision-tree implementation specification

11.1 Required input columns

```
RecordID, Age, Gender, Height, ICUType, Weight,
Albumin, ALP, ALT, AST, Bilirubin, BUN, Cholesterol, Creatinine,
DiasABP, FiO2, GCS, Glucose, HCO3, HCT, HR, K, Lactate, Mg,
MAP, MechVent, Na, NIDiasABP, NIMAP, NISysABP, PaCO2, PaO2,
pH, Platelets, RespRate, SaO2, SysABP, Temp, TropI, TropT,
Urine, WBC
```

Do **not** use `SAPS-I`, `SOFA`, `Length_of_stay`, `Survival`, or `In-hospital_death` for latent tag generation.

11.2 Derived summary features

For each time-varying variable `V`:

```
V_first
V_last
V_min
V_max
V_mean
V_median
V_count
V_missing = 1 if V_count == 0 else 0
V_delta = V_last - V_first when count >= 2
V_slope = robust slope over time when count >= 2
V_variability = IQR or standard deviation when count >= 2
```

Special derived features:

```
BMI = Weight_first / (Height / 100)^2
PF_ratio = PaO2 / FiO2 for near-paired measurements with FiO2 in (0, 1]
PF_min = minimum valid PF_ratio
Urine_24h_sum = sum urine output in each 24h block
Urine_24h_min = minimum available 24h urine sum
total_measurement_count = sum(V_count for all time-varying variables)
abg_count = count(PaO2, PaCO2, pH measurements)
lactate_measured = Lactate_count > 0
troponin_measured = TropI_count > 0 or TropT_count > 0
```

11.3 Expected binary output columns

```
LAT_CHRONIC_BASELINE_RISK
LAT_GLOBAL_SEVERITY
LAT_SHOCK
LAT_RESPIRATORY_FAILURE
LAT_RENAL_DYSFUNCTION
LAT_HEPATIC_DYSFUNCTION
LAT_COAG_HEME_DYSFUNCTION
LAT_INFLAMMATION_SEPSIS_BURDEN
LAT_NEUROLOGIC_DYSFUNCTION
LAT_CARDIAC_INJURY_STRAIN
LAT_METABOLIC_DERANGEMENT
```

11.4 Threshold table

Domain	Thresholds
Hypotension	`MAP < 70`, `NIMAP < 70`, `SysABP <= 90`, `NISysABP <= 90`, qSOFA-like systolic `<=100` for global severity
Respiratory	`PF < 300`, `SaO2 < 92`, `PaO2 < 60`, `RespRate >=22`, `PaCO2 >=50`
Renal	`Creatinine >=2.0`, `Creatinine rise >=0.3`, `BUN >=40`, `Urine_24h <500`, `K >=5.5`
Hepatic	`Bilirubin >=2.0`, `AST/ALT >=200`, `ALP >=250`, `Albumin <2.5`
Coag/heme	`Platelets <150`, severe `<100`, `HCT <25`, `WBC <4 or >20`
Inflammation	`Temp >38.3 or <36`, `WBC >12 or <4`, `HR >90`, `RR >20`, `PaCO2 <32`
Neurologic	`GCS <15`, severe `GCS <=12`, very severe `GCS <=8`
Cardiac	`TropI >0.1`, `TropT >0.01`, `HR >=130 or <50`
Metabolic	`pH <7.30 or >7.50`, critical `pH <7.20`, `HCO3 <18 or >32`, `Lactate >2`, critical `>=4`, `Na <130 or >150`, `K <3.0 or >5.5`, `Glucose <70 or >250`, `Mg <0.6 or >1.2`

11.5 Validation checks

```
required_checks = [
    "Convert -1 to missing before all summaries.",
    "Do not use mortality, survival, length of stay, SAPS-I, or SOFA in tag functions.",
```

```
    "Ensure all generated LAT_* columns are binary 0/1.",
    "Store evidence counts for each LAT_* tag.",
    "Store missingness/count features separately from biological tags.",
    "Check units match official dataset units.",
    "Flag impossible physiologic values and either winsorize or set to missing.",
    "Run DAG acyclicity check after constructing graph.",
    "Perform sensitivity analyses: first 24h vs first 48h, worst-value vs first-value tags, missingness excluded
]
```

A later preprocessing analysis of DATASET-NAME emphasized that artifacts and outliers in ICU monitor data can affect mortality models and that domain-knowledge bounds or automatic outlier rejection can improve performance. CinC

12. Rejected alternatives

Rejected item	Type	Reason
Confirmed sepsis label	Latent definition	No cultures, antibiotics, infection diagnosis, or suspected-infection timestamp
Vasopressor-dependent septic shock	Latent definition	No vasopressor medication variables
ARDS	Latent definition	No chest imaging, PEEP, bilateral infiltrates, or formal diagnosis
Chronic kidney disease	Latent definition	No pre-ICU baseline creatinine or diagnosis codes
Fluid balance latent state	Latent variable	Serial weight and urine alone are insufficient; no fluid input/diuretics
Nutritional state as separate DAG node	Latent variable	Albumin and BMI are weak and confounded by acute inflammation
Raw lab-to-raw lab dense graph	DAG structure	Confuses measurement association with causal physiology
Direct `Lactate -> Mortality` edge	DAG edge	Lactate is a proxy for shock/metabolic stress, not the primary causal state
Direct `Creatinine -> Mortality` edge	DAG edge	Creatinine measures renal dysfunction; the latent dysfunction is causal
Direct `GCS -> Mortality` edge	DAG edge	GCS measures neurologic dysfunction and sedation context
Using `SOFA` outcome descriptor to define tags	Decision rule	Training-only outcome-related file; would reduce portability and risk leakage
Using `Length_of_stay` or `Survival` in tags	Decision rule	Direct post-outcome leakage
Treating missing lactate as normal	Missingness rule	Lactate missingness may reflect clinician expectation and ordering process
Treating troponin absence as no cardiac injury	Missingness rule	Troponin is suspicion-driven
Conditioning on measurement intensity in all mortality models	Adjustment strategy	Measurement intensity is a descendant of severity and care decisions; collider risk
Adjusting for `LAT_GLOBAL_SEVERITY` when estimating total effect of sepsis-like burden	Adjustment strategy	Blocks mediated effect through organ dysfunction

13. Final confidence and limitations

The strongest latent variables are `LAT_GLOBAL_SEVERITY`, `LAT_SHOCK`, `LAT_RESPIRATORY_FAILURE`, `LAT_RENAL_DYSFUNCTION`, and `LAT_METABOLIC_DERANGEMENT`. They are supported by multiple variables and align with established ICU organ dysfunction constructs. SOFA's organ domains provide external clinical support for respiratory, coagulation, hepatic, cardiovascular, neurologic, and renal dysfunction scoring.

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The weakest approved latent variables are `LAT_CHRONIC_BASELINE_RISK`, `LAT_CARDIAC_INJURY_STRAIN`, and `LAT_INFLAMMATION_SEPSIS_BURDEN`. Chronic vulnerability lacks diagnoses and comorbidity codes; cardiac injury depends on sparse, suspicion-driven troponin testing; and sepsis-like burden lacks infection confirmation.

The most reliable trees are the respiratory, renal, shock, metabolic, and global severity trees. The most uncertain trees are hepatic, coagulation/heme, cardiac injury, and chronic baseline risk. The most uncertain DAG edges are hepatic dysfunction to coagulation dysfunction, chronic baseline vulnerability to global severity, and inflammation/sepsis-like burden to specific organ failures, because the dataset lacks diagnoses, medications, microbiology, vasopressors, and treatment timing.

Additional data that would improve identifiability include admission diagnoses, chronic comorbidity codes, vasopressor doses, antibiotics, cultures, fluid input, diuretics, transfusions, sedation, oxygen delivery mode, ventilator settings, PEEP, baseline creatinine, INR/PT/PTT, and timestamps for treatments. Without these, this graph should be treated as a **clinically plausible causal DAG**, not a proven ground-truth DAG. It is best used for transparent causal assumptions, latent proxy construction, sensitivity analysis, and avoiding naive adjustment for downstream severity, mediators, or colliders.

Sources